

Structure of the periodic table

Learning outcomes

By the end of this section you should be able to:

- Identify the positions of metals, metalloids and nonmetals in the periodic table.
- Identify the block an element belongs to.
- Identify the group and period number of an element.

If you look at the two data tables in **Figure 1** which table do you immediately notice as being more useful? Why is this?

Data table 1

130	180.5
160	97.79
200	63.38
215	39.30

Data table 2

Element	Atomic radius ($\times 10^{-12}$ m)	Melting point ($^{\circ}\text{C}$)
Li	130	180.5
Na	160	97.79
K	200	63.38
Rb	215	39.30

Figure 1. Comparison of two data tables.

In general, tables are more useful if row and column headings are included. This is certainly true for the periodic table. While the periodic table has more information within each of its entry locations, it is incredibly useful to use row and column headings too.

Groups and periods

As Dmitri Mendeleev constructed his version of the periodic table, he arranged certain elements together based on similar properties. These elements now make up the columns of the periodic table we know today and each column is referred to as a group. There are 18 groups in the periodic table, starting with group 1 (H, Li, Na, K, Rb, Cs, Fr) and ending with group 18 (He, Ne, Ar, Kr, Xe, Rn, Og).

The modern periodic table is arranged horizontally by atomic number. Each row is referred to as a period, as each subsequent row shows similar properties of the elements, hence the term periodic. **Figure 2** shows how groups and periods are deduced in the periodic table. Additionally, the periodic table (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) in your DP chemistry data booklet has the group and period numbers labelled.

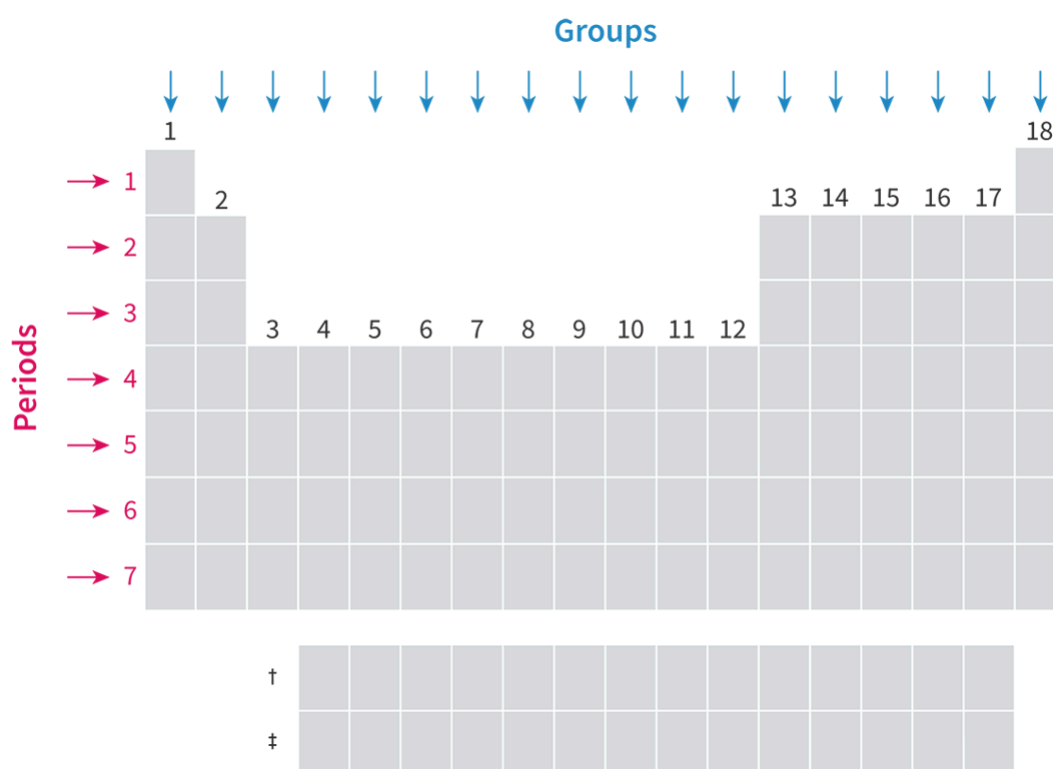


Figure 2. Group and period labels in the periodic table.

 More information for figure 2

The image is a section of the periodic table that specifically highlights the labels indicating groups and periods. Along the top of the periodic table, group numbers are labeled, which are vertical columns of the table. These labels denote the grouping of elements that share similar chemical properties. On the left side, the periods are labeled, which are horizontal rows noted for increasing atomic numbers and changing chemical properties in predictable ways. The layout visually represents the organized structure of elements in the periodic table, showing how groups and periods are organized in a grid-like format. The highlighted labels are meant to aid in understanding the positioning of elements across both dimensions: horizontally as periods and vertically as groups.

[Generated by AI]

The periodic table is found in section 7 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) of the DP chemistry data booklet. Look at the periodic table. Try to locate the element silicon, Si. Notice how it is located in the fourteenth column reading from left to right. This means silicon is in group 14. Notice how silicon is located in the third row reading from top to bottom. This means it is in period 3.

Using the periodic table to help, try to drag and drop these elements, group numbers and period numbers into the correct places.

Interactive 1. Drag and Drop Elements, Group Numbers and Period Numbers Into the Correct Places.

Metals, metalloids and non-metals

Recall from section S2.1.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) that we identified the regions of the periodic table where metals and non-metals can be found. Metals are elements on the left-hand side of the periodic table that exhibit the physical properties of malleability, lustre, high melting points, high thermal and electrical conductivities and have a tendency to lose electrons. Non-metals are elements on the right-hand side of the periodic table that are typically brittle, dull and have lower melting points, low thermal and electrical conductivities and have a tendency to gain electrons. All elements can be classified as metals, non-metals or metalloids. The majority of elements in the periodic table are metals and are found on the left and the middle of the periodic table.

Metalloids are located in a diagonal staircase that forms a boundary between the metals and non-metals, this is shown in **Figure 3**. Metalloids have properties of both metals and non-metals. They typically have high melting points, intermediate electrical and thermal conductivities, are lustrous and brittle. Silicon for example, is shiny like a metal but a poor conductor of electricity under standard conditions.

		Metal					Metalloid					Nonmetal									
H																					He
Li	Be											B	C	N	O	F	Ne				
Na	Mg											Al	Si	P	S	Cl	Ar				
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr				
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe				
Cs	Ba	La †	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn				
Fr	Ra	Ac ‡																			
		†	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu					
		‡	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr					

Figure 3. Classification of metals, non-metals and metalloids in the periodic table.

 More information for figure 3

The image shows a periodic table with elements classified into metals, nonmetals, and metalloids. The table is color-coded, with metals in blue, nonmetals in yellow, and metalloids in green. Metalloids form a diagonal boundary from boron (B) to astatine (At), positioned between metals and nonmetals. Examples of metalloids include silicon (Si), germanium (Ge), arsenic (As), antimony (Sb), tellurium (Te), and polonium (Po). These elements are often located along a staircase line in the table, illustrating their transitional properties between metals and nonmetals.

[Generated by AI]

Nature of Science

Aspect: Science as a shared endeavour

There is disagreement in the scientific community about what classifies an element as a metalloid. The elements B, Si, Ge, As, Te, Sb are widely accepted as metalloids but often Bi, Po and At are considered by some scientists as also having intermediate properties between metals and non-metals. Since the pursuit of knowledge in chemistry involves many research scientists this can cause unresolved disagreements during the classification process. These are important to highlight as they bring awareness to the limitations of the classification methods used.

Try to classify each of the elements as a metal, metalloid or non-metal using this drag and drop activity.

Metal	Metalloid	Non-metal

B C Cu I Na Pb Sb Si Xe

✓ Check

Interactive 2. Classify These Elements as a Metal, Metalloid or Non-Metal.

Blocks in the periodic table

So far, we have considered how to position elements in the periodic table using group number, period number, and the type of element. We can also position elements in the periodic table using the arrangement of their electrons. Can you think about how we might classify elements in the periodic table based on their electron configuration?

Recall from [section S1.3.3–4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/) that energy levels are divided into s, p, d and f-sublevels. Consider the electron configurations for group 1 elements (H, Li, Na, K, etc.), what sublevel do the outermost valence electrons occupy? What sublevel do the outermost valence electrons occupy for group 2? What about group 17? You might notice that the outermost valence electrons occupy sublevels in a predictable manner. **Table 2** shows the electron configuration for some elements in groups 1, 2 and 17 with the outermost valence electrons emboldened.

Table 1. Electron configurations for groups 1, 2 and

Group 1		Group 2		
H	1s¹			
Li	1s²2s¹	Be	1s²2s²	F
Na	1s²2s²2p⁶3s¹	Mg	1s²2s²2p⁶3s²	Cl
K	1s²2s²2p⁶3s²3p₆4s¹	Ca	1s²2s²2p⁶3s²3p₆4s²	Br

This means we can position elements in the periodic table based on the sublevels which the outermost valence electrons occupy. These are called blocks and specify whether an element has the outermost valence electrons occupying the s, p, d or f-sublevels. Hence the regions in the periodic table are referred to as the s-block, p-block, d-block and f-block. These regions are indicated in the periodic table in **Figure 4**. Notice how as you move across period 4 the sublevels the

outermost valence electrons are occupying move from the 4s to 3d then to 4p. This is due to the 4s-sublevel being slightly lower in energy than the 3d from the convergence of energy levels as you learned about in [section S1.3.5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>). You will notice the same thing in period 6 moving from 6s to 4f to 5d to 6p.

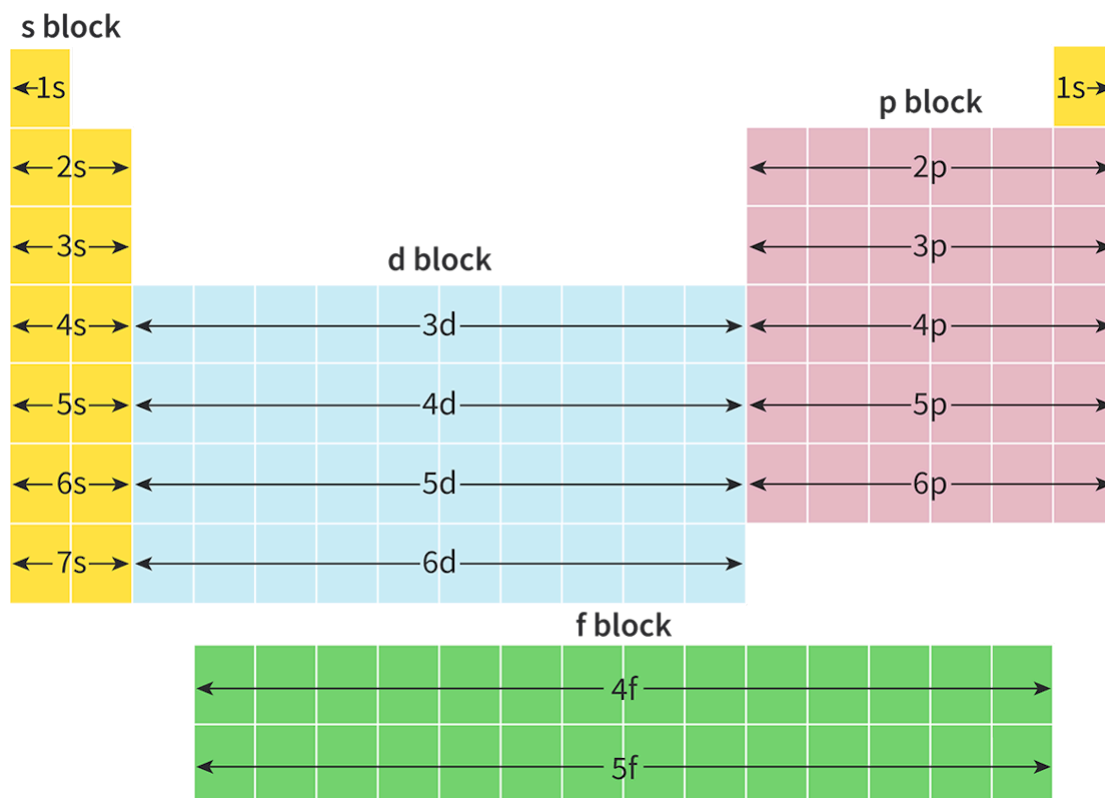


Figure 4. Group and period labels in the periodic table.

More information for figure 4

The image is a diagrammatic representation of the periodic table divided into blocks that indicate where the outermost valence electrons occupy. It shows the layout of the s-block (yellow), p-block (pink), d-block (blue), and f-block (green) across different periods.

The s-block is on the left side, including groups 1 and 2. The p-block is on the right side, spanning groups 13 to 18. The d-block is in the center, covering groups 3 to 12, and consists of transition metals. The f-block is shown as a separate section at the bottom, showcasing the lanthanides and actinides.

Arrows indicate the order in which sublevels are filled within each block, such as 1s, 2s, 3s for the s-block, and 2p, 3p, 4p for the p-block. The d-block arrows display transitions like 3d, 4d, 5d, while the f-block shows 4f and 5f. These directions highlight how electron configurations follow specific patterns when elements are arranged in periods and blocks.

[Generated by AI]

In **Interactive 3**, use the position in the periodic table to drag and drop the elements into the blocks that indicate where their outermost valence electrons occupy.

s	d	p	f

Ag

As

Be

F

Ga

Ho

K

Li

N

Ni

Pu

Sr

U

W

Yb

Zr

☒ Check

Interactive 3. Drag and Drop Table Review for Classification of Elements as s, p, d, and f-Block Elements.

Activity

- **IB learner profile attribute:** Knowledgeable

- **Approaches to learning:** Thinking skills — Asking questions based upon sensible scientific rationale
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair and whole class activity

Element BINGO

You will complete the following activity in groups of 4-6 students.

Part 1: Creation of questions for a BINGO game

- Split up the elements for elements $Z = 1$ to $Z = 54$ evenly amongst the group.
- For the elements you are assigned, you will make up a question for each element. Your question could be based on physical properties, chemical properties, structure and bonding, group and period number, electron configuration, or other information. Aim to have a different style of question for each element. For example, what element is a metalloid and has the highest melting point in period 3? If you look at the melting point data in [section 8 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet you would notice that Si has the highest melting point in period 3 and is considered a metalloid.
- Write the question for each element on an individual piece of paper.
- Each group member will fold up their questions and place them in a container. These will be chosen at random later in the BINGO game.

Part 2: Creation of a BINGO card

- Individually, create your own BINGO card.
- Place the symbol for elements of your choosing (from $Z = 1$ to $Z = 54$) into each space. You can create your own card or use the template below.

[BINGO template](https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/ChemBINGOtemplate.pdf) 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/ChemBINGOtemplate.pdf)

Part 3: BINGO game

- As a group you will play BINGO by selecting at random a question inside the container. Each person will take turns selecting the question.
- Individually, try to deduce the element from the clues in the question. Once all members in the group think they know which element it is, share with the group. The person who made up the question can confirm the identity of the element after each group member has shared their guess.
- Once the element has been confirmed, cross the element out on your BINGO (if you have it)
- When you have a full line crossed out (vertical, horizontal, diagonal) call BINGO.

Electrons and the periodic table

Learning outcomes

By the end of this section you should be able to:

- Deduce the electron configuration of an atom up to $Z = 36$ from the position of the element in the periodic table.
- Deduce the position in the periodic table of an atom of its electron configuration.
- Identify the alkali metals, halogens, transition elements and noble gases in the periodic table.

In the previous section, [S3.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) we looked at blocks in the periodic table which provided information about the sublevel that valence electrons occupy for an element. Throughout [topic S2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) we used the periodic table to make predictions about elements and the valence electrons involved during bonding between elements. Since the periodic table provides useful information about electrons for elements, how might we use it to make predictions about the specific electron configuration of each element?

Let's consider the condensed electron configuration for group 1 metals in **Table 1**. Recall from [section S3.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) that group 1 elements are in the s-block, meaning their outermost valence electrons are occupying an s-sublevel. Notice how all of the elements in group 1 have one electron occupying an s-sublevel.

Table 1. Condensed electron configuration for group 1 metals.

Element	Period number	Condensed electron configuration
Lithium	2	[He] 2s ¹
Sodium	3	[Ne] 3s ¹
Potassium	4	[Ar] 4s ¹
Rubidium	5	[Kr] 5s ¹
Caesium	6	???

What would you predict the condensed electron configuration for caesium would be? The noble gas that would represent the core electrons would be Xe and it is in period 6. Therefore, the condensed electron configuration would be [Xe] 6s¹.

Now, let's consider the condensed electron configuration for group 17 non-metals in **Table 2**. Group 17 elements are in the p-block, meaning their outermost valence electrons are occupying a p-sublevel. Notice how all of the elements in group 17 have seven valence electrons with five electrons occupying a p-sublevel.

Table 2. Condensed electron configuration for group 17 non-metals.

Element	Period number	Condensed electron configuration
Fluorine	2	$[\text{He}] 2s^2 2p^5$
Chlorine	3	$[\text{Ne}] 3s^2 3p^5$
Bromine	4	$[\text{Ar}] 4s^2 3d^{10} 4p^5$
Iodine	5	???

What would you predict the condensed electron configuration for iodine to be?

The noble gas that would represent the core electrons would be Kr and it is in period 5. Therefore, the condensed electron configuration would be $[\text{Kr}] 5s^2 4d^{10} 5p^5$.

The electron configurations of elements are easily predicted using the periodic table because all elements in a group have a common number of valence electrons and the period number shows the outer energy level that is occupied by electrons.

Using the block method to determine electron configuration

We can use the position of elements in the periodic table to deduce their electron configuration; this is known as the block method. It is an alternative method to the order of filling method you looked at in [section S1.3.3–4](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>). The block method can be used by following these guidelines:

- Identify the element of interest in the periodic table.
- Identify the period number to deduce the outermost energy level that will be occupied by electrons.
- Starting with hydrogen, move along each row, left to right, top to bottom writing out the electron configuration for each block in a period until you reach your element.

Implementation of these rules are demonstrated in **Interactive 1** for the element arsenic.

Interactive 1. Using the Block Method to Deduce the Electron Configuration for Arsenic.

 More information for interactive 1

The slideshow with 10 slides displays the Block Method to deduce the Electron Configuration for Arsenic. The slide and the accompanying text read as follows.

Slide 1. The modern periodic table with a highlighted circle on the element "As". The text on the left reads, "Identify the element of interest".

Slide 2. The modern periodic table with a highlighted circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you".

Slide 3. The modern periodic table with the first period highlighted and circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2$.

Slide 4. The modern periodic table with the second period highlighted and circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2 2s^2$.

Slide 5. The modern periodic table with the elements from groups 13 to 18 of the second period highlighted and circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2 2s^2 2p^6$.

Slide 6. The modern periodic table with the first and second elements of the third period highlighted and a circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2 2s^2 2p^6 3s^2$.

Slide 7. The modern periodic table with the elements of groups 13 to 18 of period 3 highlighted and a circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2 2s^2 2p^6 3s^2 3p^6$.

Slide 8. The modern periodic table with the elements of groups 1 and 2 of period 4 highlighted and a circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$.

Slide 9. The modern periodic table with the elements of groups 3 to 12 of period 4 highlighted and a circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you". The text below the periodic table is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$.

Slide 10. The modern periodic table with the elements of groups 13 to 15 of period 4 highlighted and a circle on the element "As". The text on the left reads, "Move along each row and write out the electron configuration for each block using the period numbers to guide you." The text below the periodic table is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$.

Test your understanding of the block method by dragging the electron configurations into their proper locations in the periodic table.

Diagram showing the periodic table blocks: s block, p block, d block, and f block.

Electron configurations to be placed:

- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$
- $1s^2 2s^2 2p^4$
- $1s^2$
- $[\text{Xe}] 6s^2 4f^4$
- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
- $[\text{Ne}] 3s^2$

Check

Interactive 1. Drag and Drop the Electron Configurations Into Their Proper Locations in the Periodic Table.

More information for interactive 1

An interactive periodic table is divided into s-block on the left, p-block on the right, d-block between s- and p-blocks, and f-block positioned at the bottom. The table is organised into rows as periods, from 1 to 7, and columns as groups, from 1 to 18. Users need to drag and drop electron configurations into their respective locations in the periodic table.

The drop zones are provided in the following positions:

s-block: Group 2, Period 3

d-block: Group 9, Period 4

p-block: Group 16, Period 2

p-block: Group 17, Period 4

s-block: Group 18, Period 1

f-block: Group 3, Period 6

Provided electron configuration options:

- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$
- $1s^2 2s^2 2p^4$
- $1s^2$
- $[\text{Xe}] 6s^2 4f^4$
- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
- $[\text{Ne}] 3s^2$

This interactivity helps users understand the relationship between the electronic configuration and the position of elements within different blocks of the periodic table. Read below for the solution:

s-block: Group 2, Period 3 - $[\text{Ne}] 3s^2$

d-block: Group 9, Period 4 - $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$

p-block: Group 16, Period 2 - $1s^2 2s^2 2p^4$

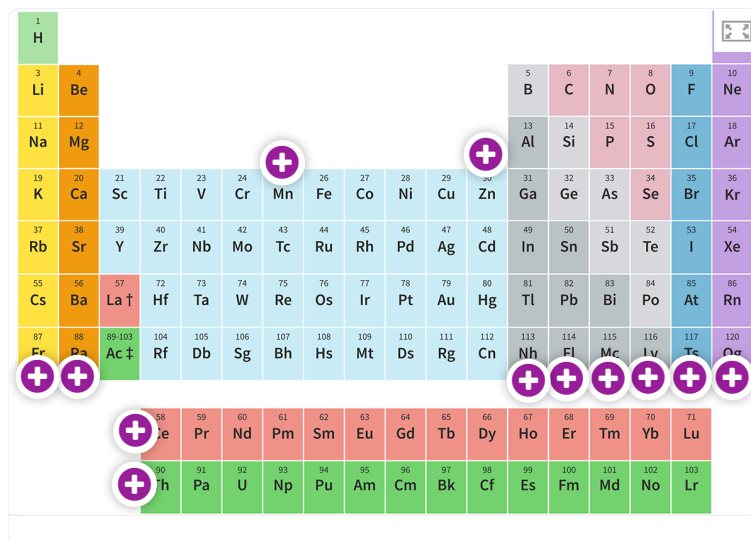
p-block: Group 17, Period 4 - $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$

s-block: Group 18, Period 1 - $1s^2$

f-block: Group 3, Period 6 - $[\text{Xe}] 6s^2 4f^4$

Further classification in the periodic table

While groups of elements with similar properties are classified according to group number, there are informal names often used to refer to a group or groups of elements. These informal names often come from historical usage. **Interactive 3** shows a periodic table. Click on each group to see the informal name assigned to that group of elements.



Interactive 3. Informal names for groups in the periodic table.

Study skills

While these informal names are often useful names to reference specific groups in the periodic table, they are not internationally recognised by IUPAC as names for groups of elements. You must be familiar with the classifications of alkali metals (group 1), transition elements (groups 3–12), halogens (group 17) and noble gases (group 18) but are not required to know the others.

Table 3 provides some additional detail about the alkali metals, transition elements, halogens and the noble gases.

Table 3. Additional detail about the alkali metals, transition elements, halogens and the noble gases.

Element Group	Category	Details
Alkali metals	Origins of name	Comes from the tendency of these metals to form alkaline solutions when they react with water.
	Properties of the elements	Soft metals that are very reactive with water and many non-metals.
	Links to further sections	The reactions of the alkali metals with water will be looked at more closely in section S3.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/) .

Element Group	Category	Details
Transition elements	Origins of name	These elements show transitional behaviour intermediate between elements in the s-block and those in the p-block.
	Properties of the elements	Can form ions with different ionic charges. Metals have high melting points and high conductivities.
	Links to further sections	HL will look more closely at formally defining transition elements in section S3.1.8—9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/).
Halogens	Origins of name	Name is Greek and means “salt producing”.
	Properties of the elements	Highly reactive and form ionic salts when they react with metals.
	Links to further sections	The reactions of halogens with halide ions will be looked at more closely in section S3.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/).
Noble gases	Origins of name	Name is German and means “immaculate”.
	Properties of the elements	Colourless monoatomic gases that have little to no reactivity with other elements.
	Links to further sections	Noble gases are often used to simplify electron configurations of other elements since they have an outer energy level that is completely filled. You learned about this condensed electron configuration in section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/).

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate how to plan and initiate a CAS experience
- Demonstrate engagement with issues of global significance

“If you can’t grow it, you have to mine it”. What do you think this expression means? It’s another way of saying that if you cannot grow a material, such as plants used for food or clothing, then you have to mine it (extract it from the ground). According to the [USGS](https://elements.visualcapitalist.com/all-the-metals-mined-in-2021/) (<https://elements.visualcapitalist.com/all-the-metals-mined-in-2021/>), 2.8 billion tonnes (2.80×10^{12} kg) of metals were mined in 2021 alone. The process of extracting these metals from their ores can have [significant negative effects](#) ([https://elements.visualcapitalist.com/all-the-metals-mined-in-2021/](#)).

([https://www.americangeosciences.org/critical-issues/faq/how-can-metal-mining-impact-](https://www.americangeosciences.org/critical-issues/faq/how-can-metal-mining-impact-environment#:~:text=If%20not%20prevented%20or%20controlled,impact%20stream%20habitats%20and%20groundv)

[environment#:~:text=If%20not%20prevented%20or%20controlled,impact%20stream%20habitats%20and%20groundv](https://www.americangeosciences.org/critical-issues/faq/how-can-metal-mining-impact-environment#:~:text=If%20not%20prevented%20or%20controlled,impact%20stream%20habitats%20and%20groundv)
the environment.

As an IB student, think about what you can do to help offset the environmental effects of mining. Perhaps you could start by investigating if any recycling programmes such as those used to recycle electrical waste (known as E-waste) or metals exist in your school or local community. Alternatively, you could raise awareness of the environmental damage caused by the mining of metals and other such materials.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication skills — Using terminology consistently and correctly
- **Time required to complete activity:** Approximately 30 minutes
- **Activity type:** Pair activity

Electron configuration battleship game

Materials (per student):

- Two laminated periodic tables. If lamination is not available paper copies will work too.
- One folder.
- One whiteboard marker. If using paper copies of the periodic table a pen or pencil would work.

[Periodic table](#) 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

Individual instructions:

- Glue or staple your two periodic tables to the inside of a folder as shown.
- One periodic table will be your gameboard, the other will be your opponents. You will need to keep track of both during this game.
- On your gameboard you will add five boats by completely colouring in the box of the elements that make up the boat. Boats must be made up of adjacent elements (vertical or horizontal). You should have the following boats:
 - One five-element boat
 - Two four-element boats
 - Two three-element boats

Pair instructions:

- Sit opposite your partner on the other side of a table.
- With your partner, connect your folders “laptop style” using a paperclip to join the top. Ensure that you are not able to view the contents of your partner’s folder.
- You will take turns trying to guess the location of your partner’s boats, you are only permitted to use electron configurations to state the position of the element. Be patient with one another, this is for practice.
- Keep track of your guesses for your opponent on the bottom periodic table, and their guesses for your boats on the top periodic table.
- Use an X for misses and colour the square in completely for a hit.
- If needed, use a periodic table such as the one shown to help guide you.

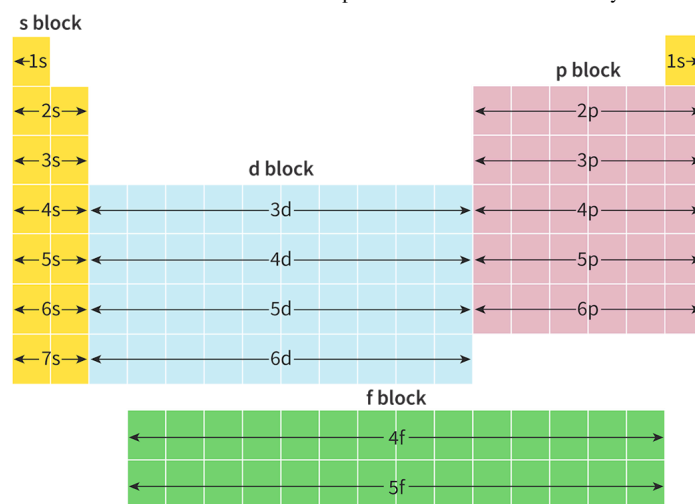


Figure 6. Periodic table blocks.

 [More information for figure 6](#)

This image shows the periodic table divided into blocks that represent different electron configurations. The left section is highlighted in yellow and labeled as the 's' block, containing arrows pointing horizontally from 1s to 7s. To the right, the blue section indicates the 'd' block, with arrow lines marked from 3d to 6d, running horizontally. On the far right, akin to the 'd' block, is the pink-colored 'p' block, with arrows for 2p to 6p extending horizontally. Below the main table, there is a green section labeled as the 'f' block, showing arrows moving horizontally from 4f to 5f. This layout illustrates the organization of elements based on their electron configurations in atomic orbitals.

[Generated by AI]

Trends in atomic and ionic radius

Learning outcomes

By the end of this section you should be able to:

- Outline what defines the atomic radius.
- Explain the periodicity of atomic and ionic radii.
- Compare the atomic and ionic radii for isoelectronic species.

Defining atomic radius

Models of the atom have evolved throughout time with the most recently updated model being the quantum model. Some of these models are shown in **Figure 1**.

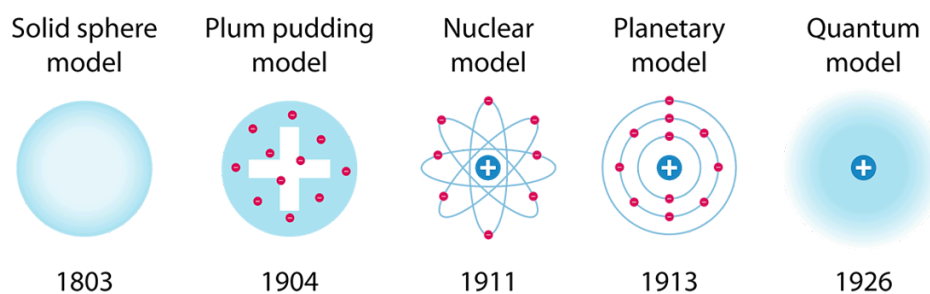


Figure 1. Models of the atom over time.

 More information for figure 1

The image shows a sequence of atomic models illustrating the evolution of the atom's understanding over time. From left to right, each model depicts a different structure:

1. **Thomson's Plum Pudding Model:** This model shows a light blue circular shape with small red dots scattered within it, representing electrons within a positively charged 'soup.'
2. **Rutherford's Model:** Consists of a dense central positive charge depicted as a cross or nucleus with small red dots (electrons) scattered around it.
3. **Bohr's Model:** Illustrates a central nucleus represented by a blue circle with electrons following distinct circular orbits around it, shown with pink curved arrows.
4. **Electron Cloud Model:** Displays a central nucleus with a surrounding blue faded circle indicating zones of high electron probability, without precise paths.

[Generated by AI]

In section S1.3.3–4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/>) you learned how electrons in atoms occupy atomic orbitals within sublevels, this would be considered the quantum model. In this section we will be looking closely into the sizes of atoms and ions. Looking at the quantum model shown, consider how you would define and describe the size of the atom. What are some ways we describe the size of spherical objects? Do you have any concerns about defining the size of an atom based on the quantum model? How would you expect the sizes of atoms to change as the number of electrons in the outermost energy level change? How would you expect the sizes of atoms to change as the number of occupied energy levels change? Radius is a common parameter used to quantify the size of circles and spheres. For both a circle and a sphere the radius is the distance between the centre to its outer edge. For an atom, what would be at its centre? What would be its outer edge? You may have noticed in the quantum model that the outer edge of an atom is poorly defined. This means that the radius of an atom cannot simply be defined as the distance between the centre to its outer edge, we must define it in another way.

Looking at the quantum model, what part of the atom has a well-defined location? The nucleus! This means it would be easy to identify the nucleus of adjacent atoms in order to deduce the atomic radius of each atom. **Figure 2** illustrates how atomic radius can be defined as half of the distance between two adjacent nuclei.

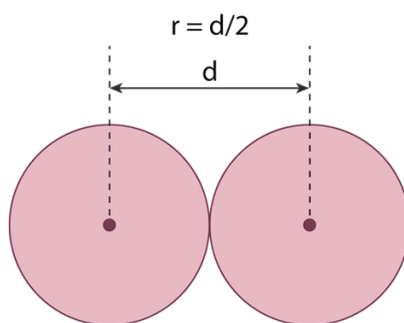


Figure 2. Defining atomic radius as half the distance between two adjacent nuclei.

 More information for figure 2

The image is a diagram illustrating the concept of atomic radius. It displays two adjacent circles, each representing an atom. At the center of each circle is a small dot representing the nucleus. Dotted lines extend from each nucleus to a point between the two atoms, labeled as 'distance between nuclei.' The atomic radius is highlighted as half of this distance, showing how it is measured from the nucleus to the outer edge of the atom. The diagram visually explains that the atomic radius is defined as half the distance between the nuclei of two adjacent atoms.

[Generated by AI]

Effective nuclear charge

The radius of an atom will depend on the interaction of the protons in the nucleus and the electrons in the outer valence shell. This interaction is due to the effective nuclear charge. Effective nuclear charge is the net positive charge experienced by outer valence shell electrons from the force of attraction between protons in the nucleus and the force of repulsion between inner core electrons. The repulsion of the inner core electrons shield the outer electrons from experiencing the full attractive force to the protons in the nucleus. These attractive and repulsive forces are illustrated in **Figure 3**.

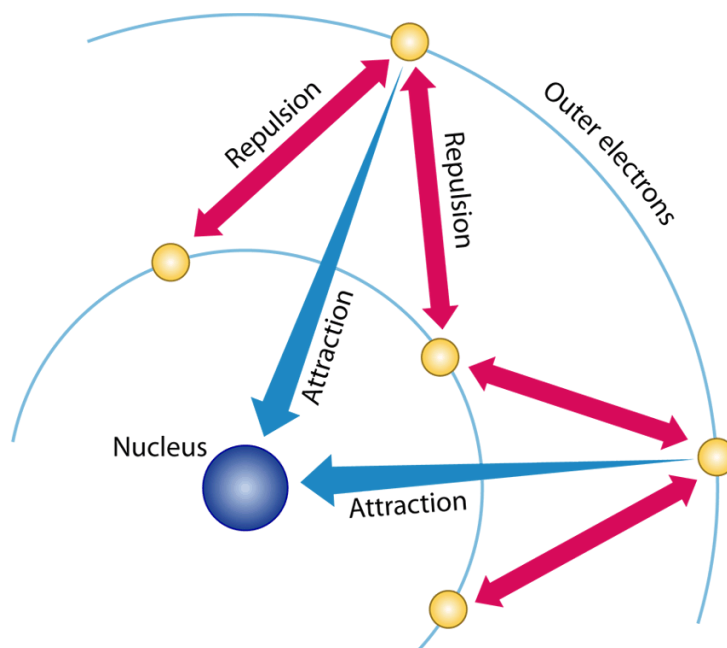


Figure 3. Interactions of the outer shell electrons with the nucleus and inner electrons.

 More information for figure 3

The diagram illustrates the interactions between the outer shell electrons and the nucleus along with the inner electrons in an atom. The central part of the diagram represents the nucleus, where protons are located, shown as a large blue sphere. Surrounding it are gold spheres representing inner core electrons. These are positioned close to the nucleus, indicating the repulsion forces that shield the outer valence electrons from the nucleus' full positive charge.

The outer area depicts pink-colored regions with black arrows indicating the path and forces acted upon the outer shell electrons (not visually represented). The diagram uses upward and diagonal arrows to show the paths of the attractive forces towards the nucleus and repulsive forces from inner electrons. The interactions are designed to display the effective nuclear charge—net positive charge experienced by outer valence electrons due to the balancing act of attraction to the nucleus and repulsion from inner electrons.

[Generated by AI]

As we move across a period, elements have a greater number of protons in the nucleus and a greater number of electrons occupying the outermost energy level. This greater number of protons and valence electrons increases the effective nuclear charge. The greater the effective nuclear charge, the stronger the force of attraction between outer valence electrons and the nucleus. The lower the effective nuclear charge, the weaker the force of attraction between valence electrons and the nucleus.

The worked example looks at the effective nuclear charge of elements in period 3.

Worked example 1

Order the following period 3 elements in order of increasing effective nuclear charge (lowest first).

Mg, Cl, P, S, Si, Na

We can compare the number of protons in the nucleus and the number of valence electrons. This can help us to deduce elements that would have stronger attractions between the nucleus and those with weaker attractions.

	Mg	Cl	P	
<i>protons</i>	12	17	15	
<i>valence electrons</i>	2	7	5	

The order of increasing effective nuclear charge for elements in the same period would correlate to the order of increasing number of protons and increasing number of valence electrons.

$\text{Na} < \text{Mg} < \text{Si} < \text{P} < \text{S} < \text{Cl}$

Trends in atomic radii

The values for the atomic radii of elements in the periodic table can be found in section 10 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-and-ionic-radii-of-the-elements-id-45112/>) of the DP chemistry data booklet. The atomic radius is plotted for the elements in the first four periods in **Figure 4**. Using this graph, identify the elements from the same group (e.g. group 1 is H ($Z = 1$), Li ($Z = 3$), Na ($Z = 11$), K ($Z = 19$)). How would you describe the trend in atomic radius going down a group? Can you think of an explanation for this trend? How would you describe the trend in atomic radius across a period? Can you think of an explanation for this?

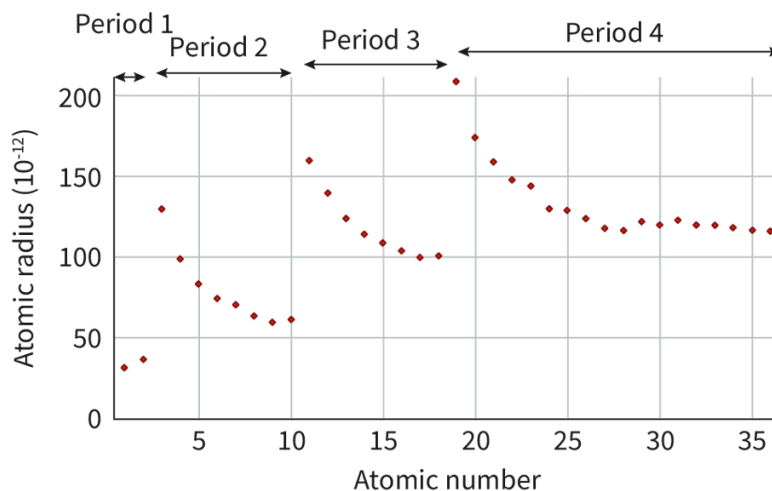


Figure 4. Atomic radii for elements in the first four periods.

 More information for figure 4

The graph displays atomic radii for elements plotted against their atomic numbers for the first four periods of the periodic table. The X-axis represents the atomic number, ranging from 0 to around 35, while the Y-axis represents the atomic radius in picometers (10^{-12}), with a scale from 0 to 200. Data points show a downward trend in atomic radius across a period, with distinct drops as atomic numbers increase, indicating how radius decreases across periods. The trend slightly increases when moving to a new period, reflecting changes in atomic structure. Periods are labeled 1 through 4 above the graph, marking transitions between them.

[Generated by AI]

Study skills

It can be helpful to create plots using atomic numbers on the x-axis and property of interest on the y-axis to study periodicity. Scatter plots and column graphs are typically used. These types of graphs are very useful to visually represent trends across a period and identify parts of the graph that correlate to the same group.

General trend in atomic radius down a group

Table 1 shows the atomic radius for elements in groups 1, 2, 17 and 18. You will notice that going down a group in the periodic table the atomic radius increases. This is because elements further down in the periodic table have electrons that occupy higher energy levels. These higher energy levels are found further from the nucleus, resulting in elements located further down in a group having larger atomic radii than those higher up in a group. This trend is illustrated using the Bohr model of the atom to visualise energy levels in **Figure 5**.

Table 1. Atomic radius for elements in groups 1, 2, 17 a

Period	Highest occupied energy level	Group 1 Atomic radius ($\times 10^{-12}$ m)		Group 2 Atomic radius ($\times 10^{-12}$ m)		Group 17 Atomic radius ($\times 10^{-12}$ m)
1	$n = 1$	H	32			
2	$n = 2$	Li	130	Be	99	F
3	$n = 3$	Na	160	Mg	140	Cl
4	$n = 4$	K	200	Ca	174	Br
5	$n = 5$	Rb	215	Sr	190	I
6	$n = 6$	Cs	238	Ba	206	At

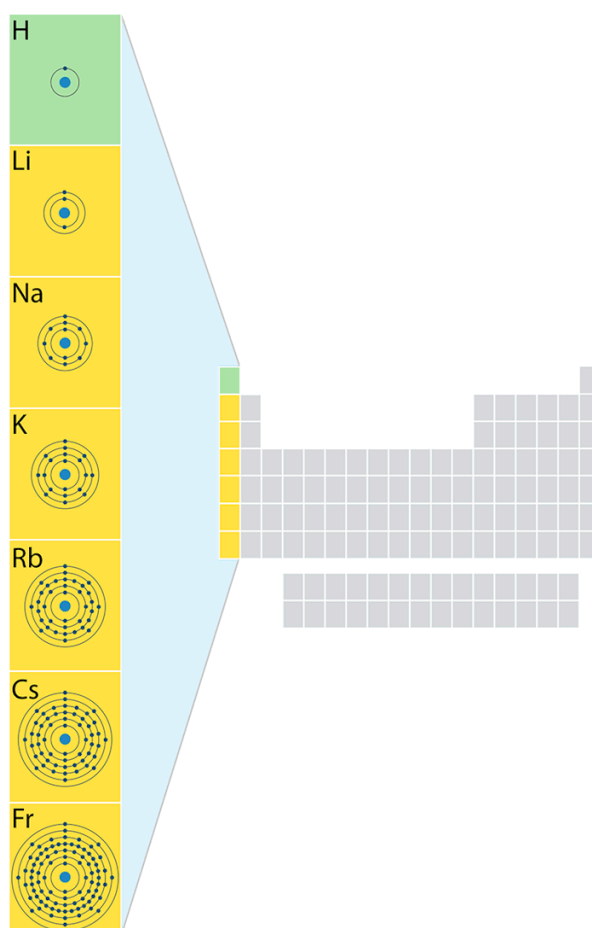


Figure 5. The atomic radius of group 1 atoms looking at the energy levels.

 More information for figure 5

The image is a diagram showing the atomic models of Group 1 elements alongside a portion of the periodic table. The elements hydrogen (H), lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and francium (Fr) are depicted with their

atomic models arranged vertically.

Each element's atomic model shows the nucleus and electron rings. Hydrogen has one electron in a single shell. Lithium has two electron shells with the first ring containing two electrons and the second shell one electron. Sodium's model has three shells, with two electrons in the first, eight in the second, and one in the outer shell. Potassium's electron configuration follows a 2-8-8-1 pattern across its rings. Rubidium, cesium, and francium follow a similar increasing pattern, each having an additional shell and increasing electron count in the inner shells.

Adjacent to these atomic models is a simplified representation of the periodic table. The section showing Group 1 is highlighted, with the elements positioned in a column. Gray squares represent other parts of the periodic table, emphasizing the place of Group 1 elements especially within periods and groups.

[Generated by AI]

General trend in atomic radius across a period

Table 2. Atomic radius for period 3 elements

Element	Na	Mg	Al	Si	P	S
Z_{eff}	1	2	3	4	5	6
Atomic radius ($\times 10^{-12}$ m)	160	140	124	114	109	104

Table 2 shows the atomic radius for period 3 elements. Looking at this table, **Figure 4** and **Table 1** you should notice the general trend of decreasing atomic radius for elements going across a period. This is in contrast to the trend going down a group where a greater number of inner core electrons resulted in a higher atomic radius. As the number of electrons in the outermost energy level increases, the radius decreases.

Why doesn't the atomic radius always increase with an increasing number of electrons? Why would the atomic radius decrease across period 3 when electrons are all occupying the same $n = 3$ energy level? This is due to the increasing effective nuclear charge increasing the attraction of the electrons in the $n = 3$ energy level to the nucleus resulting in a slight decrease of radius of the atom. This trend is illustrated in **Figure 6**.

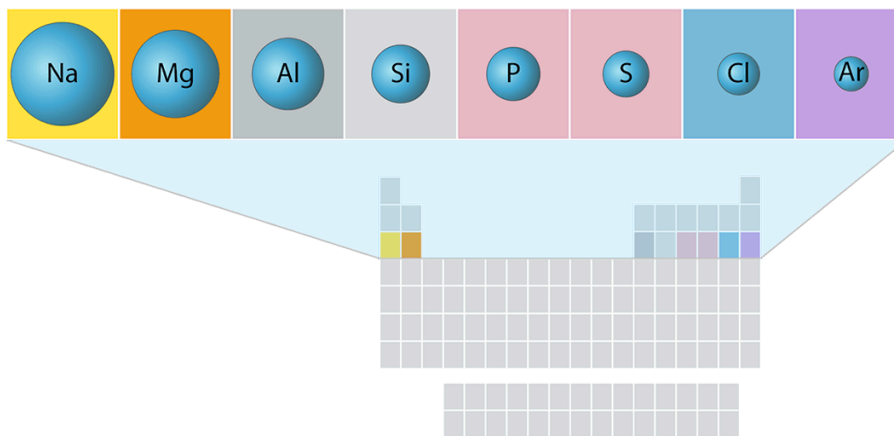


Figure 6. Atomic radius of period 3 atoms.

 More information for figure 6

The image is a diagram illustrating the atomic radius of period 3 elements in the periodic table. It highlights each element with colored blocks and circles representing their relative sizes. The elements included are Sodium (Na), Magnesium (Mg), Aluminum (Al), Silicon (Si), Phosphorus (P), Sulfur (S), Chlorine (Cl), and Argon (Ar). Each element is shown as a circle positioned within its respective block, with the size of the circle indicating the atomic radius. This visual representation helps to convey how the atomic radius decreases across period 3 as effective nuclear charge causes a greater attraction of electrons toward the nucleus. The elements are presented in order from left to right, correlating with their position in the periodic table, demonstrating the trend of decreasing atomic radius across the period.

[Generated by AI]

Study skills

It is important to note that the magnitude of trend is different going down a group than going across a period. The trend in atomic radius for elements going down a group is a **large increase** in size due to the addition of an energy level. The trend in atomic radius going across a period is a **slight decrease** in size due to the increased attraction of the nucleus to the outer shell from the same energy level.

There is periodicity of atomic radii as these general trends repeat throughout the periodic table. **Figure 7** provides an illustration to help visualise the trends in atomic radii throughout the entire periodic table.

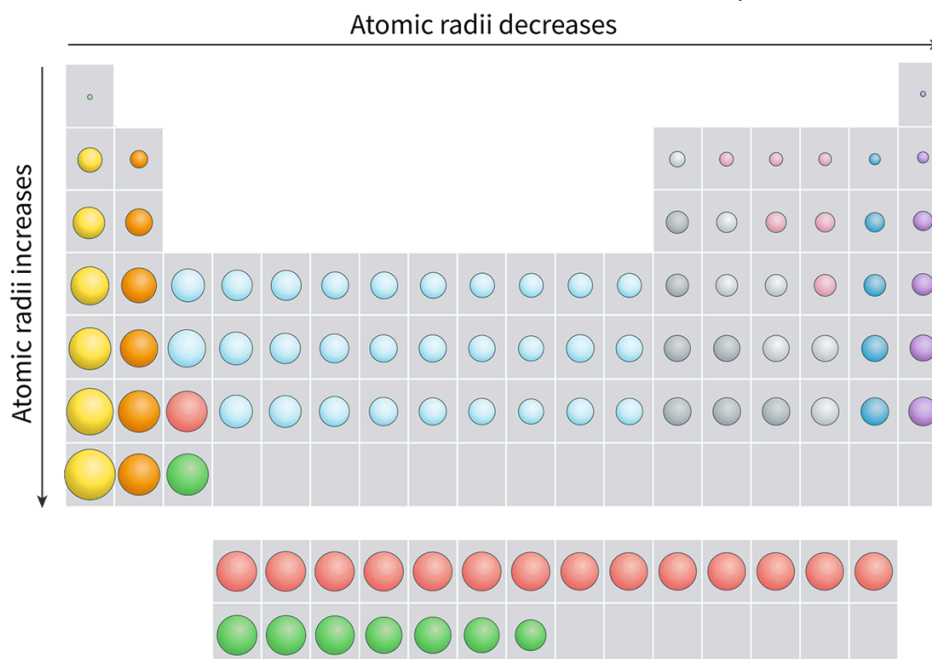


Figure 7. The atomic radii of elements in the periodic table.

 More information for figure 7

The image is a diagram of the periodic table showing the atomic radii of elements. Each element is represented by a colored circle, with the size of the circle indicating the relative atomic radius. The elements are arranged in periods and groups as in the standard periodic table layout. The top left corner has large yellow circles indicating larger atomic radii, which get progressively smaller as you move to the right across a period, then increase again when starting a new period. This pattern repeats across the table, demonstrating the periodic trend of decreasing atomic radii across a period and increasing down a group.

[Generated by AI]

Aspect: Patterns and trends

In this section we are discussing general trends. You may have noticed that some adjacent elements do not fit these trends. For example, argon has a slightly larger atomic radius than chlorine. The variations from the general trend for atomic radius is largely due to the way that atomic radius is defined in terms of half the internuclear distance between two adjacent nuclei. Since noble gases have a high tendency to exist as single atoms (monoatomic) the radius is based off of an intermolecular interaction, rather than a chemical bond. This overestimates the radius of the atom since atoms are further apart than if they were covalently or metallically bonded.

In **Interactive 1** there are 5 small clusters of elements. Each cluster contains 3 elements so we call these clusters triads. Try ordering the elements within each triad shown in the periodic table below in order of increasing atomic radii. Use (1) for the smallest radii and (3) for the largest.

Smallest radius in triad Largest radius in triad

☒ Check

Interactive 1. Drag and Drop in Order of Increasing Atomic Radii.

More information for interactive 1

This interactive image presents a section of the periodic table and highlights five small clusters of elements called 'triads'. Each triad consists of three elements and highlighted in different colors, and the users are tasked with arranging them in order of increasing atomic radius.

A set of number buttons 1, 2, 3 are given below the table, allowing users to rank the elements in each triad from smallest to largest atomic radius, by dragging and dropping the numbers. The smallest radius should be labeled as 1, and the largest radius as 3.

Triad 1 has H, Li, and Na. Triad 2 has Ca, Rb, and Sr. Triad 3 has B, Al, and Si. Triad 4 has N, O, and F. Triad 5 has Se, Br, and I.

Atomic radius is the size of an atom. According to the size of the atom, the elements are arranged in the periodic table. Down the column, atoms get bigger; across the row, atoms get smaller.

Read below for the solution:

Triad 1: H- 1, Li- 2, Na- 3.

Triad 2: Ca- 1, Sr- 2, Rb- 3.

Triad 3: B- 1, Si- 2, Al- 3.

Triad 4: F- 1, O- 2, N- 3.

Triad 5: Br- 1, Se- 2, I- 3.

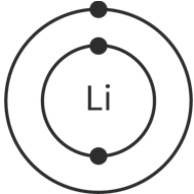
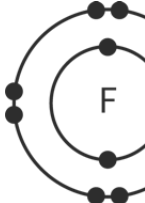
Using these patterns, users can compare atom sizes and arrange the elements in each triad from smallest to largest.

Ionic radii

Recall from section S2.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) how positively charged cations are formed through the loss of electrons and negatively charged anions are formed through the gain of electrons. **Table 3** shows the electron configuration and Bohr model for atoms and ions of lithium and fluorine. By comparing the electron configuration of Li and Li^+ how do you think the atomic radius of Li would compare to ionic radius of Li^+ ? What about F to F^- ? How do you think the addition or loss of electrons influences the force of attraction between valence electrons and protons in the nucleus? What about the force of repulsion between valence electrons and the inner core electrons?

Table 3. Electron configuration and Bohr model for lithium and flu

S p e c i e s	Li	Li^+	F
E l e c t r o n c o n f i g u r a t i o n	$1s^2 2s^1$	$1s^2$	$1s^2 2s^2 2p^5$

B o h r m o d e l	 More information	 More information	 More information
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Positive ions

- Smaller in size than their parent atom.
- Loss of electrons (typically all electrons) from outermost energy level.
- Increased attraction between nucleus and valence shell electrons.
- The greater the number of electrons removed, the smaller the ionic radius.

Negative ions

- Larger in size than their parent atom.
- Gain of electrons into outermost energy level increases repulsion between electrons.
- Decreased attraction between nucleus and valence shell electrons.
- The greater the number of electrons gained, the larger the ionic radius.

You can find data for the ionic radii of common ions for each element in [section 10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-and-ionic-radii-of-the-elements-id-45112/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-and-ionic-radii-of-the-elements-id-45112/>) of the DP chemistry data booklet.

Isoelectronic species

Can you think of anything that the species Na^+ , Mg^{2+} , Al^{3+} , N^{3-} , O^{2-} , F^- and Ne all have in common? They all have ten electrons and have the same electron configuration. Species that have the same electron configuration are said to be isoelectronic. How can we compare the sizes of species that have the same electron configuration? We can consider the effective nuclear charge. **Table 4** shows each of these species along with the number of protons and electrons they each have. How do you think the sizes of each of these species would compare?

Table 4. Comparison of protons and electrons for isoelectronic species.

Species	N^{3-}	O^{2-}	F^-	Ne	Na^+	Mg^{2+}	
Protons	7	8	9	10	11	12	
Electrons	10	10	10	10	10	10	
Atomic/ ionic radius	Decreasing radius						
	 More information						

Since each of the species in **Table 4** has the same number of electrons, the species with the greatest number of protons will have the greatest effective nuclear charge. The greater the effective nuclear charge, the stronger the force of attraction between the nucleus and the outer electrons, therefore the smaller the radius. The lower the effective nuclear charge, the weaker the force of attraction between the nucleus and the outer electrons, therefore the larger the radius. Therefore the order of decreasing radius for the species in **Table 4** would be $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Ne} > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$.

Use **Interactive 2** to order isoelectronic species from smallest to largest radius.

Interactive 2. Drag and Drop to Order Isoelectronic Species from Smallest to Largest Radius.

 More information for interactive 2

This interactive activity allows users to investigate isoelectronic species, which are atoms or ions that have the same number of electrons. The objective is to arrange these species by dragging and dropping them in the correct sequence from the smallest to the largest atomic radius.

The activity features two tables. The first table consists of three columns and four rows, with headers labeled smallest radius, in between, and largest radius. The four rows are left blank for users to fill in the appropriate species in the correct order.

The isoelectronic species in the second table are given below. This table has three rows and four columns. Each group is displayed in distinct colored sections:

Row 1. H^- , He, Li^+ (Purple Section)

Row 2. Na^+ , Mg^{2+} , Al^{3+} (Red Section)

Row 3. P^{3-} , S^{2-} , Cl^- (Orange Section)

Row 4. Ca^{2+} , Ar, P^{3-} (Yellow Section)

Read below for the solution:

The isoelectronic species for the first table in the order of smallest radius, in between, and largest radius is given below for the four rows.

Row 1: Li^+ , He, H^-

Row 2: Al^{3+} , Mg^{2+} , Na^+

Row 3: Cl^- , S^{2-} , P^{3-}

Row 4: Ca^{2+} , Ar, P^{3-}

Through this interactive, users will gain insights into how atomic radius varies among isoelectronic species and will practice correctly positioning them according to their sizes.

Activity

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:** Communication skills — Applying interpretive techniques to different forms of media
- **Time required to complete activity:** 45 minutes
- **Activity type:** Groups of 4—5 students

Create a display showing the atomic and ionic radii to scale for elements $Z = 1$ to $Z = 36$. The sizing and choice of materials is up to you. You could choose to use paper, modelling clay, balloons, pie plates, balls, yarn, etc as materials to create your display. Your scale can be either 2-D or 3-D. Be sure to clearly indicate the parent atom and the ion for each element.

After each group has created their display, perform a peer evaluation of each group's display. The group with the highest scoring peer evaluation can have their radii display permanently displayed in the classroom. Use the following rubric to evaluate the display.

	3 points	2 points
Creativity	Unique with design elements that effectively convey the radii of elements and ions.	Design elements that effectively convey the radii of elements and ions.
Visual appeal	Visually appealing, well-organised and easy to understand.	Somewhat visually appealing and/or somewhat easy to understand.
Accuracy	Sizes of atomic and ionic radii are accurate to scale.	Sizes of atomic and ionic radii are mostly accurate to scale.

Periodicity of ionisation energy, electron affinity and electronegativity

Learning outcomes

By the end of this section you should be able to:

- State the definition for the terms ionisation energy, electron affinity and electronegativity.
- Explain the periodicity of ionisation energy, electron affinity and electronegativity.

When we studied ionic bonding in [section S2.1.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) we looked at how metals have a tendency to lose electrons to form positive ions and non-metals have a tendency to gain electrons to form negative ions. But how might we quantify these tendencies? How might we compare the tendencies to lose or gain electrons between atoms?

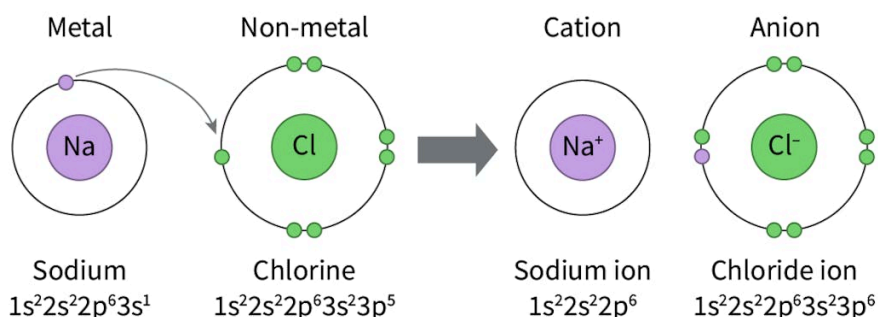


Figure 1. Metals have tendencies to lose electrons and non-metals have tendencies to gain electrons.

📄 More information for figure 1

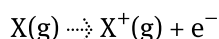
In **Figure 1** you can see the formation of the sodium cation. You will notice that the valence electron from the $n = 3$ energy level has been removed to form this cation. How is this electron removed? What is needed for the removal of an electron from an atom?

You may recall from [section S1.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) that when an electron absorbs enough energy to be promoted to the $n = \infty$ energy level, this

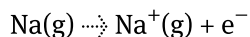
represents the point when the electron has been removed from the atom. This is when enough energy has been applied to overcome the attraction between the nucleus and that electron. The atom is said to be ionised once the removal of an electron occurs.

Ionisation energy

Ionisation energy is the energy required to remove the outermost electron from a species. A more specific form of this term is the first ionisation energy. This is the minimum energy required to lose one mole of electrons from one mole of gaseous atoms in the ground state. The qualifiers 'one mole' and 'gaseous' in this definition are used to keep variables constant to directly compare between different elements. The general equation for the first ionisation energy of an element is as follows:



Using sodium, Na, as an example, the equation is:



Data for the first ionisation energy for all elements can be found in section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet. The first ionisation energies are plotted for the first 20 elements in **Figure 2**. Using this graph, identify the elements from the same group (e.g. group 1 H (Z = 1), Li (Z = 3), Na (Z = 11), K (Z = 19)). How would you describe the trend in first ionisation energy going down a group? How would you describe the trend in first ionisation energy radius going across a period? Can you think of explanations for these trends?

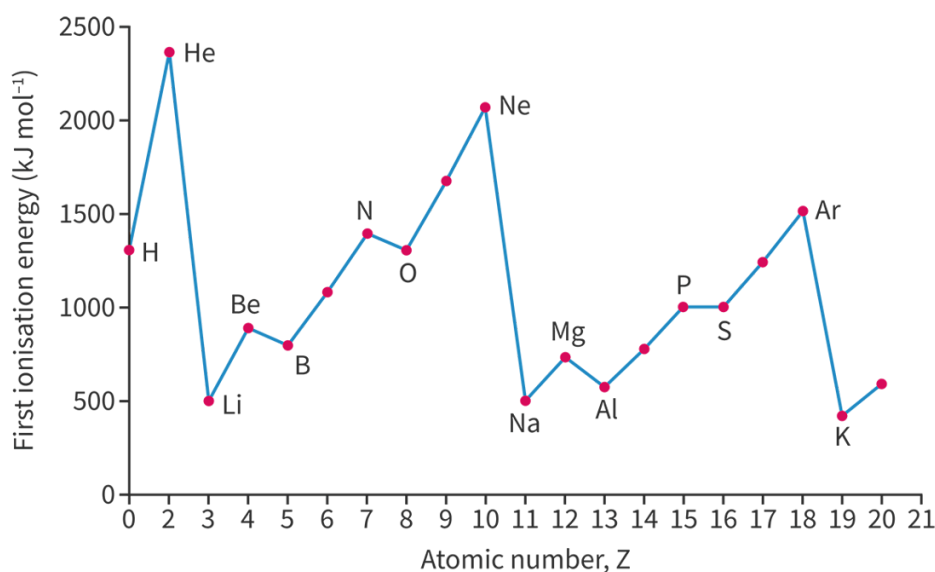


Figure 2. First ionisation energy for elements for the first 20 elements.

More information for figure 2

Table 1 shows the first ionisation energies for elements in groups 1, 2, 17 and 18. Looking at the atomic radii illustrated in **Figure 3**, how does the trend of first ionisation energies compare with that of atomic radii? What do you notice about the trend in first ionisation energy going down a group?

Table 1. First ionisation energy for elements in groups 1, 2,

Period	Highest occupied energy level	Group 1 First ionisation energy (kJ mol ⁻¹)	Group 2 First ionisation energy (kJ mol ⁻¹)	Group 17 First ionisation energy (kJ mol ⁻¹)	Group 18 First ionisation energy (kJ mol ⁻¹)
1	$n = 1$	H 1312			
2	$n = 2$	Li 520	Be 900	F 1681	Ne 2081
3	$n = 3$	Na 496	Mg 738	Cl 1251	Ar 1981
4	$n = 4$	K 419	Ca 590	Br 1140	Kr 1951
5	$n = 5$	Rb 403	Sr 549	I 1012	Xe 1980
6	$n = 6$	Cs 376	Ba 503	At 840	Rn 1936

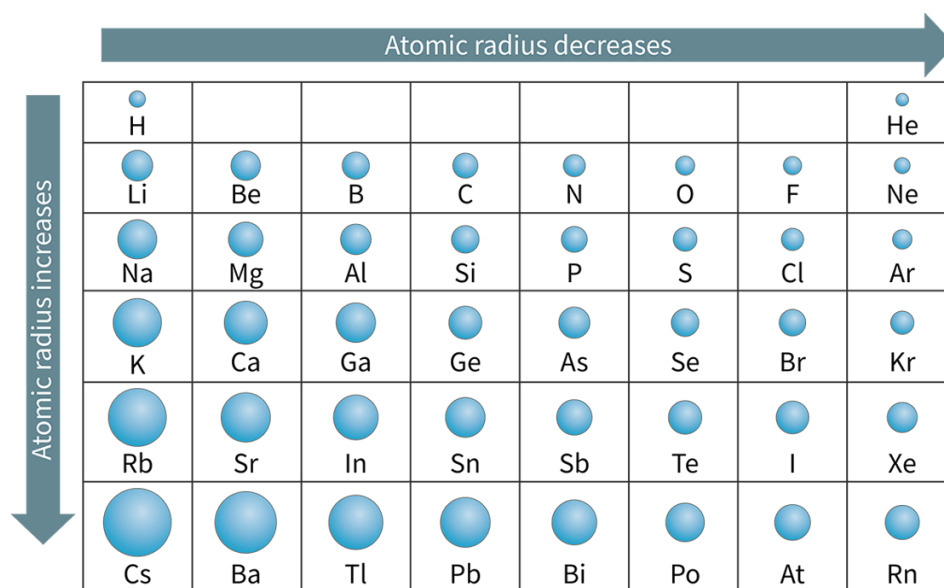


Figure 3. Atomic radii of the elements.

More information for figure 3

Table 2 shows the number of protons, atomic radius and first ionisation energy for period 3 elements. Why would the first ionisation energy increase across the period when valence electrons are all occupying the same $n = 3$ energy level?

Table 2. Number of protons, atomic radius and first ionisation energy for period 3 elements

Element	Na	Mg	Al	Si	P
# of protons	11	12	13	14	15
Atomic radius ($\times 10^{-12}$ m)	160	140	124	114	109
First ionisation energy (kJ mol^{-1})	496	738	578	786	1012

General trend in first ionisation down a group

By analysing **Table 1** and **Figure 3**, you can compile the following general trends in first ionisation energy going down a group:

- Going down a group the first ionisation energy decreases.
- Going down a group atomic radii increases and electrons occupy higher energy levels.
- Electrons in higher energy levels have a weaker attraction to the nucleus.
- Less energy is required to remove an electron.

General trend in first ionisation energy across a period

By analysing **Table 2**, you can compile the following general trends in first ionisation energy going across a period:

- Going across a period the first ionisation energy increases.
- Going across a period the nuclear charge increases (there are more protons in the nucleus).
- Going across a period the atomic radius decreases (atoms get smaller except for Ar which increases slightly compared to Cl).

- More energy is required to remove an electron.

So, to summarise, the trend in ionisation energy across a period depends on the nuclear charge and the atomic radius. The greater the nuclear charge and the smaller the atomic radius, the stronger the attraction between the nucleus and the valence electrons and the higher the ionisation energy. The trend in ionisation energy going down a group depends on the atomic radius and the number of occupied energy levels. The more occupied energy levels, the greater the atomic radius which results in a weaker attraction between the nucleus and the valence electrons. This causes the ionisation energy to decrease down a group.

Making connections

You may have noticed here that we are only evaluating general trends (i.e. the general increase or decrease across a period). There are discontinuities within these trends that we will look at more closely in [section S1.3.6](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) and [section S3.1.7](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/discontinuities-in-ionisation-energy-hl-id-44697/>).

Test your knowledge of the trends in ionisation energy by placing a > or < sign in the boxes below to indicate which element has the greater first ionisation energy. For example, $\text{Mg} > \text{Ca}$ would be used to indicate that Mg has a greater first ionisation energy than Ca.

Interactive 1. Identification of which element has the greater first ionisation energy.

More information for interactive 1

Electron affinity

The term ionisation energy is specific to the absorption of energy to remove an electron to form positive cations. The quantitative term that is specific to the gain of electrons to form negative ions is called electron affinity. If you think about the term affinity, what do you think it might mean? A non-scientific understanding of the word affinity would be to have a natural liking for. You can think of electron affinity in a similar manner, elements with a large absolute value for electron affinity are those with the greatest tendency to gain electrons and those with the lowest absolute value for electron affinity are those with the lowest tendency to gain electrons.

Electron affinity is the energy change that occurs when an electron is added to a species. When a neutral atom gains an electron, energy is released. This is called the first electron affinity and is illustrated in **Figure 4** for hydrogen. Energy is released due to the small force of attraction between the nucleus and the incoming electron.

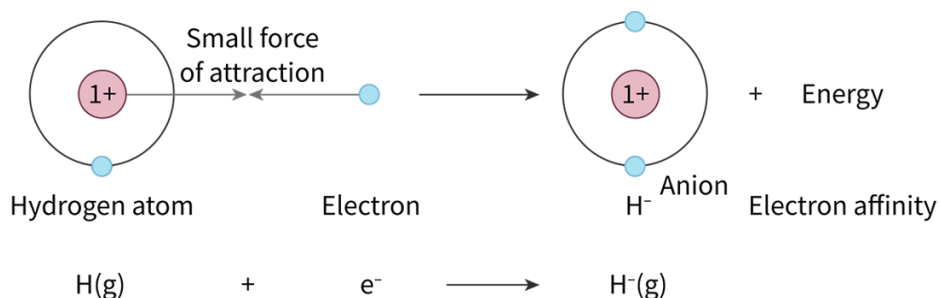


Figure 4. Hydrogen releasing energy as it gains an electron.

More information for figure 4

Section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet has values for the first electron affinity of most elements. You will notice that these values are negative, indicating the release of energy. **Table 3** shows the first electron affinity for group 17. How would you describe the general trend of electron affinity down a group?

Table 3. First electron affinity values for group 17 elements.

Group 17	
Element	Electron affinity (kJ mol^{-1})
F	—328
Cl	—349
Br	—325
I	—295
At	—270

General trend in electron affinity going down a group

- The absolute value for first electron affinity generally decreases going down a group (lower affinity for gaining an electron).
- Elements further down have a larger atomic radius, with outer valence shells further away from the nucleus.
- Weaker attraction between the added electron and the nucleus.

Table 4 shows the first electron affinity for period 3 elements. How would you describe the general trend of electron affinity across a period? Why do you think Mg and Ar have no values for electron affinity?

Table 4. Electron affinity values for period 3 elements.

Element	Na	Mg	Al	Si	P	S	Cl
Electron affinity (kJ mol^{-1})	—53		—42	—134	—72	—200	—329

General trend in electron affinity going across a period

- Absolute value for first electron affinity generally increases going across a period (higher affinity for gaining an electron).

- Elements across the period have a higher effective nuclear charge.
- Stronger attraction between the added electron and the nucleus.

You may have noticed that there are no values for first electron affinity listed for Mg and Ar. This is because many elements with completely filled subshells do not tend to gain electrons, therefore they do not have an associated electron affinity value.

Test your knowledge of the trends in electron affinity by placing a > or < sign in the boxes below to indicate which element has the greater first electron affinity.

Identify which element has the greater first electron affinity.

Use < or > to fill in the blanks.

First electron affinity

S Na

O F

Li F

Na K

Cl Br

 Check

Interactive 2. Identification of which element has the greater first electron affinity.

 More information for interactive 2

Electronegativity

Recall from section S2.2.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>) that electronegativity is the tendency of an atom to attract a shared pair of electrons in a covalent bond. Each atom has an assigned electronegativity value that is relative to fluorine. They can be found in section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet. Of the quantities discussed in this section, which quantity

would you expect the trends in electronegativity to be most similar to? What do you think the general trend for electronegativity would be going down a group? What about going across a period?

Table 5 shows the electronegativity values for group 17. How would you describe the general trend of electronegativity going down a group?

Table 5. Electronegativity values for group 17 elements.

Group 17	
Element	Electronegativity
F	4.0
Cl	3.2
Br	3.0
I	2.7
At	2.2

General trend in electronegativity going down a group

- Electronegativity values generally decrease down a group
- Atomic radius increases down a group from additional energy levels
- Weaker attraction between the nucleus and the shared pair of electrons

Table 6 shows the electronegativity values for period 2 elements. How would you describe the general trend across a period? Why do you think the noble gases do not have an electronegativity value assigned to them?

Table 6. Electronegativity values for period 2 elements

Element	Li	Be	B	C	N
Electronegativity	1.0	1.6	2.0	2.6	3.0

General trend in electronegativity going across a period

- Electronegativity values generally increase going across a period.
- Effective nuclear charge increases going across a period.
- Stronger attraction between the nucleus and the shared pair of electrons.

Aspect: Evidence

Electronegativity values are relative and are a 'made up' quantity. These values are calculated by considering the energy it takes to break the bonds between two elements. We assign a lot of value to these numbers (think back to the bonding triangle) but the fact that they are not measurable quantities certainly has its limitations. When scientists encounter evidence that does not fit a model, it calls for both reexamination of the evidence and reexamination of the model. In April 2021, scientists proposed an updated scale for electronegativities in [this](https://www.nature.com/articles/s41467-021-22429-0)  (<https://www.nature.com/articles/s41467-021-22429-0>) *Nature Communications* paper.

The noble gases do not have an assigned value for electronegativity as they do not form shared pairs of electrons with other atoms.

Test your knowledge of the trends in electronegativity by placing a > or < sign in the boxes below to indicate which element has the greater electronegativity.

Interactive 3. Identification of which element has the greater electronegativity.

👁 More information for interactive 3

Periodicity of atomic radius, ionisation energy, electron affinity and electronegativity

In both this and the previous [section 3.1.3a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>), we have been looking at general trends in the periodic table for atomic radius, ionisation energy, electron affinity and electronegativity. **Figure 5** is a periodic table summarising these general trends.

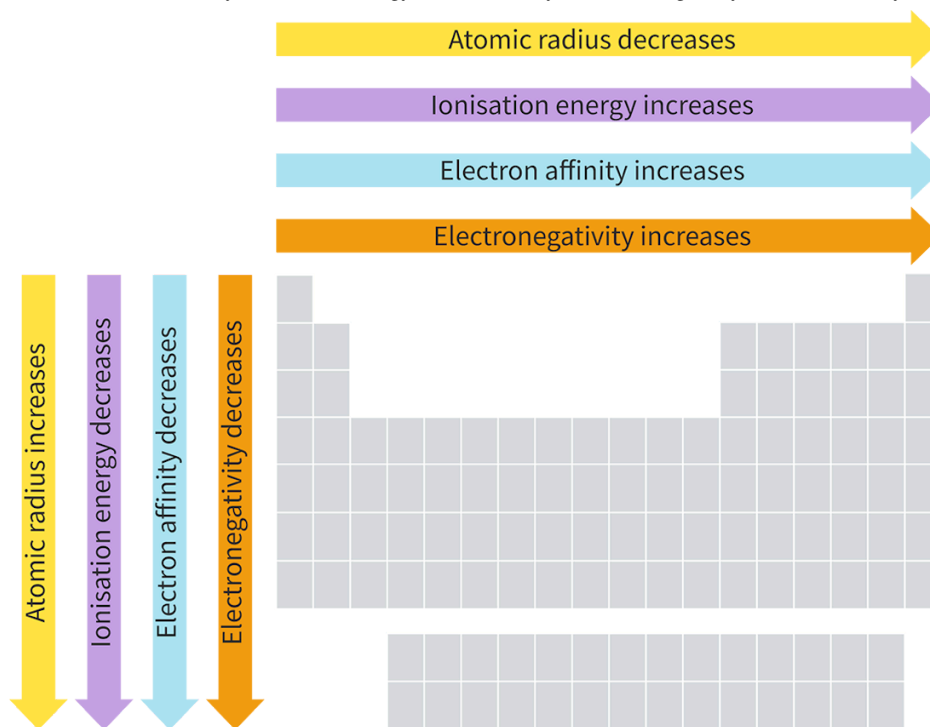


Figure 5. General trends for atomic radius, ionisation energy, electron affinity and electronegativity in the periodic table.

© More information for figure 5

Study skills

Periodicity refers to the general trends we observe in the periodic table for different properties. While it is helpful to describe the trends in terms of increasing/decreasing from left to right and top to bottom it is important to note that these are not explanations. Explanations for these trends involve unpacking the attractive forces between the nucleus and the outer valence shell, along with the repulsion between electrons. The strength of the attractive forces is always influenced by the number of occupied energy levels and the effective nuclear charge of an atom.

Activity

- **IB learner profile attribute:** Balanced
- **Approaches to learning:** Research skills — Evaluating information sources for accuracy
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Group activity

Pasta periodic table

Materials needed (for each student):

- Dry spaghetti (alternatively can use bamboo skewers or pipe cleaners)
- Ruler
- Glue
- Printed copy of the periodic table
- Two pieces of cardboard the same size as your periodic table
- Thumb tack (to poke holes into paper and cardboard)

As a group of four, each group member will select one of the following periodic trends:

- Atomic radius
- Ionisation energy
- Electron affinity
- Electronegativity

You will create a three-dimensional bar graph for your trend using spaghetti to represent the bars. You will need to scale the length of each spaghetti stick to represent value of your chosen parameter. Below provides steps to complete this task.

Three-dimensional bar graph

- Glue both pieces of cardboard together to create one thick piece.
- Glue the periodic table on the top side.
- Using the thumb tack poke a hole through the periodic table and the first cardboard layer for each element.
- Measure the length of your spaghetti stick.
- Decide what measurement you would like each unit to represent. For example, if a spaghetti stick measured 25 cm, then your largest value can only be 25 cm. Let's say the largest value you are representing is 9567. This would be represented by the full spaghetti stick. Now if you wanted to represent 8653 using spaghetti sticks you would simply multiply 25 by $\frac{8653}{9567}$.
- Measure and cut your spaghetti to scale for each value using the following formula:

$$\text{length of full spaghetti} \times \frac{\text{value you are representing}}{\text{largest value from your data set}}$$

- Insert each stick of spaghetti into the appropriate element punched hole in your periodic table.

Once you are finished with your periodic table, help your group members with their spaghetti sticks until all group members are finished. Place all periodic tables side by side.

As a group discuss and complete the following:

- Identify any periodic trends that are evident.
- Discuss the ease to which you were able to identify trends. Were any properties particularly difficult to identify a clear trend? Share why you think these are difficult.
- In ChatGPT ask the AI chatbot to explain each trend down a group and across a period. The AI chatbot will likely produce a reasonable but incomplete response. As a group evaluate the response. What parts of the

response were well crafted? What parts of the response are missing for a high-level chemistry explanation?

Periodicity of ionisation energy, electron affinity and electronegativity

Learning outcomes

By the end of this section you should be able to:

- State the definition for the terms ionisation energy, electron affinity and electronegativity.
- Explain the periodicity of ionisation energy, electron affinity and electronegativity.

When we studied ionic bonding in [section S2.1.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) we looked at how metals have a tendency to lose electrons to form positive ions and non-metals have a tendency to gain electrons to form negative ions. But how might we quantify these tendencies? How might we compare the tendencies to lose or gain electrons between atoms?

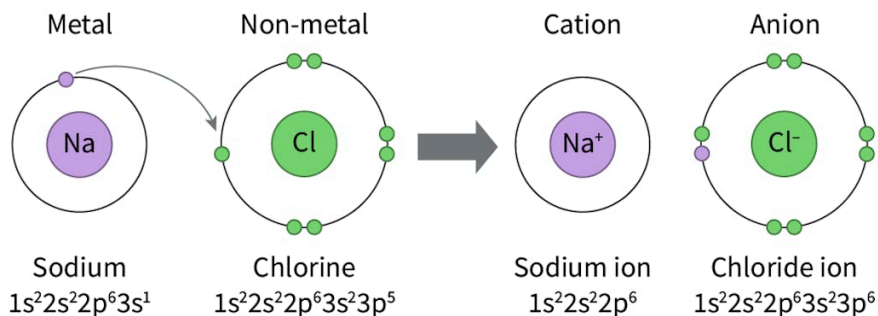


Figure 1. Metals have tendencies to lose electrons and non-metals have tendencies to gain electrons.

📄 More information for figure 1

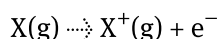
In **Figure 1** you can see the formation of the sodium cation. You will notice that the valence electron from the $n = 3$ energy level has been removed to form this cation. How is this electron removed? What is needed for the removal of an electron from an atom?

You may recall from [section S1.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) that when an electron absorbs enough energy to be promoted to the $n = \infty$ energy level, this

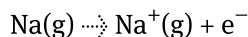
represents the point when the electron has been removed from the atom. This is when enough energy has been applied to overcome the attraction between the nucleus and that electron. The atom is said to be ionised once the removal of an electron occurs.

Ionisation energy

Ionisation energy is the energy required to remove the outermost electron from a species. A more specific form of this term is the first ionisation energy. This is the minimum energy required to lose one mole of electrons from one mole of gaseous atoms in the ground state. The qualifiers 'one mole' and 'gaseous' in this definition are used to keep variables constant to directly compare between different elements. The general equation for the first ionisation energy of an element is as follows:



Using sodium, Na, as an example, the equation is:



Data for the first ionisation energy for all elements can be found in [section 9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet. The first ionisation energies are plotted for the first 20 elements in **Figure 2**. Using this graph, identify the elements from the same group (e.g. group 1 H (Z = 1), Li (Z = 3), Na (Z = 11), K (Z = 19)). How would you describe the trend in first ionisation energy going down a group? How would you describe the trend in first ionisation energy radius going across a period? Can you think of explanations for these trends?

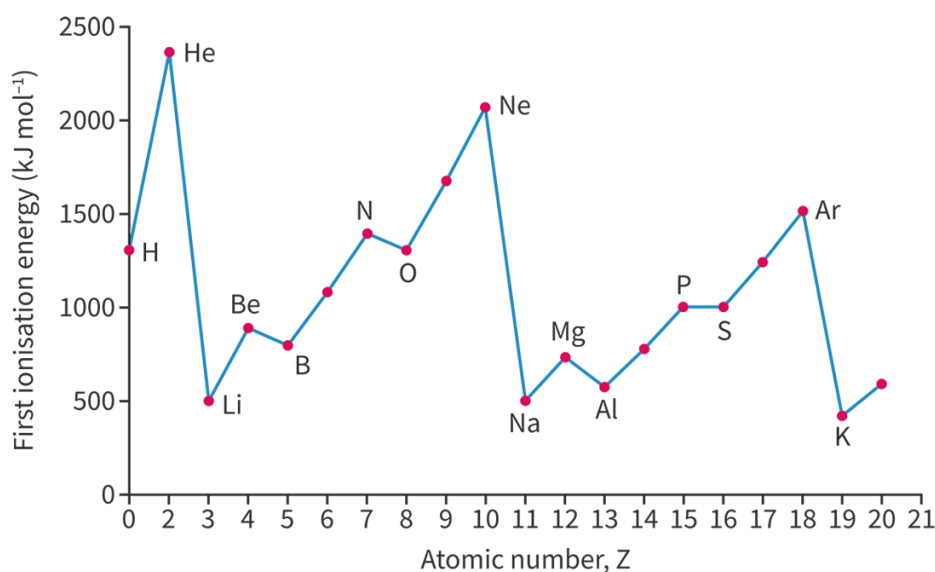


Figure 2. First ionisation energy for elements for the first 20 elements.

More information for figure 2

Table 1 shows the first ionisation energies for elements in groups 1, 2, 17 and 18. Looking at the atomic radii illustrated in **Figure 3**, how does the trend of first ionisation energies compare with that of atomic radii? What do you notice about the trend in first ionisation energy going down a group?

Table 1. First ionisation energy for elements in groups 1, 2,

Period	Highest occupied energy level	Group 1 First ionisation energy (kJ mol ⁻¹)	Group 2 First ionisation energy (kJ mol ⁻¹)	Group 17 First ionisation energy (kJ mol ⁻¹)	Group 18 First ionisation energy (kJ mol ⁻¹)
1	$n = 1$	H 1312			
2	$n = 2$	Li 520	Be 900	F	
3	$n = 3$	Na 496	Mg 738	Cl	
4	$n = 4$	K 419	Ca 590	Br	
5	$n = 5$	Rb 403	Sr 549	I	
6	$n = 6$	Cs 376	Ba 503	At	

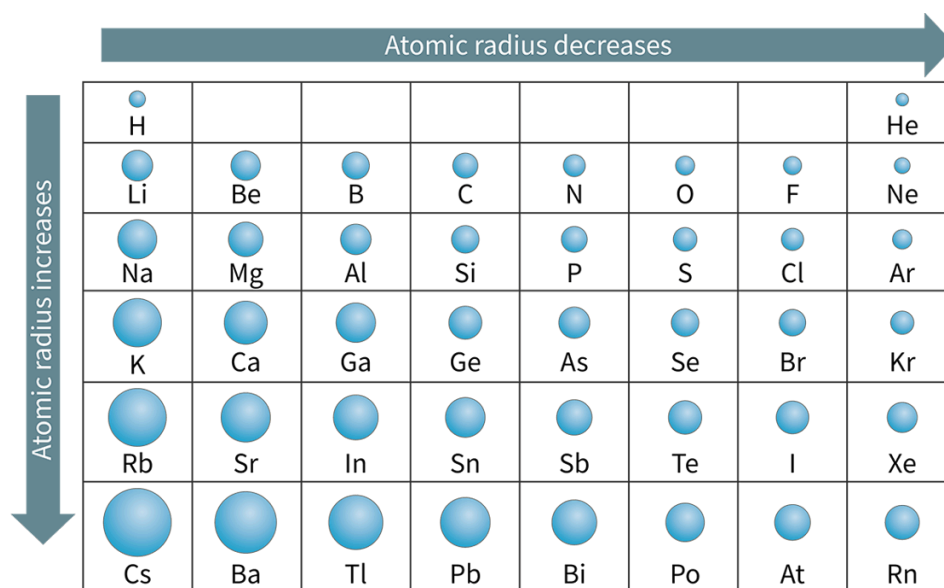


Figure 3. Atomic radii of the elements.

More information for figure 3

Table 2 shows the number of protons, atomic radius and first ionisation energy for period 3 elements. Why would the first ionisation energy increase across the period when valence electrons are all occupying the same $n = 3$ energy level?

Table 2. Number of protons, atomic radius and first ionisation energy for period 3 elements

Element	Na	Mg	Al	Si	P
# of protons	11	12	13	14	15
Atomic radius ($\times 10^{-12}$ m)	160	140	124	114	109
First ionisation energy (kJ mol^{-1})	496	738	578	786	1012

General trend in first ionisation down a group

By analysing **Table 1** and **Figure 3**, you can compile the following general trends in first ionisation energy going down a group:

- Going down a group the first ionisation energy decreases.
- Going down a group atomic radii increases and electrons occupy higher energy levels.
- Electrons in higher energy levels have a weaker attraction to the nucleus.
- Less energy is required to remove an electron.

General trend in first ionisation energy across a period

By analysing **Table 2**, you can compile the following general trends in first ionisation energy going across a period:

- Going across a period the first ionisation energy increases.
- Going across a period the nuclear charge increases (there are more protons in the nucleus).
- Going across a period the atomic radius decreases (atoms get smaller except for Ar which increases slightly compared to Cl).

- More energy is required to remove an electron.

So, to summarise, the trend in ionisation energy across a period depends on the nuclear charge and the atomic radius. The greater the nuclear charge and the smaller the atomic radius, the stronger the attraction between the nucleus and the valence electrons and the higher the ionisation energy. The trend in ionisation energy going down a group depends on the atomic radius and the number of occupied energy levels. The more occupied energy levels, the greater the atomic radius which results in a weaker attraction between the nucleus and the valence electrons. This causes the ionisation energy to decrease down a group.

Making connections

You may have noticed here that we are only evaluating general trends (i.e. the general increase or decrease across a period). There are discontinuities within these trends that we will look at more closely in [section S1.3.6](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) and [section S3.1.7](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/discontinuities-in-ionisation-energy-hl-id-44697/>).

Test your knowledge of the trends in ionisation energy by placing a > or < sign in the boxes below to indicate which element has the greater first ionisation energy. For example, Mg > Ca would be used to indicate that Mg has a greater first ionisation energy than Ca.

Interactive 1. Identification of which element has the greater first ionisation energy.

More information for interactive 1

Electron affinity

The term ionisation energy is specific to the absorption of energy to remove an electron to form positive cations. The quantitative term that is specific to the gain of electrons to form negative ions is called electron affinity. If you think about the term affinity, what do you think it might mean? A non-scientific understanding of the word affinity would be to have a natural liking for. You can think of electron affinity in a similar manner, elements with a large absolute value for electron affinity are those with the greatest tendency to gain electrons and those with the lowest absolute value for electron affinity are those with the lowest tendency to gain electrons.

Electron affinity is the energy change that occurs when an electron is added to a species. When a neutral atom gains an electron, energy is released. This is called the first electron affinity and is illustrated in **Figure 4** for hydrogen. Energy is released due to the small force of attraction between the nucleus and the incoming electron.

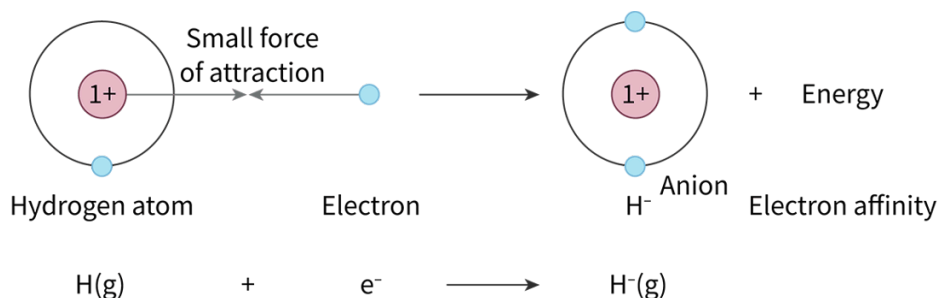


Figure 4. Hydrogen releasing energy as it gains an electron.

More information for figure 4

Section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet has values for the first electron affinity of most elements. You will notice that these values are negative, indicating the release of energy. **Table 3** shows the first electron affinity for group 17. How would you describe the general trend of electron affinity down a group?

Table 3. First electron affinity values for group 17 elements.

Group 17	
Element	Electron affinity (kJ mol^{-1})
F	—328
Cl	—349
Br	—325
I	—295
At	—270

General trend in electron affinity going down a group

- The absolute value for first electron affinity generally decreases going down a group (lower affinity for gaining an electron).
- Elements further down have a larger atomic radius, with outer valence shells further away from the nucleus.
- Weaker attraction between the added electron and the nucleus.

Table 4 shows the first electron affinity for period 3 elements. How would you describe the general trend of electron affinity across a period? Why do you think Mg and Ar have no values for electron affinity?

Table 4. Electron affinity values for period 3 elements.

Element	Na	Mg	Al	Si	P	S	Cl
Electron affinity (kJ mol^{-1})	—53		—42	—134	—72	—200	—329

General trend in electron affinity going across a period

- Absolute value for first electron affinity generally increases going across a period (higher affinity for gaining an electron).

- Elements across the period have a higher effective nuclear charge.
- Stronger attraction between the added electron and the nucleus.

You may have noticed that there are no values for first electron affinity listed for Mg and Ar. This is because many elements with completely filled subshells do not tend to gain electrons, therefore they do not have an associated electron affinity value.

Test your knowledge of the trends in electron affinity by placing a > or < sign in the boxes below to indicate which element has the greater first electron affinity.

Identify which element has the greater first electron affinity.

Use < or > to fill in the blanks.

First electron affinity

S Na

O F

Li F

Na K

Cl Br

 Check

Interactive 2. Identification of which element has the greater first electron affinity.

 More information for interactive 2

Electronegativity

Recall from section S2.2.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>) that electronegativity is the tendency of an atom to attract a shared pair of electrons in a covalent bond. Each atom has an assigned electronegativity value that is relative to fluorine. They can be found in section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet. Of the quantities discussed in this section, which quantity

would you expect the trends in electronegativity to be most similar to? What do you think the general trend for electronegativity would be going down a group? What about going across a period?

Table 5 shows the electronegativity values for group 17. How would you describe the general trend of electronegativity going down a group?

Table 5. Electronegativity values for group 17 elements.

Group 17	
Element	Electronegativity
F	4.0
Cl	3.2
Br	3.0
I	2.7
At	2.2

General trend in electronegativity going down a group

- Electronegativity values generally decrease down a group
- Atomic radius increases down a group from additional energy levels
- Weaker attraction between the nucleus and the shared pair of electrons

Table 6 shows the electronegativity values for period 2 elements. How would you describe the general trend across a period? Why do you think the noble gases do not have an electronegativity value assigned to them?

Table 6. Electronegativity values for period 2 elements

Element	Li	Be	B	C	N
Electronegativity	1.0	1.6	2.0	2.6	3.0

General trend in electronegativity going across a period

- Electronegativity values generally increase going across a period.
- Effective nuclear charge increases going across a period.
- Stronger attraction between the nucleus and the shared pair of electrons.

Aspect: Evidence

Electronegativity values are relative and are a 'made up' quantity. These values are calculated by considering the energy it takes to break the bonds between two elements. We assign a lot of value to these numbers (think back to the bonding triangle) but the fact that they are not measurable quantities certainly has its limitations. When scientists encounter evidence that does not fit a model, it calls for both reexamination of the evidence and reexamination of the model. In April 2021, scientists proposed an updated scale for electronegativities in [this](https://www.nature.com/articles/s41467-021-22429-0)  (<https://www.nature.com/articles/s41467-021-22429-0>) *Nature Communications* paper.

The noble gases do not have an assigned value for electronegativity as they do not form shared pairs of electrons with other atoms.

Test your knowledge of the trends in electronegativity by placing a > or < sign in the boxes below to indicate which element has the greater electronegativity.

Interactive 3. Identification of which element has the greater electronegativity.

👁 More information for interactive 3

Periodicity of atomic radius, ionisation energy, electron affinity and electronegativity

In both this and the previous [section 3.1.3a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>), we have been looking at general trends in the periodic table for atomic radius, ionisation energy, electron affinity and electronegativity. **Figure 5** is a periodic table summarising these general trends.

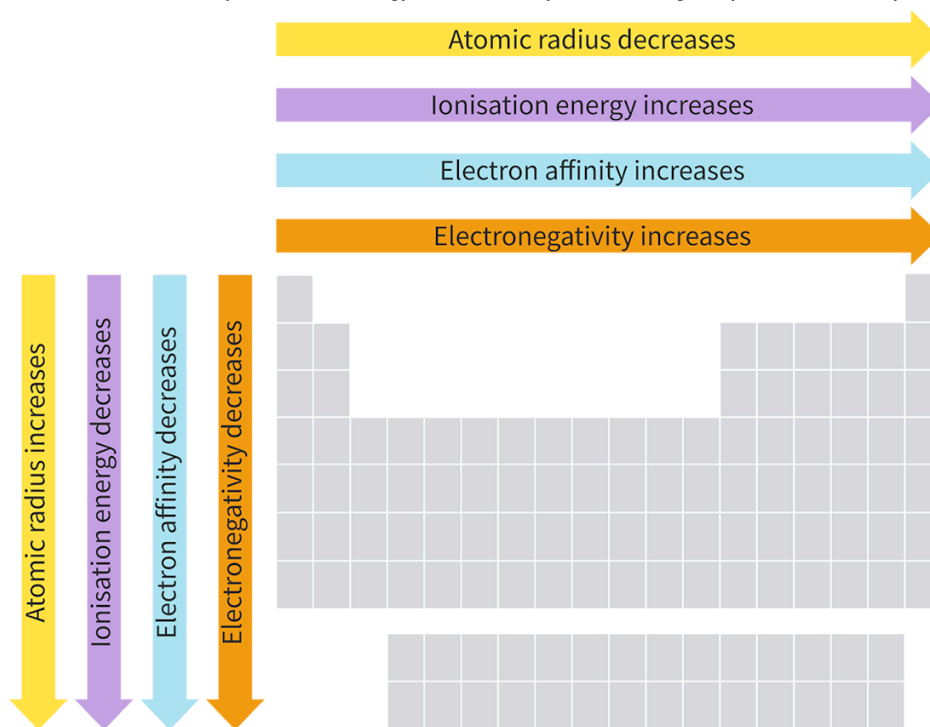


Figure 5. General trends for atomic radius, ionisation energy, electron affinity and electronegativity in the periodic table.

© More information for figure 5

Study skills

Periodicity refers to the general trends we observe in the periodic table for different properties. While it is helpful to describe the trends in terms of increasing/decreasing from left to right and top to bottom it is important to note that these are not explanations. Explanations for these trends involve unpacking the attractive forces between the nucleus and the outer valence shell, along with the repulsion between electrons. The strength of the attractive forces is always influenced by the number of occupied energy levels and the effective nuclear charge of an atom.

Activity

- **IB learner profile attribute:** Balanced
- **Approaches to learning:** Research skills — Evaluating information sources for accuracy
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Group activity

Pasta periodic table

Materials needed (for each student):

- Dry spaghetti (alternatively can use bamboo skewers or pipe cleaners)
- Ruler
- Glue
- Printed copy of the periodic table
- Two pieces of cardboard the same size as your periodic table
- Thumb tack (to poke holes into paper and cardboard)

As a group of four, each group member will select one of the following periodic trends:

- Atomic radius
- Ionisation energy
- Electron affinity
- Electronegativity

You will create a three-dimensional bar graph for your trend using spaghetti to represent the bars. You will need to scale the length of each spaghetti stick to represent value of your chosen parameter. Below provides steps to complete this task.

Three-dimensional bar graph

- Glue both pieces of cardboard together to create one thick piece.
- Glue the periodic table on the top side.
- Using the thumb tack poke a hole through the periodic table and the first cardboard layer for each element.
- Measure the length of your spaghetti stick.
- Decide what measurement you would like each unit to represent. For example, if a spaghetti stick measured 25 cm, then your largest value can only be 25 cm. Let's say the largest value you are representing is 9567. This would be represented by the full spaghetti stick. Now if you wanted to represent 8653 using spaghetti sticks you would simply multiply 25 by $\frac{8653}{9567}$.
- Measure and cut your spaghetti to scale for each value using the following formula:

$$\text{length of full spaghetti} \times \frac{\text{value you are representing}}{\text{largest value from your data set}}$$

- Insert each stick of spaghetti into the appropriate element punched hole in your periodic table.

Once you are finished with your periodic table, help your group members with their spaghetti sticks until all group members are finished. Place all periodic tables side by side.

As a group discuss and complete the following:

- Identify any periodic trends that are evident.
- Discuss the ease to which you were able to identify trends. Were any properties particularly difficult to identify a clear trend? Share why you think these are difficult.
- In ChatGPT ask the AI chatbot to explain each trend down a group and across a period. The AI chatbot will likely produce a reasonable but incomplete response. As a group evaluate the response. What parts of the

response were well crafted? What parts of the response are missing for a high-level chemistry explanation?

Group 1 and 17 elements

Learning outcomes

By the end of this section you should be able to:

- Describe the trend in metallic and non-metallic behaviour in the periodic table.
- Describe and explain the reactions of group 1 metals with water.
- Describe and explain the reactions of group 17 elements with halide ions.

In section S3.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/>) we looked at metals, metalloids and non-metals along with some of their properties. **Table 1** shows images and properties of aluminium, silicon and sulfur, all elements in period 3. Using **Table 1** how would you classify the type of each element (metal, metalloid, non-metal)? What clues did you use to classify them?

Table 1. Images and properties of aluminium, silicon and sulfur

		Aluminium	Silicon	
Image		 Credit: MirageC, Getty Images (https://www.gettyimages.com/detail/photo/crumpled-aluminium-foil-roll-royalty-free-image/1192681753)	 Credit: Oliver Helbig, Getty Images (https://www.gettyimages.com/detail/photo/high-purity-polycrystalline-silicon-from-germany-royalty-free-image/1395000196)	(s)

		Aluminium	Silicon	
Pr op er tie s	Me ltin g poi nt (°C)	660.3	1414	
	Co nd uct ivit y	High	Moderate	

Did you correctly identify aluminium as a metal, silicon as a metalloid and sulfur as a non-metal? While we as humans like to place elements into distinct categories, the reality is these metallic and non-metallic characteristics are a spectrum. How might we describe the trend of metallic or non-metallic character of an element as we travel down a group or across a period? How might you expect the metallic or non-metallic behaviour of an element to correlate with ionisation energy? What about electron affinity?

Metallic character

Metallic character is the tendency of an element to lose electrons to form positive ions. What quantity from section S3.1.3b

([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/)

[753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/)) would provide an indicator for the tendency of electron loss? Recall that ionisation energy is the energy required to remove the outermost electron from a neutral atom. The lower the ionisation energy, the greater the tendency that element has for electron loss and therefore the greater metallic character. This means that elements lower down to the left of the periodic table have the greatest metallic character. Which element do you think would have the most metallic character out of Mg, K and Ca? Check the box below to confirm your answer.

Interactive 1. Metallic Character.

🔗 More information for interactive 1

Non-metallic character

Non-metallic character is the tendency of an element to gain electrons to form negative ions. What quantity from [section S3.1.3b](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/>) would provide an indicator for the tendency of electron gain? Recall that electron affinity is the energy change that occurs when an element gains an electron. The higher the electron affinity, the greater the tendency that element has for electron gain and therefore the greater non-metallic character. This means that elements higher up to the right of the periodic table have the greatest non-metallic character. Which element do you think would have the most non-metallic character out of F, S and Cl? Check the box below to confirm your answer.

Interactive 2. Non-metallic Character.

🔗 More information for interactive 2

Figure 1 labels the trend in metallic and non-metallic character in the periodic table.

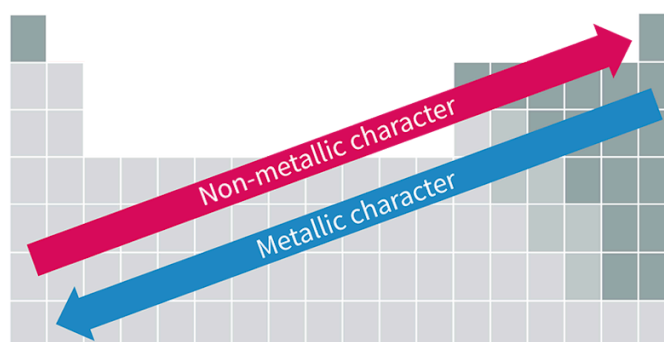


Figure 1. Trend in metallic and non-metallic character.

🔗 More information for figure 1

Study skills

We will be using metallic and non-metallic character to help explain the chemical reactivity of group 1 metals with water and group 17 elements with halide ions. In these examples we will be looking at chemical reactions in terms of electron transfer (electrons transferred from one species to the other). This type of reaction will be the most familiar to you at this time, particularly after looking at the formation of ionic compounds in [section S2.1.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>). It is important to note that there are other mechanisms for

chemical reactions that do not involve electron transfer. We will look at these in the R3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>) subtopics.

Reactions of group 1 metals with water

While we often hear of group one metals such as sodium and potassium in foods, we most commonly encounter these elements as ions (Na^+ and K^+) where they have already lost their outer valence shell electron. Why is it uncommon to find these elements as pure metals? Why don't we see metallic solids made of group 1 metals as building structures or making up a surface in our kitchen?

The reason we cannot use group 1 metals for solid materials is that they are highly reactive with substances such as water. Since water is in our atmosphere it would not be possible to create solid materials out of these metals without a reaction occurring. This is why these substances are typically stored in oil to prevent reactivity with water.

Why are group 1 metals highly reactive with water? What do you think is occurring? What do you think the trend in reactivity down group 1 would be with water? **Video 1a** and **Video 1b** show the reactivity of the group 1 metals sodium and potassium with water. Watch each of them, record your observations and compare the reactivity of sodium and potassium with water. Which substance was the most reactive? Which was the least? Can you think of an explanation for why one would be more reactive than the other? Can you predict any of the products formed in this chemical reaction?

Reaction of Sodium and Water



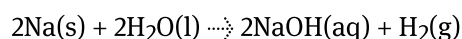
Video 1a. Reaction of sodium and water.

Video 1b. Reaction of potassium and water.**Practical skills****Tools 1:** Experimental techniques — Addressing safety

The reaction of group 1 metals with water poses significant safety concerns. Notice in the clips in **Video 1** only a very small piece of the metal was used in the container of water. This is to minimise the risks to the observer.

Handling of group 1 metals should only ever be done by an experienced adult with a chemistry background. Many research chemists only work with large quantities of group 1 metals in an inert environment such as a glovebox where all atmospheric air has been replaced with an inert gas like argon. These metals should never be left out in the open air and should always be stored in oil. It is important that any tools used to handle the metal pieces be completely dry to avoid any reaction with the metal in the storage container.

The group 1 metals all react similarly with water. The reaction for sodium with water is:



You may have noticed bubbling in the videos showing these reactions. These bubbles produced show the production of a gas, in this case hydrogen gas. Since hydrogen is flammable it can ignite, you would have observed this in the reaction of potassium with water.

The other product NaOH is the formation of a compound containing hydroxide. You might recall from [section S2.1.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) that hydroxide is the polyatomic ion, OH^- . This means an ionic compound is formed from Na^+ and OH^- that is soluble in water, hence the aqueous state. Compounds containing

hydroxides form basic solutions with a high pH. This is why you can see the presence of a pink colour forming in the videos as an acid-base indicator was used to show this change. You will learn more about the pH scale and bases in section R3.1.4-5 ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

Concept

The reactivity of group 1 metals with water is one of the first mentions of chemical reactions so far. As you might expect, reactivity is a very important concept in chemistry. In chemistry chemical equations are symbolic representation of chemical reactions that show reactants (starting materials) on the left-hand side and products (what is formed) on the right-hand side, separated by an arrow. Coefficients are used to represent the relative amounts of each substance. These can be deduced by ensuring the total number of atoms of each element are the same on the reactants side as the products side.

In order to form the ionic hydroxide compound, the originally neutral metal needed to lose its valence electron to water. We can evaluate the ease at which a metal loses its outer valence electron by comparing the values of first ionisation energy. **Table 2** shows the reaction, observations and first ionisation energies for Li, Na and K.

Table 2. Reactions, observations and first ionisation energies for Li, Na and K reacting with water that contains phenolphthalein indicator.

Element	First ionisation energy (kJ mol ⁻¹)	Reaction with water	Qualitative observations of reaction with water
Li	520	$2\text{Li(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{LiOH(aq)} + \text{H}_2\text{(g)}$	Solid floats and moves slowly on surface, gentle fizzing. Water begins to turn pink during reaction.
Na	496	$2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$	Solid melts to form a molten ball that floats and moves more quickly on surface, vigorous fizzing. Water begins to turn pink during reaction.
K	419	$2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$	Solid melts to form a molten ball that floats and moves very quickly on surface, very vigorous fizzing. Water begins to turn pink during reaction.

Element	First ionisation energy (kJ mol^{-1})	Reaction with water	Qualitative observations of reaction with water
			quickly ignites produces a purple flame. Water boils to turn pink due to reaction.

General trend of reactivity for group 1 metals with water

- Increasing reactivity going down group 1.
- Going down group 1, elements have lower first ionisation energies due to the presence of an additional energy level.
- Easier for with lower ionisation energy to transfer the outer valence electron to water increasing reactivity with water.

How would you predict the reactivities of Rb and Cs would compare with Li, Na and K? Watch the video to confirm your predictions.

Alkali Metals - 20 Reactions of the alkali metals with water



Video 2. Reactivity of the group 1 metals with water.

Credit: Twenty47studio, Getty Images

(<https://www.gettyimages.com/detail/photo/sulfur-mining-at-mount-kawah-ijen-crater-in-east-royalty-free-image/1319947829>)

Try **Interactive 3** as a drag and drop activity to review your knowledge of the general trend for number of occupied energy levels, first ionisation energy and reactivity with water for group 1 metals.

Number of occupied energy levels

First ionisation energy

Reactivity with water

Li, Na, K, Rb, Cs, Fr

INCREASES

DECREASES

Check

Interactive 3. Test your knowledge.

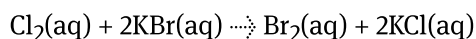
More information for interactive 3

Reactions of group 17 elements with halide ions

The halogens are non-metallic elements in group 17 that exist as diatomic molecules. These elements are highly reactive with many substances since they have a tendency to gain electrons. We can quantify this tendency to gain electrons by looking at the values for electron affinity for these elements. Recall from [section S3.1.3b \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/) that the general trend for electron affinity is that it decreases down a group. What do you think this means for reactivity? How do you think the general trend for reactivity will be for group 17 elements when they react in a manner to gain an electron?

A common type of reaction that the halogens readily undergo is called a displacement reaction. This is where part of a reactant replaces or displaces part of another reactant. These reactions are common to occur between halogens and

halide salts. An example of this reaction is Cl_2 with KBr :



In this reaction, Cl_2 is the diatomic halogen and KBr is the ionic compound made up of the metal ion K^+ and a non-metal halide ion Br^- . In this type of displacement reaction, the extra electron that the non-metal ion Br^- has is transferred to the halogen to form the Cl^- ion. This results in each halogen atom in Cl_2 , gaining an electron and each halide ion, Br^- , losing an electron.

Table 3 shows a comparison of the reactions involving chlorine and bromine along with their electron affinity values. In the first example, chlorine is reacting with sodium bromide. Since chlorine has a higher electron affinity than bromine, chlorine gains the extra electron from the bromide ion to form the chloride ion (Cl^-) and bromine (Br_2). Since sodium remains as the Na^+ ion throughout it is often referred to as a spectator ion and can be omitted from the reaction.

Notice how there is no reaction between Br_2 and NaCl . This is because bromine has a lower electron affinity than chlorine and therefore unable to gain an electron from the chloride ion. Only one combination of halogen and halide ions can result in a reaction, otherwise, the products would be able to react to form reactants over and over again (for example, $\text{Cl}_2(\text{aq})$ will react with $\text{NaBr}(\text{aq})$ but $\text{Br}_2(\text{aq})$ will not react with $\text{NaCl}(\text{aq})$).

Table 3. Electron affinity and reactions for chlorine and bromine.

Halogen	Electron affinity (kJ mol^{-1})	Reaction
Cl_2	−349	$\text{Cl}_2(\text{aq}) + 2\text{NaBr}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{Br}_2(\text{aq})$ or $\text{Cl}_2(\text{aq}) + 2\text{Br}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{aq})$
Br_2	−325	$\text{Br}_2(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{no reaction}$

Use **Video 3** to make observations of the reactions of aqueous solutions of Cl_2 , Br_2 and I_2 with aqueous solutions of KCl , KBr and KI . Identify which combination of halogens and halides will lead to a reaction using this video.

Video 3. Reactivity of halogens with halide ions.

Worked example 1

State the reactions of aqueous solutions of Cl_2 , Br_2 and I_2 as they react with KCl , KBr , KI . Explain why some combinations have no reaction.

Halogen (Cl ₂ , Br ₂ , I ₂)	Reactions with KCl, KBr, and KI — Displacement
Cl ₂	Cl₂ + KCl → no reaction (no element to displace) Cl₂ + 2KBr → 2KCl + Br₂ (chlorine has a higher electron affinity than bromine) Cl₂ + 2KI → 2KCl + I₂ (chlorine has a higher electron affinity than iodine)
Br ₂	Br₂ + KCl → no reaction (bromine has a lower electron affinity than chlorine) Br₂ + KBr → no reaction (no element to displace) Br₂ + 2KI → 2KBr + I₂ (bromine has a higher electron affinity than iodine)
I ₂	I₂ + KCl → no reaction (iodine has a lower electron affinity than chlorine) I₂ + KBr → no reaction (iodine has a lower electron affinity than bromine) I₂ + KI → no reaction (no element to displace)

Try Interactive 4 a drag and drop activity to review your knowledge of the general trend for number of occupied energy levels, electron affinity and reactivity of the halogens with halide ions.

Interactive 4. The general trend of the number of occupied energy levels, electron affinity and reactivity of the halogens with halide ions.

🕒 More information for interactive 4

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual first followed by paired discussion

Identifying unknown substances

You find six substances and suspect they might be the metals Li, Na, K, and aqueous solutions of Cl_2 , Br_2 and I_2 . You make observations of the substances and their reactions and record your observations in **Table 4**.

Table 4. Observations of unknown substances and the

Substance	Appearance	
AA	Stored in oil and very soft solid.	Vigorous reaction with water
BB	Aqueous solution light orange-brown in colour.	Reaction with water produces a gas
CC	Stored in oil and soft solid.	Molten base reacts with water
DD	Aqueous solution dark brown in colour.	No reaction with water
EE	Aqueous solution very pale green in colour almost colourless.	Reaction with water produces a gas
FF	Stored in oil. Firm.	Solid floats on water

Individually, complete the following:

- Identify the six substances AA, BB, CC, DD, EE, and FF.
- For each substance explain how you came to your conclusion using both evidence and reasoning.
- Write out a chemical equation to represent the reaction described for each of the group 1 metal substances.

With a partner:

- Compare your identities of AA, BB, CC, DD, EE and FF. For any you and your partner do not agree on discuss your reasoning with one another and come to a consensus about the identity. Check in with your teacher.
- Discuss explanations for the trends in reactivity for:
 - Group 1 with water.
 - Group 17 with halide ions.
- Ensure your explanations include the following:
 - Description of general trend.
 - A quantitative property that supports that trend.
 - What is occurring in the reaction that connects to that quantitative property.
 - Discussion of the strength of the interaction between the nucleus and outer valence shell.

Metallic and non-metallic continuum

Learning outcomes

By the end of this section you should be able to:

- Describe the trend from basic metal oxides through amphoteric to acidic non-metal oxides.
- Deduce equations for the reactions with water of the oxides of group 1 and group 2 metals, carbon and sulfur.

You will notice in chemistry that many of the categories used to label substances actually show a continuum across the categories. We saw this in [section S2.4.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/>) as metallic and non-metallic elements showed a continuum in their bonding. Consider the elements in period 3 (Na, Mg, Al, Si, P, S, Cl). How would you classify the type of each of these elements? How would you classify the bonding each of these elements have?

Compounds of metallic and non-metallic elements also show a continuum in their bonding. Consider the bonding triangle for the oxides of period 3 elements (Na_2O , MgO , Al_2O_3 , SiO_2 , P_4O_{10} , SO_2/SO_3 , Cl_2O), what type of bonding do you expect each oxide to have? Compare your expectations to the bonding triangle in **Figure 1**.

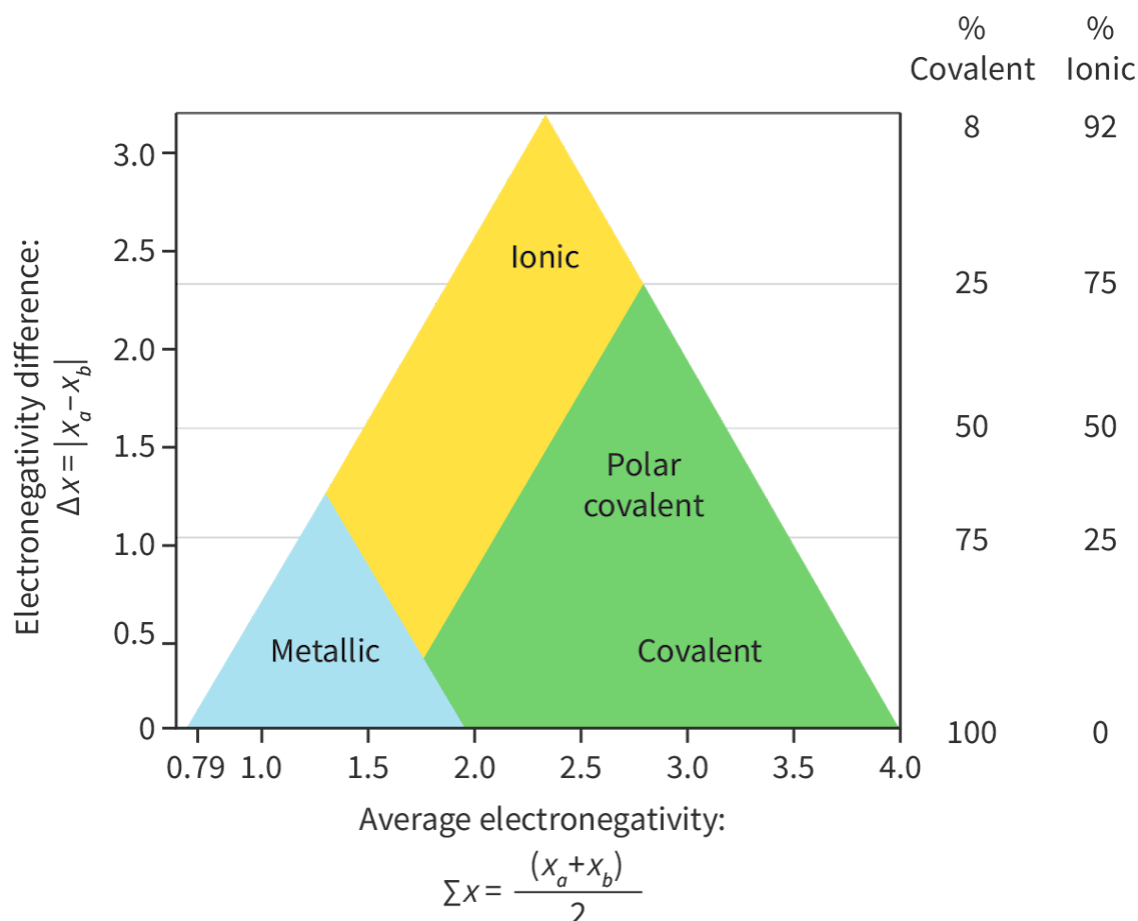


Figure 1. Continuum of bonding of the period 3 oxides.

More information for figure 1

The image is a triangle diagram that categorizes chemical bonds according to average electronegativity and electronegativity difference. The diagram is divided into three sections: metallic, polar covalent, and ionic.

- The X-axis represents average electronegativity ranging from 0.79 to 4.0. The formula ($\Sigma x = \frac{(x_a + x_b)}{2}$) is shown on this axis.
- The Y-axis shows electronegativity difference, ranging from 0 to 3.0, noted as ($\Delta x = |x_a - x_b|$).
- The bottom left triangle is labeled "Metallic," covering lower values on both axes. This area suggests bonds with low average electronegativity and small differences.
- The upper triangle is labeled "Ionic," indicating high electronegativity differences, regardless of average values.
- The middle section is labeled "Polar covalent," bridging the gap between metallic and ionic areas.

There is also additional information on the right indicating the percentage composition of covalent vs. ionic character as a gradient, with extremes marked at the top as 8% covalent and 92% ionic, respectively.

[Generated by AI]

What do you expect about the structure and properties of metallic and non-metallic elements and compounds? The structure of metallic and non-metallic elements and compounds also show a continuum. The structure of the elements and oxides of period 3 are shown in **Table 1**.

Table 1. Structure of the elements and oxides of period 3

Element s & Their Structure	Element s	Na	Mg	Al	Si	P
	Structure	Metallic lattice	Metallic lattice	Metallic lattice	Covalent network lattice	Covalent molecule (P ₄)
Oxides & Their Structure	Oxides	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀
	Structure	Ionic lattice	Ionic lattice	Ionic lattice	Covalent network lattice	Small cage- like structure

Acid-base nature of metallic and non-metallic oxides

Another property that shows a continuum is the acid-base nature of metallic and non-metallic oxides. Can you think of any acids you have encountered before in your personal experience? Lemons and vinegar are sour and acidic because of the

acid molecules present within them that form H^+ ions in solution. Cleaning products and soaps are slippery and basic because of the substances within them that can form OH^- ions in solution. Note that this section is considering acids and bases more generally. You will look at them in more detail in [subtopic 3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>).

A quantitative measurement to determine the concentration of H^+ ions in solution is called the pH scale. Acids are classified as solutions with a pH between 0 and 7 where bases are classified as solutions with a pH between 7 and 14. **Figure 2** shows common substances you may have encountered and where they fall on the pH scale. You will look more closely at the pH scale in [section R3.1.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

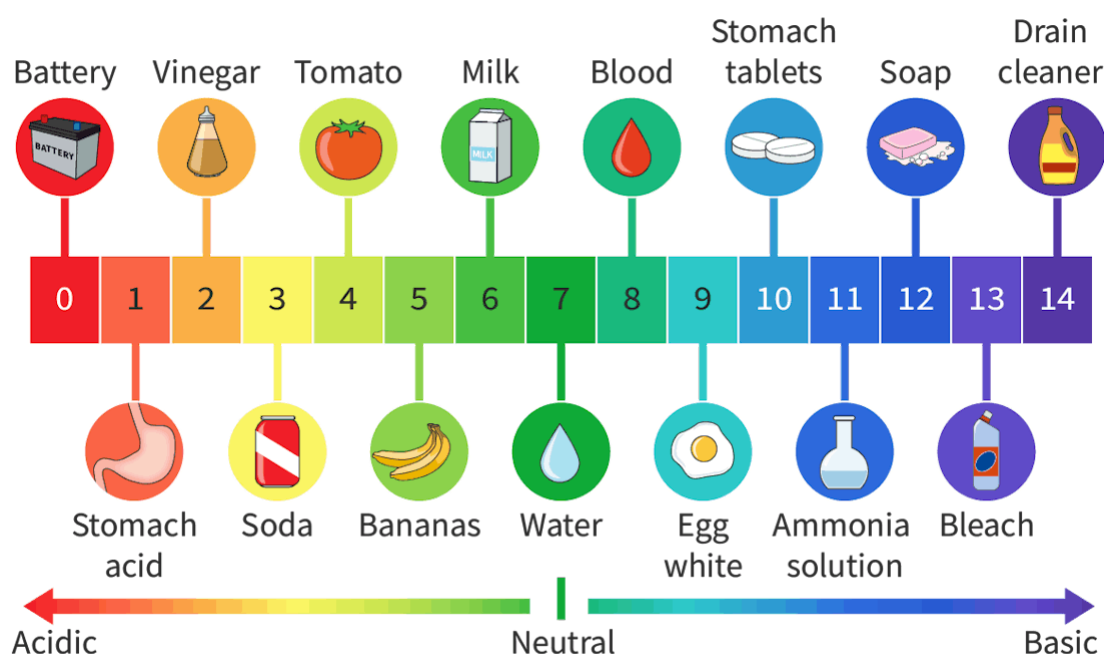


Figure 2. pH of common substances.

 More information for figure 2

This diagram illustrates the pH scale, which ranges from 0 to 14, with various common substances placed along the scale according to their acidity or basicity. The scale begins with 0, marked with a battery icon, indicating highly acidic. Moving along the scale to 7, it shows neutral substances like water. Substances are depicted with images: battery (pH 0), stomach acid (pH 1), vinegar (pH 2), tomato (pH 4), milk (pH 6), blood (pH 7), stomach tablets (pH 10), and soap (pH 12), among others, culminating in drain cleaner at pH 14, which is highly basic. An arrow runs beneath the scale, pointing from acidic to basic, with a central point labeled "Neutral."

[Generated by AI]

Period 3 oxides show a continuum in their acid base nature. **Table 2** shows the acid base character of the period 3 oxides. You can see the continuum from basic metal oxides to acidic non-metal oxides. You might notice that Al_2O_3 is listed as **amphoteric**. The term amphoteric means that a substance has the ability to act as either an acid or a base depending on the surrounding environment. Notice in **Figure 1** how Al_2O_3 also falls right at the ionic/covalent cusp of the bonding triangle providing further evidence for why it can sometimes act like a basic metal oxide and sometimes act like an acidic non-metal oxide.

Table 2. Acid-base character of the period 3 oxide

Oxides	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}
Acid-base character	Basic		Amphoteric		
pH	High (>7)		Powder (>7)) in water (<7)		

Chemical reactions for oxides

We can look at some of the reactions of period 3 oxides with water to identify the acids and bases produced in **Figure 3** and **Figure 4**.

Basic oxides

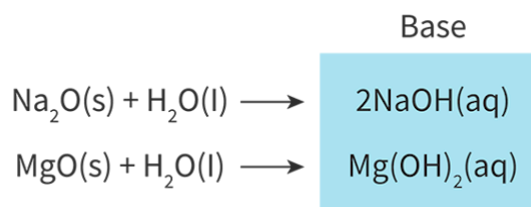


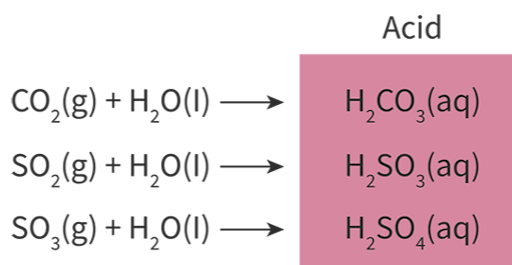
Figure 3. Reactions of metal oxides with water forming a strong base. More information for figure 3

The diagram illustrates the reactions between metal oxides and water, resulting in the formation of metal hydroxides. Two reactions are depicted with chemical formulas and directional arrows indicating the relationship between reactants and products. The first reaction shows a metal oxide reacting with water to form $2\text{NaOH}(\text{aq})$, suggesting the involvement of sodium. The second reaction similarly depicts an oxide reacting with water to produce $\text{Mg}(\text{OH})_2(\text{aq})$, indicating the formation of magnesium hydroxide. The diagram highlights that these reactions lead to the formation of strong bases, as metal hydroxides are released in aqueous form. This visual representation underscores the chemical transformation of metal oxides into basic solutions upon interaction with water.

[Generated by AI]

When metal oxides react with water, a metal hydroxide is formed. Metal hydroxides are ionic compounds and the chemical formula of the metal hydroxide formed can be deduced by considering the charge on the metal ion and the 1– charge on the hydroxide ion as you learned in [section S2.1.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>). The metal hydroxide formed is a base leading to the high pH of aqueous solutions containing metal oxides. You will learn more about bases in [section R3.1.6](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>).

Acidic oxides

**Figure 4.** Reactions of non-metal oxides with water forming an acid. More information for figure 4

The diagram illustrates the reactions of various non-metal oxides with water resulting in the formation of acids. Starting from the top, carbon dioxide (CO_2) is shown reacting with water (H_2O) to produce carbonic acid (H_2CO_3). Next, sulfur dioxide (SO_2)

combines with water to form sulfurous acid (H_2SO_3). Finally, sulfur trioxide (SO_3) reacts with water, giving rise to sulfuric acid (H_2SO_4). Each reaction is depicted with an arrow pointing from the non-metal oxide and water towards the resulting acid. The chemical formulas are clearly displayed next to each reaction arrow.

[Generated by AI]

Non-metal oxides react with water to produce acids. For example, carbonic acid, H_2CO_3 , is produced when carbon dioxide reacts with water. And when sulfur dioxide or sulfur trioxide react with water, sulfurous acid, H_2SO_3 or sulfuric acid, H_2SO_4 are produced. You will learn more about acids in [section R3.1.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>).

Study skills

Note: you do need to be able to deduce the equations for the reactions with water of the oxides of group 1 and group 2 metals, carbon and sulfur. Most should be fairly straightforward as the products formed are examples of ionic formulas using polyatomic ions you encountered in [section S2.1.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>). SO_2 is the less intuitive example and worth spending additional time reviewing to familiarise yourself with it.

Implications of acidic non-metal oxides

The ability of non-metal oxides to form acids when they react with water has some major implications for our ecosystems, our health and our infrastructure. The two major global concerns of these acidic non-metal oxides are:

1. ocean acidification
2. acid rain.

Where might you have heard of these terms before? What sort of impact do you expect these to have on ecosystems?

Ocean acidification

Did you notice in **Figure 4** how when CO_2 interacts with water an acid, H_2CO_3 is formed? Can you think about why this might be a global concern? How is CO_2 impacting our earth's atmosphere? What are some major sources of CO_2 ?

The accumulation of CO_2 in the Earth's atmosphere, primarily due to the combustion of fossil fuels, has major implications for our environment as shown in **Figure 5**. Not only does CO_2 act as a greenhouse gas, but it also can dissolve in bodies of water, acidifying it as H_2CO_3 is formed. This has major implications for our oceans in particular as marine organisms have difficulty adapting to minor pH changes of their environment. Ocean acidification can affect societies as marine food chains and storm protection from reefs are negatively impacted.

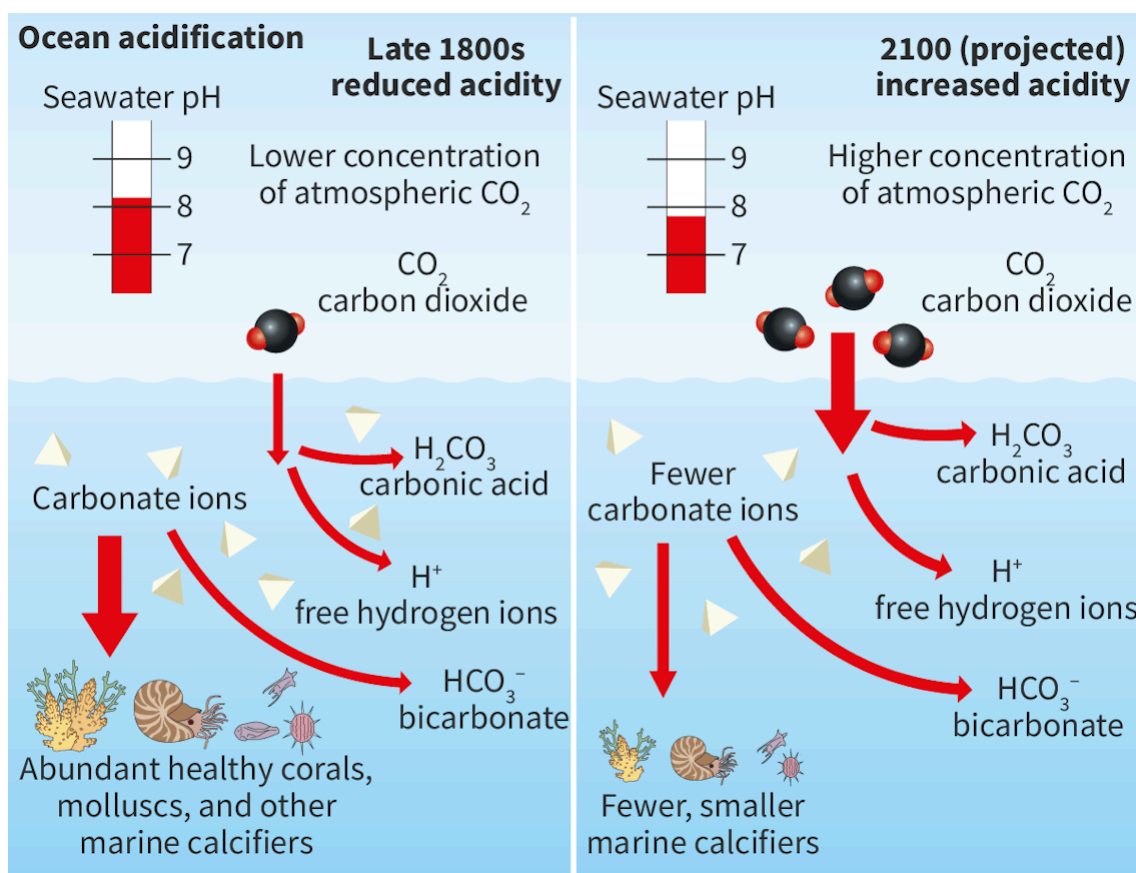


Figure 5. Effects of ocean acidification on marine life.

 More information for figure 5

The diagram illustrates the effects of ocean acidification on marine life comparing two time periods: the late 1800s, representing reduced acidity, and the projected 2100, showing increased acidity. On the left, the late 1800s section shows a lower concentration of atmospheric CO_2 . Carbon dioxide dissolves in seawater, forming carbonic acid (H_2CO_3), which then dissociates into free hydrogen ions (H^+) and carbonate ions. This environment supports abundant, healthy corals, mollusks, and other marine calcifiers. The pH scale on the left indicates seawater pH ranging from 9 to 7, where lower acidity is indicated by a higher pH.

On the right, the 2100 (projected) increased acidity section depicts a higher concentration of atmospheric CO₂. More carbon dioxide leads to more carbonic acid formation, resulting in more free hydrogen ions, thereby reducing carbonate ions available to marine organisms. Consequently, fewer and smaller marine calcifiers, such as corals and mollusks, are evident. The pH scale also ranges from 9 to 7, but this time illustrates increased acidity as represented by a lower pH. Arrows in the diagram trace the transformation of CO₂ to carbonic acid and further to bicarbonate, highlighting the decreasing carbonate availability in the future scenario.

[Generated by AI]

Compare the two photos in **Interactive 1** showing thriving healthy coral and coral bleached from rising temperatures and acidified oceans.

Interactive 1. Comparison of Healthy and Bleached Coral.

 More information for interactive 1

The interactive has two slides that display coral reefs grown under two different environmental conditions, showing the impact of climate change on marine ecosystems.

Slide 1: In this interactive, the coral reef is healthy. The corals are brightly colored in shades of orange, pink, yellow, and green, indicating that they are alive and covered with algae. These algae provide food to the corals through photosynthesis, giving the reef its vibrant colors. A variety of marine species depend on this coral reef for shelter and survival. The water appears clear, suggesting a balanced and stable ecosystem. The text below the image reads, thriving healthy coral.

Slide 2: In this interactive, the coral reef is completely bleached. The colorful corals

have turned white. This phenomenon, known as coral bleaching. The text below the image reads, bleached coral caused by rising temperature and acidification of the ocean.

Comparing the two interactives highlights the devastating impact of climate change and environmental stressors on coral reefs. Coral bleaching is a warning sign of ocean health deterioration, emphasizing the need for conservation efforts to protect marine life and mitigate climate change effects.

Acid rain

Have you heard of acid rain before? What do you think it means? Where do you think the acid in acid rain comes from? Two gaseous non-metal oxides are the major contributors to acid rain. Sulfur and nitrogen containing oxides readily produce the acids H_2SO_4 and HNO_3 , when they react with water. Any form of deposition (rain, sleet, snow, fog) that contains H_2SO_4 and/or HNO_3 is referred to as acid rain. Since H_2SO_4 and HNO_3 form easily when gaseous oxides of sulfur and nitrogen interact with water, it is events producing these gases that are the root causes of acid rain.

Sulfur oxides can be produced naturally from volcanoes and nitrogen containing oxides can be produced naturally in a thunderstorm, but these quantities are small to negligible in comparison to the volumes produced by industrial activity. Sulfur is an impurity in fossil fuels extracted from the ground. When these are burned SO_2 (which easily reacts with O_2 in the air to form SO_3) are released into the atmosphere. Nitrogen oxides are produced when NO (which easily reacts to form NO_2) is formed when N_2 in the air reacts with oxygen during a combustion reaction of fossil fuels.

The formation of acid rain has significant environmental concerns on both land and aquatic ecosystems. These range from impacting crop yields, to damaging infrastructure, impacting tree growth and the negatively impacting the respiratory systems of mammals. These are illustrated in **Figure 6**.

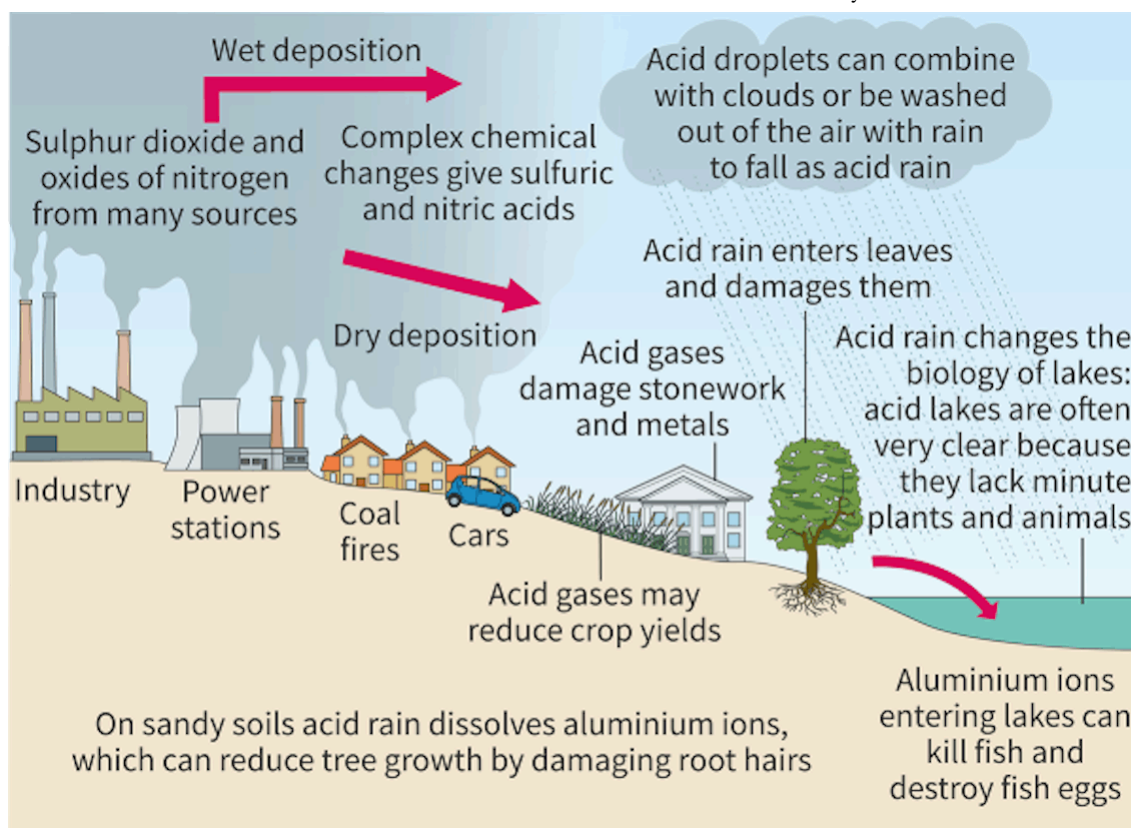


Figure 6. Causes and effects of acid rain.

 More information for figure 6

The diagram presents the causes and effects of acid rain with various labeled elements and arrows indicating the flow of causes to effects. The left side shows industrial factories, power stations, coal fires, and cars emitting sulfur dioxide and nitrogen oxides, leading to complex chemical changes that produce sulfuric and nitric acids. These emissions are transported as wet or dry deposition. Arrows labeled "wet deposition" and "dry deposition" point toward the right, indicating how these acids fall to the ground as acid rain.

On the right side, the effects of acid rain are shown. Arrows indicate that acid rain can enter leaves and damage them, affect the biology of lakes by making them clear due to lack of organisms, and harm stonework and metals. Text specifies that acid gases may reduce crop yields and notes that acid rain's effects on soil can inhibit tree growth by dissolving aluminum ions, which damage root hairs. Additionally, aluminum ions entering lakes can kill fish and destroy fish eggs. The diagram captures the overall flow from pollution sources to environmental impacts.

[Generated by AI]

It is easy to observe roughened surfaces and loss of detail in carving in buildings and statues as a result of acid rain. **Figure 7** takes a closer look at the effects of acid rain on a statue.



Figure 7. Effects of acid rain on statues.

Credit: RapidEye, [Getty Images](https://www.gettyimages.co.uk/detail/photo/sad-young-woman-mourning-on-a-1932-grave-monument-royalty-free-image/1159180665)

(<https://www.gettyimages.co.uk/detail/photo/sad-young-woman-mourning-on-a-1932-grave-monument-royalty-free-image/1159180665>)

International Mindedness

There have been many global efforts to reduce the production of nitrogen and sulfur containing gases. Many industrial facilities reduce sulfur oxides by adding CaCO_3 during the process to produce CaSO_4 either before or after the combustion process. This drastically reduces the sulfur oxides in the atmosphere. The combustion engines of modern vehicles all contain catalytic converters. These aim to return nitrogen containing oxides back to inert nitrogen. The addition of catalytic converters to vehicle combustion engines have had a dramatic impact on the nitrogen oxides in the atmosphere.

Acid rain was a considerable global environmental concern in the 1970s and 1980s and, through collaborative policy making between many nations, effects have been significantly reduced. While still an environmental concern, it is not the urgent environmental crisis it once was. How might societies use lessons learned from acid rain to tackle urgent pressing matters in today's climate fight?

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Designing procedures
- **Time required to complete activity:** 45 minutes
- **Activity type:** Individual activity

Part A: Design lab

You have two mystery elements, one metal and one non-metal. Below is a list of lab equipment you have access to as well as two videos showing the formation of oxides from magnesium and sulfur. Using this information, design an experiment to determine which element is the metal and which is the non-metal. Ensure to carefully plan safety considerations and waste disposal.

- Element 1
- Element 2
- pH or litmus paper
- Bunsen burner
- Tongs
- Test tube holder
- Test tube rack
- Pyrex test tube
- Beaker
- Water
- Fume hood
- Safety glasses

Burning Magnesium Ribbon



Video 1. Burning of magnesium in air.

 More information for video 1

Sulphur Dioxide forms an acid | Acids & Bases | Chemis...



Video 2. Burning of sulfur in air.

Part B: Chemical equations

Write chemical equations for the following:

- Formation of magnesium oxide from magnesium.
- Formation of sulfur dioxide from sulfur.

- Reactions showing both magnesium oxide and sulfur dioxide reacting with water.

Part C: Reading through an article

Read through the abstract, introduction, conclusion and **Figure 1** of this open access article  (<https://link.springer.com/content/pdf/10.1007/s13280-019-01244-4.pdf?pdf=button>). Write down your responses to the following questions:

- What period in time was acid rain of most concern?
- Discuss the progress made in the reduction of nitrogen and sulfur oxide emissions.
- Do you think we should still be concerned about acid rain today?
- Discuss how important you think science-policy interactions are to addressing climate change.
- What are some of the limitations of science-policy interactions?

Oxidation states

Learning outcomes

By the end of this section you should be able to:

- Deduce the oxidation states of an atom or an ion in a compound.
- Explain why the oxidation state of an element is zero.

Metallic and non-metallic character is the tendencies of atoms to lose or gain electrons. Can you think of any ways that we can track how many electrons are transferred or distributed when a chemical bond is formed? How did we keep track of this when we looked at ionic bonding in [section S2.1.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>)? When we think of a substance like NaCl it is easy to identify that sodium has transferred an electron to chlorine forming Na^+ and Cl^- ions.

But what about in a substance like H_2O ? There is no electron transfer occurring between oxygen and hydrogen, but the two atoms also do not share electrons equally in their polar covalent bond. These electrons are unevenly distributed. So how might we track this electron distribution of atoms in any substance, regardless of bonding type?

In ionic compounds, we tracked how electrons were transferred by considering the ionic charges of the positive and negative ions. Ionic charges however are only specific to ionic compounds. A similar but more inclusive electron tracking method involves assigning an oxidation state or oxidation number to each atom in a substance. Oxidation state is the numerical, hypothetical charge assigned to an atom in a substance based on the degree of electron transfer between atoms. These values are based on the relative electronegativities of elements.

Study skills

Oxidation state and oxidation number are two commonly interchangeable terms with the same meaning.

You might be wondering how is an oxidation state different from an ionic charge? Oxidation states are hypothetical and a method to track how electrons are distributed. They may be used in substances where electron transfer has not occurred at all. Ionic charges on the other hand indicate a numerical quantity of electrons that has been either lost or gained from a species. To differentiate the two terms a slightly different notation is used. **Table 1** shows the different notations used for the ion Na^+ . Note that since sodium has lost an electron in Na^+ the numerical value is the same for both the oxidation state and ionic charge.

Table 1. Comparing ionic charge and oxidation state of Na^+ .

Na^+	
Ionic charge	Oxidation state
1+	+1

Rules for assigning oxidation states

In order to assign the oxidation states for substances, the following rules can be applied. For each rule we will look at examples involving application of the rule.

1. A pure element will have an oxidation state of 0. This is because the distribution of electrons between atoms is the same for atoms of the same element. Atoms of the same element will have the same tendency to attract or gain electrons, therefore an oxidation number of zero is used to indicate no degree of electron transfer.

Element	Oxidation State
Na(s)	Oxidation state of Na: 0
$\text{Cl}_2(\text{g})$	Oxidation state of each Cl: 0
$\text{P}_4(\text{s})$	Oxidation state of each P: 0
C(s)	Oxidation state of C: 0

2. Monoatomic ions have an oxidation state equal to the charge of the ion.

Element	Oxidation State
Cu^{2+}	Oxidation state of Cu: +2

Element	Oxidation State
Cl^-	Oxidation state of each Cl: -1
S^{2-}	Oxidation state of each S: -2
Fe^{3+}	Oxidation state of Fe: $+3$

3. Oxidation state of all atoms in a neutral compound (ionic, molecular or both) must sum up to give zero overall.

4. Oxidation states of atoms in polyatomic ions must add up to give the overall charge of the ion.

OH^-	
O	Oxidation number on O: -2
H	Oxidation number on H: $+1$
Sum	$-2 + 1 = -1$
To sum we would add the oxidation number of the oxygen atom and the hydrogen atom	

5. Oxidation numbers of atoms are typically the charge of the most common ion. The typical oxidation numbers for the following groups are:

- Group 1 metals: $+1$ oxidation number.
- Group 2 metals: $+2$ oxidation number.
- Group 17 elements: -1 oxidation number (unless bonded to another more electronegative halogen).
- Oxygen: -2 oxidation number (unless in a peroxide, O_2^{2-} , then it is -1)
- Hydrogen: $+1$ oxidation number (unless in a metallic hydride such as NaH).

LiCl		MgF ₂	
Li	Oxidation number on Li: $+1$	Mg	Oxidation number on Mg: $+2$
Cl	Oxidation number on Cl: -1	F	Oxidation number on each F: -1

LiCl		MgF ₂	
Sum	$+1 + (-1) = 0$	Sum	$+2 + 2(-1) = 0$
Compounds are neutral so the sum of the oxidation states must be equal to zero.			

6. Assign positive oxidation numbers to the most metallic elements first and negative oxidation numbers to the most non-metallic elements first. Since metallic character is defined as the tendency of an atom to lose electrons and non-metallic character as the tendency of an atom to gain electrons these are the elements that will have the most predictable oxidation state. In most cases, this involves working from the elements on the outside of the chemical formula and working inwards.

KMnO ₄		Na ₃ PO ₄	
K is the most metallic element and O is the most non-metallic element.		Na is the most metallic element and O is the most non-metallic element.	
K	Oxidation number on K: +1	Na	Oxidation number on Na: +1
O	Oxidation number on each O: -2	O	Oxidation number on each O:
Sum	$+1 + x + 4(-2) = 0$	Sum	$3(+1) + x + 4(-2) = 0$
Can use the variable x as a placeholder to solve for the oxidation state of the last element. the overall substance is neutral the oxidation states must sum to zero.			
Mn	Oxidation number on Mn: +7	P	Oxidation number on P: +5

Study skills

When a substance is made up of more than two elements it is helpful to set up a mathematical equation in order to algebraically solve for the oxidation number of the element of interest.

Worked examples

Try assigning oxidation states following the three worked examples shown below.

Worked example 1

Deduce and compare the oxidation states of each element in NO and NO₂.

Both NO and NO₂ are made up of only nitrogen and oxygen. Since both elements are non-metals, we can assign the oxidation number to oxygen first since it has a higher non-metallic character and a greater tendency to gain electrons.

NO			
O	Oxidation state of O: -2	O	Oxidation state of O: -2
Sum	$-2 + x = 0$	Sum	$2(-2) + x = 0$
Can use the variable x as a placeholder to solve for the oxidation state of N. Since the overall substance is neutral the oxidation states must sum to 0.			
N	+2	N	+4

Therefore the oxidation states of nitrogen and oxygen are +2 and -2 in NO and +4 and -2 in NO₂.

In both substances, the oxidation states of nitrogen are positive. In NO₂ the oxidation state of nitrogen is a greater positive number since it is bonded to electronegative oxygen atoms

Worked example 2

Assign oxidation numbers to each element in Al₂(SO₄)₃.

Looking at each of the elements in Al₂(SO₄)₃, aluminium is the most metallic element and oxygen is the most non-metallic element. This means we will assign the oxidation numbers to Al and O first and then solve for S last.

Since Al is in group 13 and has 3 valence electrons its oxidation number would be +3.

Since O is in group 16 and has 6 valence electrons, its oxidation number would be -2.

We can now create a mathematical equation summing the oxidation numbers using the variable x as a placeholder for S. The sum of all the oxidation numbers will equal 0 since Al₂(SO₄)₃ is an overall neutral substance. Be sure to account for the subscripts in your mathematical equation noting that there are 2 Al atoms, 3 S atoms and 12 O atoms.

$$2(+3) + 3(x) + 12(-2) = 0$$

Rearranging and solving for x we find:

$$x = \frac{+24 - 6}{3}$$

$$x = +6$$

Therefore the oxidation numbers of each element in $\text{Al}_2(\text{SO}_4)_3$ are:

- Al: +3
- S: +6
- O: -2

Below is a tricky worked example. Try answering it yourself first and then read through the solution.

Worked example 3

Assign oxidation numbers to each element in hydrogen peroxide, H_2O_2 .

This is a tricky example. And the reason it is tricky is that you will get a different answer depending on which element you assign the oxidation number to first. Let's look at both possible options and reason through which answer makes more sense.

H_2O_2			
Assigning H first		Assigning O first	
H	+1	O	-2
Sum	$2(+1) + 2x = 0$	Sum	$2x + 2(-2) = 0$
O	-1	H	+2

You will notice that if we assign hydrogen first oxygen will have an oxidation number of -1 and if we assign oxygen first hydrogen will have an oxidation number of +2. Can you identify an issue with one of these answers?

The oxidation number of +2 does not make sense for hydrogen. This is because hydrogen only has one electron. It is not possible for hydrogen to have an oxidation number of +2 because there are not two electrons to lose.

Therefore the oxidation numbers of each element in H_2O_2 are:

- H: +1
- O: -1

This is an example of a peroxide. The peroxide anion also has this same exception of oxidation number for oxygen. This can happen when metals are bonded to the peroxide anion, such as in the example of Na_2O_2 . Oxygen in this compound will have the -1 oxidation state in comparison to Na_2O we are more familiar with where oxygen will have the -2 oxidation state.

Making connections

Oxidation numbers are very useful for analysis of redox reactions. By assigning oxidation numbers to each element in the reactant and products it is easy to analyse how electrons have been redistributed throughout a chemical reaction. Redox reactions are a type of electron transfer reactions that you will learn more about in [section R3.2.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/).

Naming conventions using oxidation numbers

When naming substances containing elements that can have variable oxidation states, Roman numerals are used to indicate the oxidation state. You have already looked at examples that do this in [section S2.1.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) when you looked at naming ionic compounds with cations made up of a transition element. For example in CuCl_2 , Cu has the +2 oxidation state therefore the name of the ionic compound is copper(II) chloride.

Roman numerals can also be used to show the oxidation states of elements within oxyanions. Oxyanions are negatively charged polyatomic ions involving oxygen bonded with another element. When naming oxyanions, the oxidation state of the element bonded to oxygen is indicated using Roman numerals. For example in [section S3.1.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/) when we looked at the formation of H_2SO_3 and H_2SO_4 acids from oxides of water dissolving in water, these acids can also be named sulfuric(IV) acid and sulfuric(VI) acid. The Roman numeral indicates the oxidation number on the sulfur since that is the element bonded to oxygen making up the anion part of the substance. Can you identify which name belongs to which acid?

Table 2 shows some common examples of substances involving transition metals and oxyanions that use Roman numerals to indicate oxidation states. The generic names are also included as these are names that do still persist in chemistry

despite not being recognised as proper naming by IUPAC.

Table 2. Common examples of substances involving transition metals and oxyanions.

Name of compound	Defining oxidation number	IUPAC name (Roman numerals)	Generic name
FeCl_2	Fe: +2	iron(II) chloride	ferrous chloride
FeCl_3	Fe: +3	iron(III) chloride	ferric chloride
$\text{Na}_2\text{Cr}_2\text{O}_7$	Cr: +6	sodium dichromate(VI)	sodium dichromate
KMnO_4	Mn: +7	potassium manganate(VII)	potassium permanganate
Na_2SO_4	S: +6	sodium sulfate(VI)	sodium sulfate
Na_2SO_3	S: +4	sodium sulfate(IV)	sodium sulfite
$\text{Ca}(\text{NO}_3)_2$	N: +5	calcium nitrate(V)	calcium nitrate
$\text{Ca}(\text{NO}_2)_2$	N: +3	calcium nitrate(III)	calcium nitrite

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills – Using terminology, symbols and communication conventions consistently
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Oxidation number pixel art

Complete the following [oxidation number pixel art](https://docs.google.com/spreadsheets/d/1jSJBUTtypl-cGmtEixhzNYBNfB3ACNNEenuAViFVZql/copy?) 

([https://docs.google.com/spreadsheets/d/1jSJBUTtypl-](https://docs.google.com/spreadsheets/d/1jSJBUTtypl-cGmtEixhzNYBNfB3ACNNEenuAViFVZql/copy?)

[cGmtEixhzNYBNfB3ACNNEenuAViFVZql/copy?](https://docs.google.com/spreadsheets/d/1jSJBUTtypl-cGmtEixhzNYBNfB3ACNNEenuAViFVZql/copy?)) on Google sheets. You will need to make a copy of the document for yourself. There are 20 substances where you will need to correctly identify the oxidation number of the specified element. For each substance you identify correctly part of an image will be revealed.

When all 20 oxidation numbers have been identified the entire pixel art image will be revealed.

Discontinuities in ionisation energy (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to explain how discontinuities in first ionisation energy provide evidence for the existence of energy sublevels.

In section S3.1.3b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/>) we looked at the general trends in ionisation energy, the energy required to remove an electron. However, we only looked generally at these trends down a group and across a period. **Figure 1** shows the first ionisation energies for elements in periods 1 to 3. Looking at this, does the first ionisation energy always increase as you travel across a period? Are there any elements that stand out as anomalies to this trend? Why do these elements fall outside of the general trend? Why is it important to analyse data that fall outside of general trends?

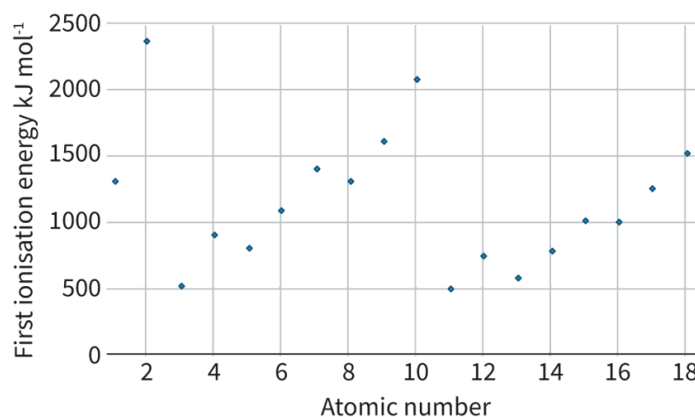


Figure 1. First ionisation energies for elements in periods 1 to 3.

 More information for figure 1

The scatter plot illustrates the first ionisation energies for elements in periods 1 to 3 of the periodic table. The X-axis represents the atomic number of the elements, ranging from 0 to 18. The Y-axis represents the first ionisation energy in kilojoules per mole (kJ mol⁻¹), ranging from 0 to 2500.

Data points indicate the ionisation energies of specific elements. The plot shows a trend of increasing ionisation energy across the periods, but there are noticeable fluctuations and anomalies. Not all elements show a consistent increase in ionisation energy, with some dips and spikes occurring, representing the discontinuities or anomalies in the general trend.

[Generated by AI]

You may recall from [section S1.3.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>) that these anomalies are called discontinuities. **Figure 2** highlights these discontinuities and takes a closer look at what elements fall near them. Looking at **Figure 2**, consider the following questions: What groups do these discontinuities fall between? What are the electron configurations of the valence shell for these elements? Why might these discontinuities exist?

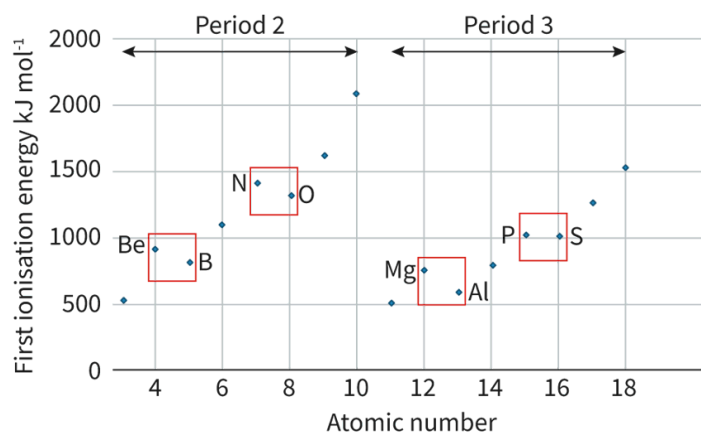


Figure 2. Discontinuities in ionisation energies in periods 2 and 3.

 More information for figure 2

The graph shows the first ionisation energy in kJ mol^{-1} on the Y-axis, ranging from 0 to 2000, against atomic numbers on the X-axis, ranging from 4 to 20, which correspond to elements across periods 2 and 3 of the periodic table. Data points indicate the ionisation energies for elements like Be, B, N, O, Mg, Al, P, and S. There are highlighted sections with red boxes showing discontinuities between certain elements such as Be and B, N and O, Mg and Al, and P and S. The graph includes horizontal lines marking energy levels and vertical lines indicating atomic numbers, assisting in identifying these discontinuities.

[Generated by AI]

These discontinuities appear in two distinct regions moving across a period on the periodic table:

- Between groups 2 and 13
- Between groups 15 and 16

These discontinuities observed, based on decreasing ionisation energy values, indicate that it is easier to remove an electron from elements in group 13 than in group 2, and to remove from elements in group 16 than group 15. The lower than expected ionisation energy values suggest that group 13 and group 16 elements have a weaker attraction between the nucleus and outer valence shell electrons than we would expect, given their higher effective nuclear charge than group 12 and group 15 elements. So why might this attraction be weaker? Let's take a closer look at the electron configuration for elements in these groups in **Table 1**.

Table 1. General electron configurations for valence electrons for elements in groups 2 and 13

Period	Outer energy level	Group 2	Group 13
2	$n = 2$	Be [He] $2s^2$	B [He] $2s^2 2p^1$
3	$n = 3$	Mg [Ne] $3s^2$	Al [Ne] $3s^2 3p^1$

First discontinuity

What do you notice about the sublevel that the outer electron occupies in group 13 in comparison to in group 2? Let's look at an orbital diagram to compare the sublevels of the outer electrons for beryllium (Be) and boron (B) in **Figure 3**.

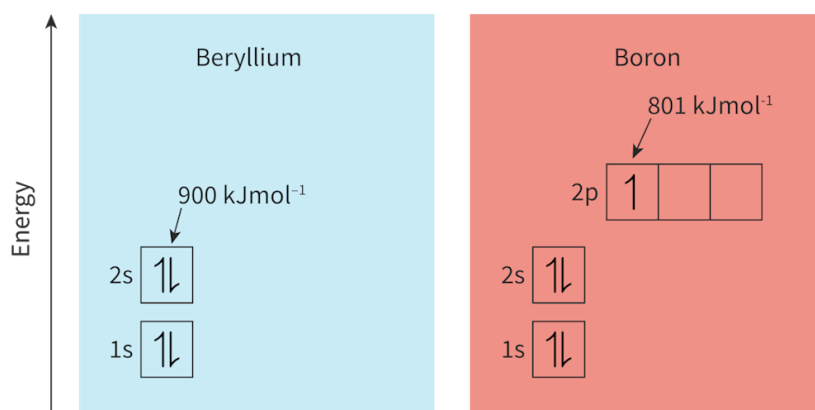


Figure 3. Orbital diagrams for beryllium and boron.

 More information for figure 3

The image displays two orbital diagrams side-by-side, comparing the electron configurations of beryllium and boron.

On the left, the diagram for beryllium shows two electron shells. The 1s sublevel contains two electrons with opposite spins, represented as arrows pointing in opposite directions within a box labeled "1s." The 2s sublevel also contains two electrons with opposite spins in a box labeled "2s." The energy level is marked as 900 kJmol⁻¹.

On the right, the diagram for boron displays three sublevels. Like beryllium, the 1s and 2s sublevels each contain two electrons with opposite spins in their corresponding boxes. The new 2p sublevel contains one electron in the first of three available slots, represented by a single upward arrow in a box labeled "2p." The energy level is marked as 801 kJmol⁻¹.

These diagrams illustrate that beryllium has its outer valence electron in the 2s sublevel, while boron has its outer valence electron entering the higher energy 2p sublevel.

[Generated by AI]

Group 2 elements, like beryllium, have an outer valence electron occupying an s-sublevel, whereas group 13 elements boron have an outer valence electron occupying a p-sublevel. Since p-sublevels are slightly higher in energy than s-sublevels, the nucleus has a weaker attraction to these electrons. This is why the first ionisation energy of elements in group 13 is lower than for elements in group 2 as we move across a period.

This discontinuity observed in the general trend of first ionisation energy provides evidence for the existence of a sublevel that is slightly higher in energy for the third valence electron to occupy.

Second discontinuity

What do you notice about the sublevel that the outer electron occupies in group 15 in comparison to in group 16? Let's look at an orbital diagram to compare the sublevels of the outer electrons for nitrogen (N) and oxygen (O) in **Figure 4**.

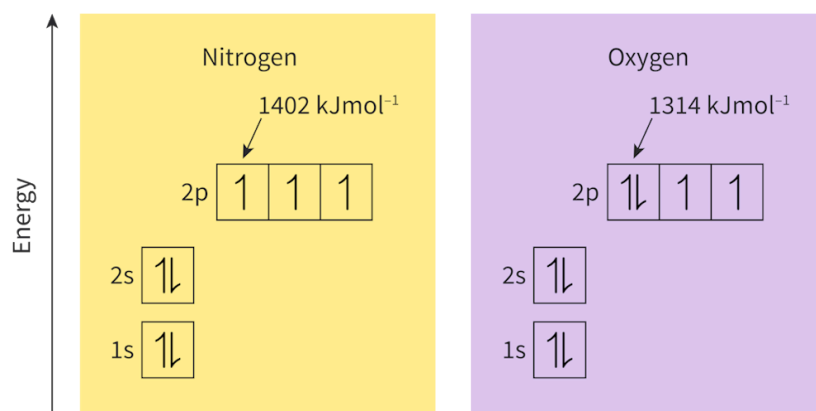


Figure 4. Orbital diagrams for nitrogen and oxygen.

 More information for figure 4

The image displays two orbital diagrams comparing the electron configurations of nitrogen and oxygen. On the left, the nitrogen diagram is presented with a yellow background. It shows a vertical axis labeled "Energy." Below are three horizontal boxes labeled 1s, 2s, and 2p. The 1s and 2s orbitals each have two arrows in opposite directions, denoting paired electrons. The 2p sublevel is shown with three boxes, each containing a single arrow, indicating singly occupied orbitals. The text "1402 kJmol⁻¹" is highlighted, reflecting nitrogen's ionization energy.

On the right, the oxygen diagram is on a purple background. It also includes three horizontal boxes labeled 1s, 2s, and 2p. Similar to nitrogen, the 1s and 2s orbitals are fully occupied with paired arrows. However, in the 2p sublevel, the first box has two arrows in opposite directions, noting double occupancy, while the other two boxes contain a single arrow each. The ionization energy for oxygen is specified as "1314 kJmol⁻¹."

This comparison illustrates the difference in electron configuration between the two elements, particularly in the 2p sublevel where nitrogen has singly occupied orbitals, whereas oxygen has one doubly occupied orbital.

[Generated by AI]

Both of these elements have outer valence electrons occupying a p-sublevel. Group 15 elements, such as nitrogen, have valence electrons singly occupying each orbital within the p-sublevel, whereas group 16 elements, such as oxygen, have one orbital in the p-sublevel with double occupancy. Due to the orbital's double occupancy, there will be added repulsion between the two electrons in the same orbital. This increased repulsion decreases the attraction between the valence electrons and the nucleus making it easier to remove an electron. Hence, this explains the slightly lower first ionisation energy of elements in group 16 in comparison to group 15.

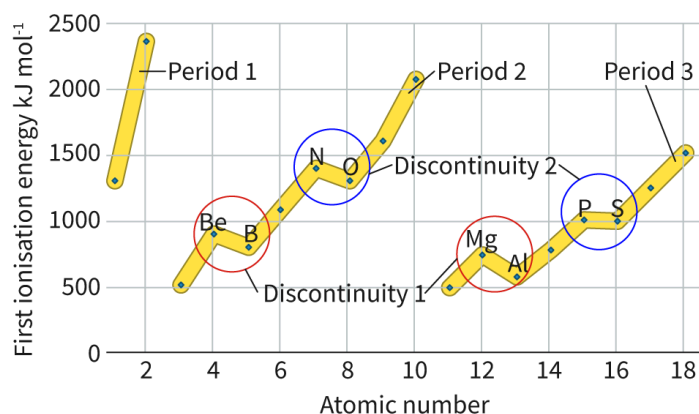
This discontinuity observed in the general trend of first ionisation energy provides evidence for the existence of orbitals able to hold a maximum of two electrons that make up each sublevel.

Nature of Science

Aspect: Evidence

Since atomic orbitals are just a region of space where there is a high probability of finding an electron, evidence of their existence is quite limited (i.e. we cannot just see an orbital with a microscope). First ionisation energies are quantitative data that we can measure. Trends in ionisation energies, along with their discontinuities provide a lot of evidence to show how electrons are arranged within an atom. The general trend in ionisation energy provides evidence for the existence of energy levels, but it is the discontinuities that provide evidence for the existence of sublevels and atomic orbitals.

Interactive 1 is an animation showing a closer view of these discontinuities, by looking at the way that electrons are arranged within their sublevels.



Interactive 1. A Closer Look at the Discontinuities of First Ionisation Energy.

 More information for interactive 1

An interactive slideshow with five slides presents the discontinuity of the first ionisation energy across elements. Each slide presents a scatterplot exhibiting first ionisation energy in kJ mol^{-1} on the y-axis and atomic number on the x-axis. The y-axis ranges from 0 to 2500 kJ mol^{-1} , and the x-axis ranges from 0 to 18.

Slide 1:

This slide depicts ionisation energy trends for Periods 1, 2, and 3. In each period, a line connects the data points for its elements. The following table presents the approximated data:

All periods show an overall increasing trend, with the discontinuities in Periods 2 and 3.

Discontinuity 1:

Atomic number	First ionisation energy (KJmol ⁻¹)
Period 1	
1	1350
2	2400
Period 2	
3	500
4	900
5	850
6	1150
7	1400
8	1300
9	1650
10	2100
Period 3	
11	500
12	750
13	600
14	800
15	1050
16	1000
17	1250
18	1500

Beryllium, Be (Atomic number $\backslash (= 4 \backslash)$ and Ionisation energy $\backslash (= 900 \backslash)$) and Boron, B (Atomic number $\backslash (= 5 \backslash)$ and Ionisation energy $\backslash (= 850 \backslash)$). Boron has a lower ionisation energy than Beryllium.

Magnesium, Mg (Atomic number $Z = 12$) and Ionisation energy $I = 750$) and Aluminum, Al (Atomic number $Z = 13$) and Ionisation energy $I = 600$). Aluminum has a lower ionisation energy than Magnesium.

Discontinuity 2:

Nitrogen, N (Atomic number $Z = 7$) and Ionisation energy $I = 1400$) and Oxygen, O (Atomic number $Z = 8$) and Ionisation energy $I = 1300$). Oxygen has a lower ionisation energy than Nitrogen.

Phosphorus, P (Atomic number $Z = 15$) and Ionisation energy $I = 1050$) and Sulfur, S (Atomic number $Z = 16$) and Ionisation energy $I = 1000$). Sulfur has a lower ionisation energy than Phosphorus.

Slide 2:

This slide highlights discontinuity 1 in detail.

Beryllium and Magnesium are from group 2, and Boron and Aluminum are from group 13. Their electronic configuration and their orbital filling are as follows:

Group 2 elements:

Be: $1s^2 2s^2$

1s orbital has 2 electrons, 2s orbital has 2 electrons, whereas 2p, 3s, and 3p orbitals are empty.

Mg: $1s^2 2s^2 2p^6 3s^2$

1s, 2s, and 3s orbitals each have 2 electrons and 2p orbital has six electrons, whereas 3p orbital is empty.

Group 13 elements:

B: $1s^2 2s^2 2p^1$

1s and 2s orbitals each have 2 electrons and 2p orbital has one electron, whereas 3s and 3p orbitals are empty.

Al: $1s^2 2s^2 2p^6 3s^2 3p^1$

1s, 2s, and 3s orbitals each have 2 electrons, 2p orbital has six electrons, and 3p orbital has one electron.

These valence electrons are slightly higher in energy than the valence electrons in group 2 elements.

Slide 3:

Therefore electrons from Boron and Aluminum are easier to remove than the outer electron from group 2 elements.

Slide 4:

This slide highlights discontinuity 2 in detail.

Nitrogen and Phosphorus are from group 15, and Oxygen and Sulfur are from group 16. Their electronic configuration and their orbital filling are as follows:

Group 15 elements:

N: $1s^2 2s^2 2p^3$

1s and 2s orbitals each have 2 electrons and 2p orbital has 3 electrons, whereas 3s and 3p orbitals are empty.

P: $1s^2 2s^2 2p^6 3s^2 3p^3$

1s, 2s, and 3s orbitals each have 2 electrons, 2p orbital has six electrons, and 3p orbital has 3 electrons.

Only single occupancy.

Group 16 elements:

O: $1s^2 2s^2 2p^4$

1s and 2s orbitals each have 2 electrons and 2p orbital has four electrons, whereas 3s and 3p orbitals are empty.

S: $1s^2 2s^2 2p^6 3s^2 3p^4$

1s, 2s, and 3s orbitals each have 2 electrons, 2p orbital has six electrons, and 3p orbital has four electrons.

Double occupancy causes repulsion of electrons in the orbital.

Slide 5:

Repulsion between the electrons from Oxygen and Sulfur reduces attraction between nucleus and electrons. Easier to remove and lower IE.

This slideshow helps users identify the periodic trends, relate electronic

configuration and orbital occupancy, and recognise the discontinuities in ionisation energy.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking — Asking questions and framing hypotheses based upon sensible scientific rationale
- **Time required to complete activity:** 45 minutes
- **Activity type:** Group activity

Model of the atom mystery box

Materials needed per group:

- Box (shoebox size or smaller)
- 5 marbles
- Assortment of craft materials (cardboard, paper, toilet paper rolls, felt, glue, stapler, tape, etc.)

Creating a model of the atom (groups of 3—4 people)

- Your task is to create a model of an atom with 5 electrons, these electrons will be represented by marbles.
- You are going to come up with your own model of how these electrons are arranged in your atom. Your atom will be represented by the inside of the box.
- Your electrons could be fixed, move in an orbit, in a line, have free motion over certain regions of the atom (box). The choice is yours.
- You will use the craft materials you have at your disposal to design the arrangement of electrons within your atom.
- When complete your box will be sealed shut. Your peers will have to use evidence they can collect by moving, shaking and listening to come up with their own theory about how atoms are arranged in other students' boxes.

Creating models using evidence

- Swap your box with another group's box.
- Make observations (moving, listening, light shaking, rocking side to side) to try and figure out how the 5 electrons are arranged within the atom.
- Create a sketch to show how you think the electrons are arranged. Note all the evidence you have to support this arrangement.

Sharing models with class

- Each group will present their sketches to the class for how they think electrons are arranged in other group's boxes, making explicit mention of what evidence they have to support each part of their sketch.

- Students can then decide whether to reveal what the inside of the original box looked like or not, keeping in mind that we never really have the opportunity to see inside the atom to confirm the models we construct from evidence.

Class discussion

- Discuss how this activity relates to:
 - Models of the atom.
 - The importance of using different types of evidence when constructing models, particularly when visual observations are extremely limited.

Characteristic properties of transition elements (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Recognise properties including variable oxidation state, high melting points, magnetic properties, catalytic properties, formation of coloured compounds and formation of complex ions with ligands.
- Explain why transition elements have variable oxidation states.
- Deduce the electron configurations of ions of the first-row transition elements.

Long before the development of our modern periodic table, it was known that the addition of certain metal salts during the production of glass would form coloured glass. You may have encountered coloured glass before in coloured glass bottles, artwork or even on windows. Metal salts containing transition elements are responsible for their colour. **Figure 1** shows an example of coloured glass and some transition elements and ions responsible for the apparent colour.



Figure 1. Coloured glass from transition elements.

Credit: Fred de Noyelle, [Getty Images](#)

(<https://www.gettyimages.com/detail/photo/heiligenkreuz-abbey-chapterhouse-stained-glass-royalty-free-image/538211406>)

When we looked at the formation of ions earlier in [section S2.1.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>), we identified transition elements as metals in the middle region of the periodic table that were capable of forming different ionic charges (**Figure 2**). But why are they called transition elements? What are they transitioning from? What makes these elements unique in comparison to other elements?

Figure 2. Region of the periodic table where transition metals are found.

 More information for figure 2

The image is a diagram of a periodic table highlighting the transition metals. The transition metals are marked by an oval outline encompassing the middle portion of the table, labeled as groups 3 to 12. It includes rows 4 to 7 horizontally, indicating the location where transition metals are typically found. The image features a grid-like structure representing the periodic table layout, with the label "Transition metals" inside the oval. The typical shape of the periodic table is maintained, showing different groups and periods as rows and columns. The elements themselves are not individually labeled, focusing instead on the general location of the transition metals within the table's framework. Asterisks are denoted below and to the left, likely indicating notation or additional groupings, though specifics are not provided in the image.

[Generated by AI]

Furthermore, when we looked at transition elements in [section S2.3.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/transition-metals-hl-id-43565/>), we considered these as metals from groups 3–12 on the periodic table. This description we provided was incomplete. There is one group of elements that does not meet the criteria the International Union of Pure and Applied Chemistry (IUPAC  (<https://iupac.org/>)) has set to define transition elements. Using [section 8](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/>) of the DP chemistry data booklet, look at the melting points for groups 3–12 in period 4. Can you predict which element within the d-block of period 4 may not meet the criteria of a transition element?

A transition element is an element with an incomplete d-sublevel that is able to form cations with an incomplete d-subshell. They are called transition elements or transition metals because they exhibit transitional behaviour acting as somewhere

between s-block metals and p-block non-metals. The incomplete d-sublevels of transition elements give them characteristic properties.

If we consider the electron configuration of all the d-block elements in period 4 it becomes clear that the element zinc does not meet the criteria for a transition element. Zinc has the electron configuration $[\text{Ar}] 4s^2 3d^{10}$ and since it has a complete d-sublevel, it does not have the same characteristic properties as the other elements in this region of the periodic table. It is therefore not considered a transition element.

Characteristic properties of transition elements

1. Transition elements have variable oxidation states

Recall the formation of ions from transition elements from [section S2.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>). Since transition elements have valence electrons occupying both s and d-sublevels, electron loss can occur from the outer d-sublevel, after loss from the outer s-sublevel occurs. This results in a variable number of electrons that are able to be easily removed from transition elements. Let's look at common ions of iron and their electron configurations in **Table 1**.

Table 1. Oxidation number and electron configuration for common species of iron.

Species	Oxidation number	Electron configuration
Fe	0	$[\text{Ar}] 4s^2 3d^6$
Fe^{2+}	+2	$[\text{Ar}] 3d^6$
Fe^{3+}	+3	$[\text{Ar}] 3d^5$

We will look more closely as to why transition elements have variable oxidation states in the next [section S3.1.9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-variable-oxidation-states-id-44698/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-variable-oxidation-states-id-44698/>).

2. High melting points

Recall from [section S2.3.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/>) that metallic bonding is the electrostatic attraction between a lattice of cations and delocalised electrons. We considered in [section S2.3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/>) how this strong attraction between cations in the lattice and delocalised valence electrons led to metals having high melting points. **Table 2** shows the melting points for period 4 metals. What do you notice about the melting points of metals from group 1 and 2 in comparison to the transition metals? What do you notice about the melting point of zinc in comparison to the transition metals?

Table 2. Melting point temperatures for period 4 metals.

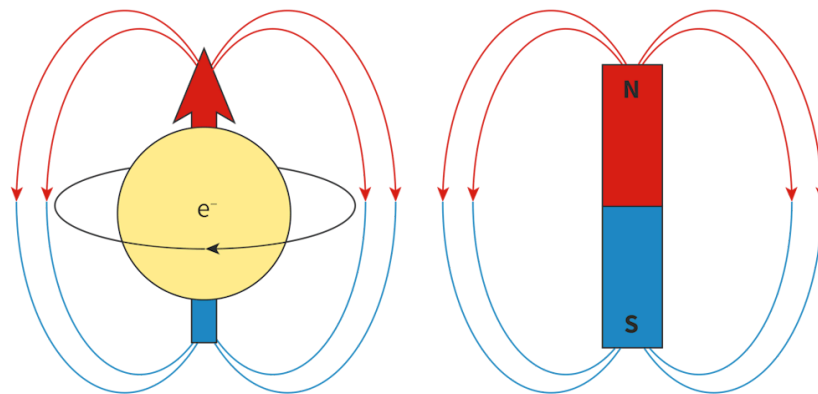
Element	Melting point (°C)
K	63.38
Ca	842.0
Sc	1541
Ti	1670
V	1910
Cr	1907
Mn	1246
Fe	1538
Co	1495
Ni	1455
Cu	1085
Zn	419.5

Since transition metals can form ions of varying oxidation states, this leads to a large number of delocalised valence electrons surrounding cations in the metallic lattice. The increased charge with a large number of valence electrons increases the strength of the metallic bond. These stronger metallic bonds require more energy to break, leading to higher melting points. This is why many everyday items you have encountered (e.g. cookware, coins, jewellery, etc.) are made from transition metals.

Since zinc can only form cations with a 2+ charge there are not as many delocalised valence electrons in comparison to transition metals. This results in a lower melting point of zinc in comparison to transition metals.

3. Magnetic properties

Did you know that electrons act as tiny magnets in atoms and molecules? The magnetic behaviour of electrons is the property called spin that you learned about in [section S1.3.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/). In that section, you learned that two electrons can occupy the same orbital, provided they have opposite spin. When elements have sublevels with unpaired electrons, magnetism can be observed in the substance. Since transition metals and ions have at least one unpaired electron in the d-sublevel, these metals and ions form a weak interaction with a magnetic field (which is called paramagnetism). This is shown in **Figure 3**.



Electron spin can be considered similar to a small magnet.

 More information

The image shows a comparison between electron spin and a small magnet. On the left side, a diagram displays an electron represented by a yellow circle labeled with 'e-', surrounded by arrows indicating its spin and magnetic field lines in blue and red. The right side shows a bar magnet, divided into red and blue sections indicating the north (N) and south (S) poles, respectively, with field lines similar to the electron's. The caption reads, "Electron spin can be considered similar to a small magnet."

[Generated by AI]

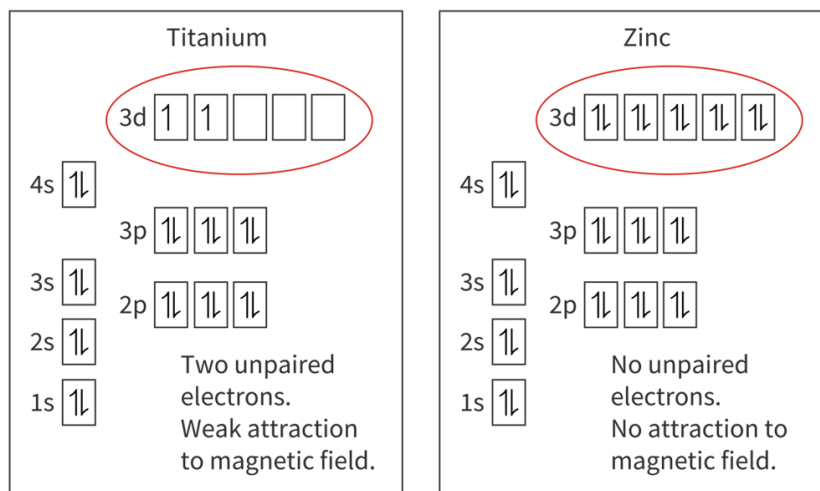


Figure 3. Unpaired d-electrons responsible for the magnetic behaviour of transition metals.

 More information for figure 3

The image shows electron configuration diagrams for two transition metals, titanium and zinc, illustrating their magnetic behavior. Both diagrams include the distribution of electrons in various shells and subshells (1s, 2s, 2p, 3s, 3p, 3d, 4s).

On the left side, the diagram for titanium displays the 3d subshell highlighting the presence of two unpaired electrons. Text under the diagram notes, "Two unpaired electrons. Weak attraction to magnetic field," indicating that these

unpaired electrons contribute to titanium's magnetic properties.

On the right side, the zinc diagram shows a fully filled 3d subshell with no unpaired electrons. The accompanying text states, "No unpaired electrons. No attraction to magnetic field," explaining why zinc does not exhibit magnetic properties.

Overall, the image explains how the presence or absence of unpaired d-electrons in these transition metals affects their magnetic characteristics.

[Generated by AI]

Some of the transition metals form a metallic lattice structure where there is long range alignment of these unpaired electrons. This causes a strong interaction with a magnetic field (called ferromagnetism). The three elements that are ferromagnetic are iron, nickel and cobalt. **Figure 4** shows iron nails and the long-range alignment of unpaired electrons that leads to the strong magnetic behaviour of elemental iron.

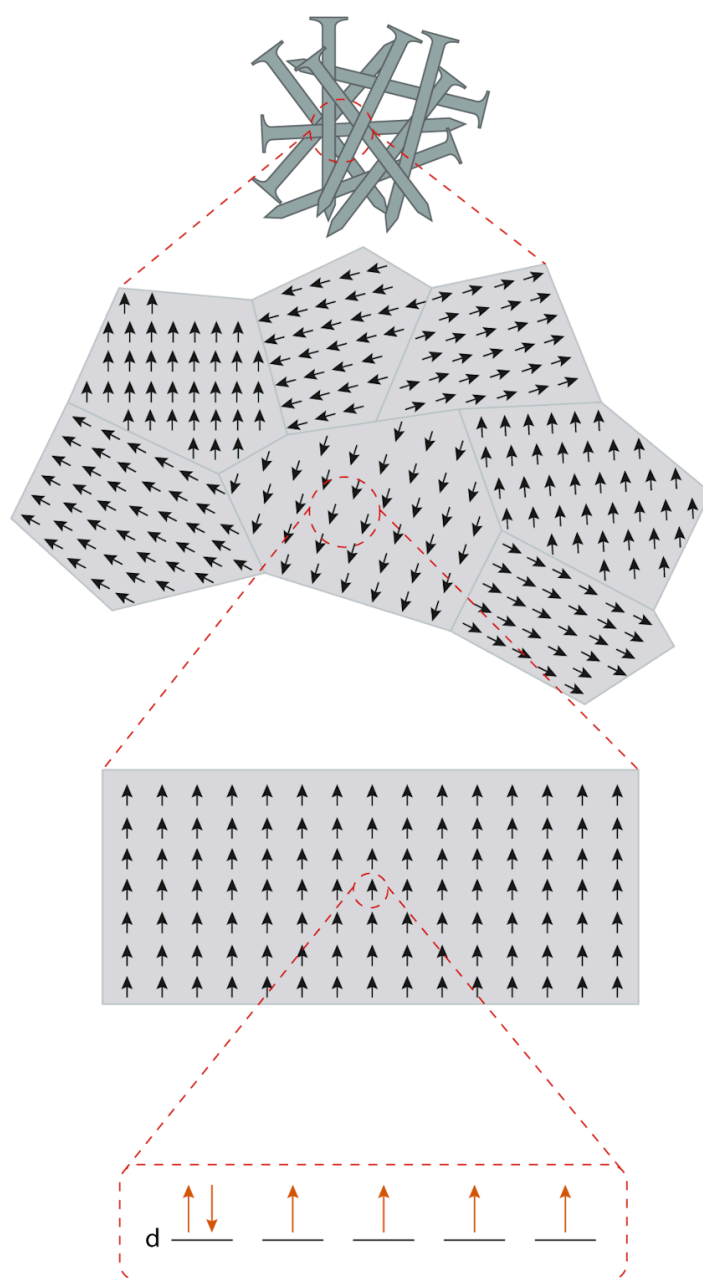


Figure 4. Strong magnetic behaviour of iron.

 More information for figure 4

The image is a diagram illustrating the magnetic alignment in iron. At the top, it shows a collection of gray iron nails grouped together. Below the nails, the diagram zooms in to reveal the atomic structure, represented as a series of pentagonal and hexagonal patterns filled with arrow symbols. These arrows indicate the direction of alignment of the unpaired electrons, demonstrating regions of magnetic domains. A closer view at the bottom part of the diagram focuses on a rectangular section of aligned arrows, indicating ferromagnetic alignment. The detailed inset at the bottom magnifies this alignment, showing arrows pointing mostly upwards, signifying parallel alignment contributing to magnetism. Red dashed lines are used to indicate transitions and focuses on different levels of the illustration.

[Generated by AI]

Study skills

You do need to be familiar with the concept that transition elements have magnetic properties, but the different types of magnetism (paramagnetic or ferromagnetic) will not be assessed.

4. Catalytic properties

Have you heard the term catalyst before? If so, where have you heard it used? You may be more familiar with enzymes, biological catalysts. Do you know what enzymes do? Catalysts are substances that speed up chemical reactions while remaining unchanged themselves. You will learn about these in more detail in section R2.2.2-3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>).

Transition metals and transition metal ions are often used as catalysts in chemical reactions. Transition metals can provide a surface for the reaction to take place and ions lose or gain electrons to form variable oxidation states. Ions of transition elements can react with one reactant (reactant A) to increase its oxidation state while, afterwards reacting with the other reactant (reactant B) returning to its original oxidation state. The two reactants have either gained or lost electrons allowing them to easily react with one another, forming a product (product C).

Figure 5 shows this in a simplified version of a catalytic cycle diagram for transition metals.

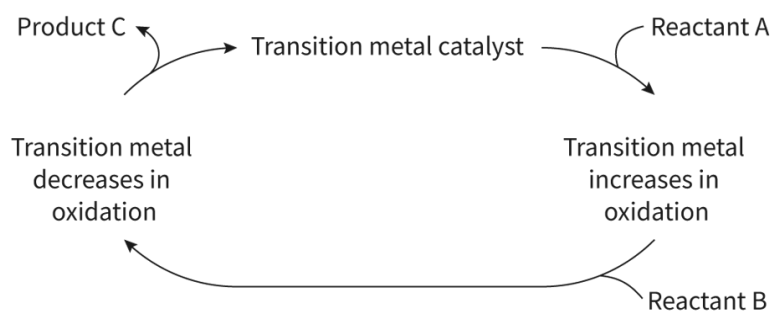


Figure 5. General catalytic cycle diagram for transition metals.

 More information for figure 5

The diagram illustrates a general catalytic cycle for transition metals used in chemical reactions. The cycle begins with Reactant A, which interacts with a transition metal catalyst, increasing the oxidation state of the metal. The transition metal then interacts with Reactant B, allowing the metal to return to its original oxidation state. Product C is formed as a result of this cycle.

Key elements of the diagram include: - "Reactant A" is positioned at the top right and points to the "Transition metal catalyst." - The "Transition metal increases in oxidation" label is on the right side, pointing downward to "Reactant B." - "Reactant B" is at the bottom of the cycle, pointing to "Transition metal decreases in oxidation" on the left. - "Product C" is at the top left, completing the cycle back to the "Transition metal catalyst."

[Generated by AI]

Since the transition metal is in its original form with its original oxidation state at the end of the reaction, it is not considered a reactant, but rather a catalyst. It simply facilitated the reaction without experiencing a net change.

Haemoglobin is a protein responsible for delivering oxygen in the bloodstream. The haem group within haemoglobin contains the Fe^{2+} ion at its centre. Fe^{2+} can bind to oxygen undergoing a change in oxidation state to Fe^{3+} . When it releases oxygen, it returns to Fe^{2+} . This is illustrated in **Figure 6**.

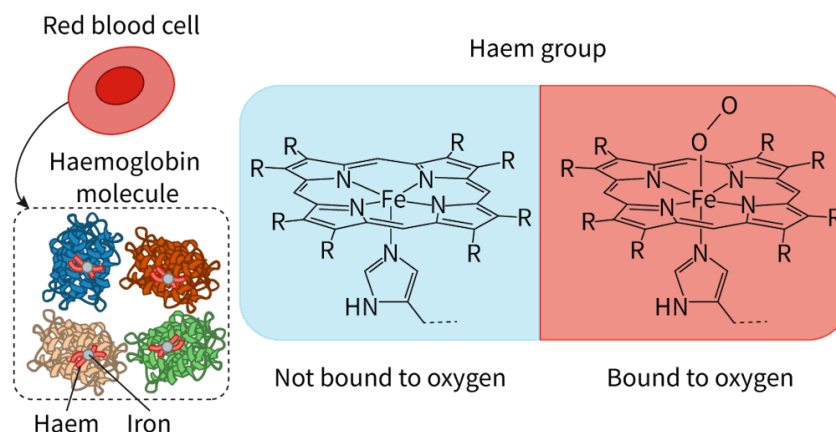


Figure 6. Iron in haem changing oxidation state to bind to iron.

[More information for figure 6](#)

The diagram illustrates the structure of the haem group within haemoglobin, highlighting changes in the iron (Fe) oxidation state. On the left side, the structure is shown without oxygen, where iron is in the Fe^{2+} state. It consists of a porphyrin ring structure bonded with iron at the center, depicted in a blue background. On the right side, the structure is shown bonded with an oxygen molecule, indicating the oxidized Fe^{3+} state, depicted in a red background. The transition between these states is crucial for oxygen binding and release in the bloodstream.

[Generated by AI]

5. Formation of coloured compounds

Have you encountered any vibrant coloured solutions in chemistry or science so far? Do you remember what solute made up the solution? Chances are you have encountered a coloured solution like the vibrant blue copper(II) sulfate or purple potassium permanganate. These are both shown in **Figure 7**.



Credit: GIPhotoStock, Getty Images

(<https://www.gettyimages.com/detail/photo/pipetting-sodium-carbonate-to-copper-sulfate-both-royalty-free-image/565785491>)



Credit: GIPhotoStock, Getty Images

(<https://www.gettyimages.com/detail/photo/erlenmeyer-flask-containing-a-solution-of-potassium-royalty-free-image/702545749>)

Figure 7. Solutions of copper(II) sulfate (left) and potassium permanganate (right).

Transition elements form coloured compounds. The colours can vary depending on the transition element, the oxidation state, the other species making up the compound, etc. **Figure 8** shows some common transition metal ions and the colour that they typically appear. Notice how Sc^{3+} and Zn^{2+} are colourless. We will look more closely at the theory behind why solutions with transition elements are coloured in the next [section S3.1.10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coloured-complexes-of-transition-element-hl-id-44716/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coloured-complexes-of-transition-element-hl-id-44716/>).

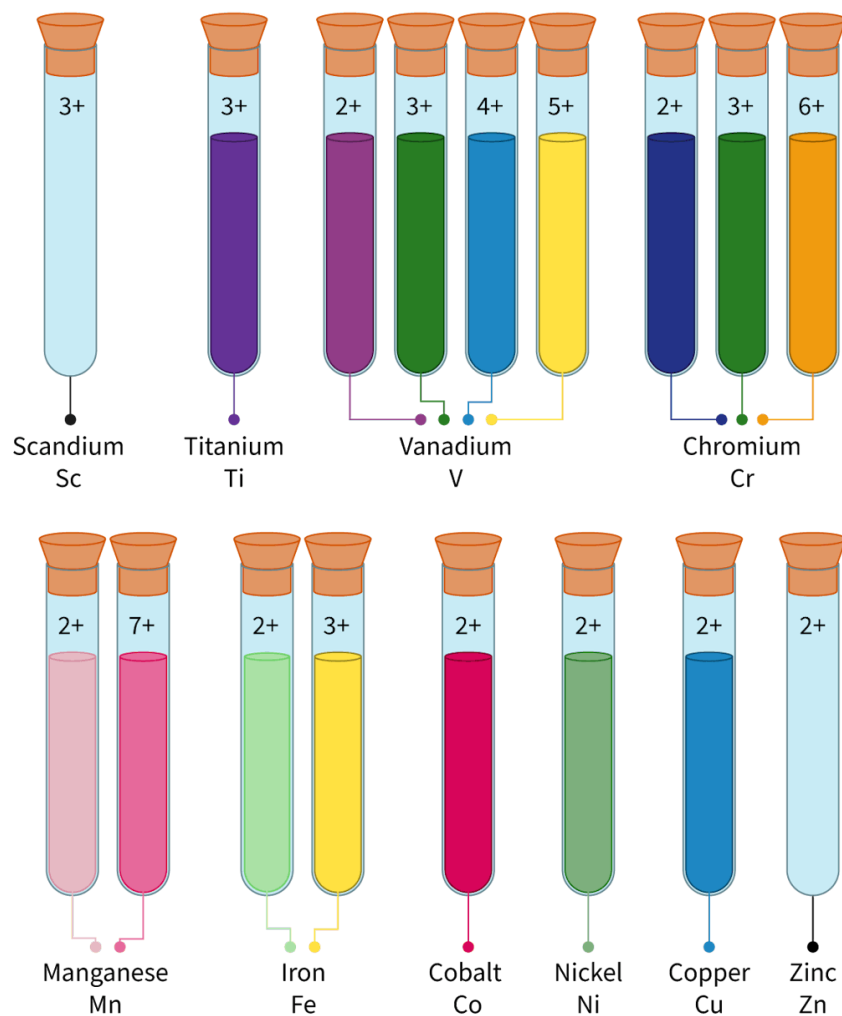


Figure 8. Coloured solutions of some common ions of transition elements.

 More information for figure 8

The image depicts an array of test tubes filled with solutions of different colors, each labeled with a charge representing various transition metal ions. The tubes are organized in two rows.

In the first row: 1. A colorless solution with a "3+" label. 2. A purple solution with a "3+" label. 3. A purple solution with a "2+" label. 4. A green solution with a "3+" label. 5. A blue solution with a "4+" label. 6. A yellow solution with a "5+" label. 7. One more purple solution with a "2+" label. 8. A green solution with a "3+" label. 9. A blue solution with a "6+" label.

In the second row: 1. A pink solution with a "2+" label. 2. A dark pink solution labeled "7+". 3. A light green solution with a "2+" label. 4. A yellow solution labeled "3+". 5. A reddish-pink solution with a "2+" label. 6. A green solution labeled "2+". 7. A blue colored solution with a "2+" label. 8. A colorless solution with a "2+" label.

The labels denote different oxidation states of the ions, and their colors indicate the typical appearance in solution as referenced in discussions about transition metal elements and their compounds.

[Generated by AI]

6. Formation of complex ions

We mentioned earlier that the name 'transition' comes from the fact that these elements show transitional behaviour, somewhere between elements in the s-block and p-block. The bonding of transition metal ions to other substances is one important reason why these elements show transitional behaviour. **Figure 9** shows what structure is formed when a Cu^{2+} ion forms a complex with water. Take a look at the structure drawn. What do you notice? Did you expect the structure to look like this? What type of bonds do the lines drawn indicate? Would you expect that type of bond for a transition element or its ion?

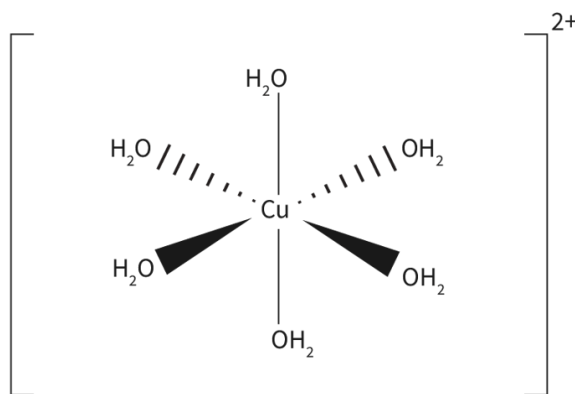


Figure 9. Complex ion of Cu^{2+} and water.

 More information for figure 9

The image shows a diagram of a complex ion where a central copper ion (Cu^{2+}) is surrounded by water molecules. The copper ion is at the center and is connected to six water molecules (H_2O) arranged in an octahedral configuration. The diagram uses solid lines and wedge shapes to indicate the spatial arrangement of the molecules. Each water molecule donates a lone pair of electrons to the copper ion, forming coordination bonds. The copper ion is depicted in the center with a positive charge indicated by the "2+" sign on the right side. The water molecules are labeled with "H₂O" and are drawn with different orientations to illustrate their spatial arrangement around the copper ion.

[Generated by AI]

To form this complex, the lone pairs on the water molecules attract to the positively charged ion and donate a lone pair of electrons to Cu^{2+} forming a coordination bond. You learned about coordination bonds in section S2.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>). Water does this multiple times with Cu^{2+} forming a complex ion. This is illustrated in **Figure 10**.

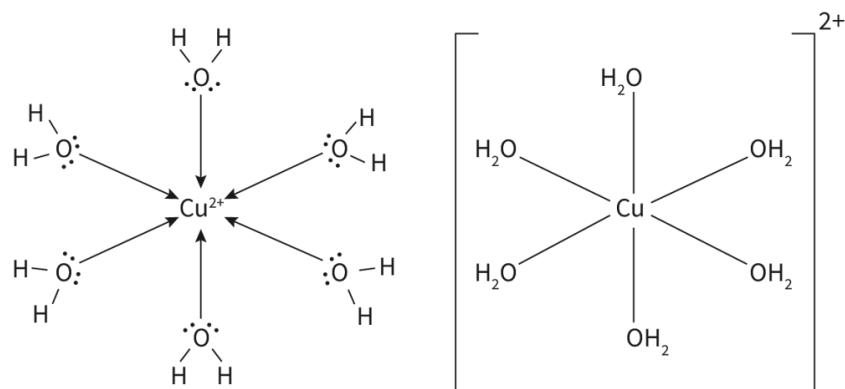


Figure 10. The formation of coordination bonds in the complex ion of Cu^{2+} and water.

 More information for figure 10

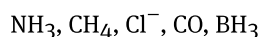
The image shows a diagram illustrating the formation of coordination bonds between a copper ion (Cu^{2+}) and water molecules. On the left side, there is a central Cu^{2+} ion surrounded by six water (H_2O) molecules. Each water molecule is represented with an oxygen atom, depicted with two lone pairs of electrons, bonding with the copper ion. The oxygen atoms point towards the Cu^{2+} ion with arrows indicating the direction of electron donation. The image displays a geometric structure where the water molecules are positioned symmetrically around the Cu^{2+} ion, suggesting an octahedral geometry typical of such complexes. On the right side, the diagram simplifies the complex ion structure, showing the copper ion at the center, surrounded symmetrically by water molecules labeled as H_2O , with lines connecting them to the copper, all enclosed within brackets denoting the complex ion state and the $2+$ charge indication on the top right corner.

[Generated by AI]

The surrounding species that donates a lone pair of electrons to form a coordinate bond with a transition metal ion is known as a ligand. In the previous example, water acted as a ligand by donating one of its lone pairs of electrons to the Cu^{2+} ion. A species can only act as a ligand if it has a lone pair of electrons on at least one atom in a molecule.

Worked example 1

Identify which of the following species can act as a ligand.



A species can only act as a ligand if it has a lone pair of electrons on at least one atom.

If you are not able to identify which substances have lone pairs or not, you will need to draw a Lewis formula.

NH_3 , Cl^- and CO all have at least one lone pair and therefore can act as a ligand.

CH_4 and BH_3 do not have any lone pairs and therefore cannot act as a ligand.

Making connections

The formation of complexes involves a ligand donating an electron pair and a transition metal ion accepting an electron pair in order to form a coordinate bond. In [section R3.4.6](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lewis-acids-and-bases-id-44715/>), you will learn how species that are electron-pair donors act as a specific type of base known as a Lewis base and how species that are electron-pair acceptors act as a specific type of acid known as a Lewis acid. This means that ligands act as a Lewis base and transition metal ions as a Lewis acid during the formation of a complex.

We can write out the chemical formula of complex ions by using square brackets around the entire complex ion and round brackets around the ligand. For example, in the complex ion we have looked at here, we could write the chemical formula as $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$. This shows the overall charge of the ion, the six bonded water ligands and the transition metal.

Worked example 2

Deduce the oxidation state of the transition metal ion and the charge on the complex ion in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$.

In a complex ion, the entire complex is always enclosed inside of the square brackets. Since there is another element written after the square brackets, this would indicate this is an ionic compound. The positive ion would be the complex, with the contents outside the brackets representing the negative ion. We know that bromine (Br) is in group 17 and likely to form the Br^- ion. This means that the positive ion must have the 1+ charge in order for the ionic compound to be neutral overall.

This means the complex ion is $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ with a charge of 1+.

To deduce the oxidation state on chromium we now need to look at the ligands and their charge. There are four NH_3 ligands and two Cl^- ligands. We can create an algebraic equation using x to represent the oxidation state of Cr and summing up the charges of each ligand to equal 1+, the charge of the complex ion.

$$\begin{aligned}x + 4(0) + 2(-1) &= +1 \\x &= +3\end{aligned}$$

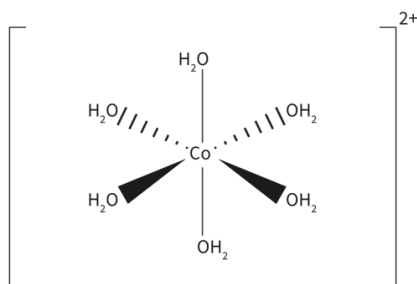
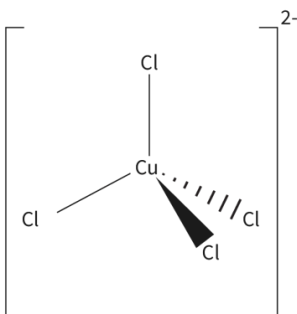
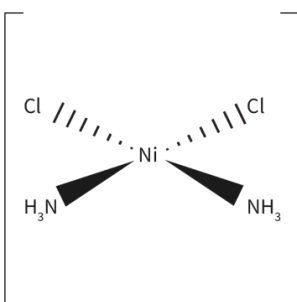
Therefore, the oxidation state on Cr is +3.

In this example we have looked at six bonded ligands, each with one coordinate bond, giving a coordination number of 6. This is a common arrangement of ligands around a transition metal ion, resulting in an octahedral geometry. Other coordination numbers and geometries also exist. **Table 3** shows common coordination numbers and shapes of transition metal complexes.

Table 3. Common shapes of transition metal co

C
o
m
p
l
e
x

Structure

[C
o
(
H₂
O
)
₆]
2+
[More information](#)
[C
u
Cl
₄]
2-
[More information](#)
[N
iCl
₂(
N
H₃
)₂]
[More information](#)

Complex	Structure
$[\text{Ag}(\text{NH}_3)_2]^+$	$\left[\text{H}_3\text{N} - \text{Ag} - \text{NH}_3 \right]$ 👁 More information

It is also possible for one ligand to form multiple coordinate bonds with the transition metal ion. This requires the ligand to have more than one lone pair available for coordinate bonding. Two common ligands that form multiple coordinate bonds with transition metal ions are ethylenediamine and ethylenediaminetetraacetic acid (EDTA). **Figure 11** shows two octahedral complexes formed from formation of coordinate bonds with these two ligands. Notice that one ethylenediamine ligand forms two coordinate bonds, whereas EDTA forms six. Three ethylenediamine ligands, each with two coordinate bonds, forms a complex with a coordination number of 6. Only one EDTA ligand bonded will have a coordination number of 6.

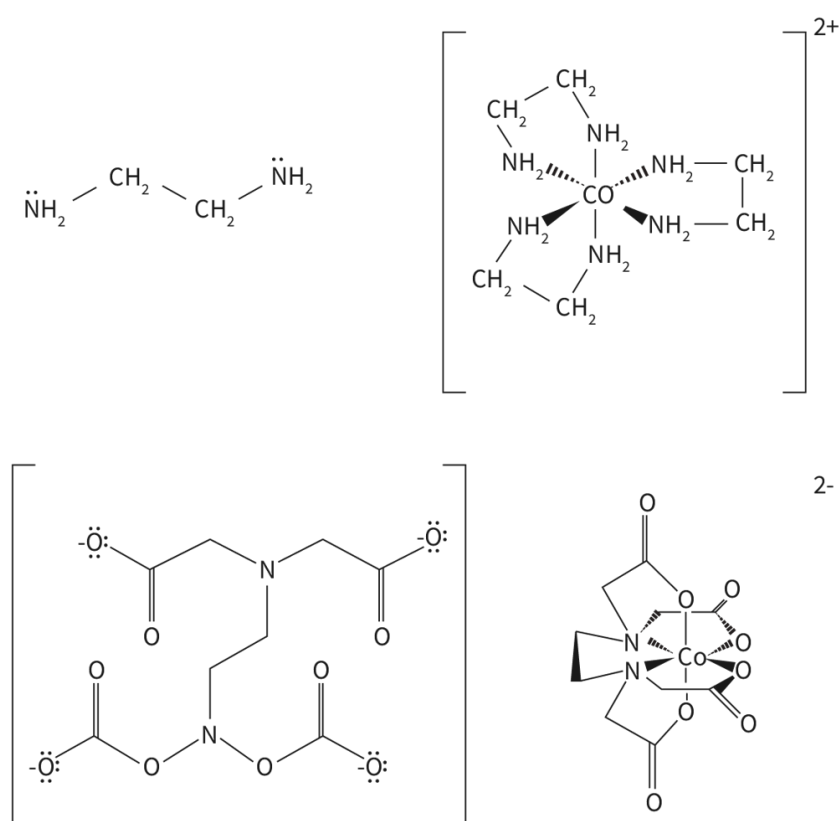


Figure 11. Cobalt complexes with ligands that can form more than one coordinate bond.

 More information for figure 11

The image presents two cobalt complexes with different ligands showing multiple coordinate bonds. The top part shows the structure of ethylenediamine as a ligand, labeled with nitrogen atoms (NH₂) linked by carbon atoms (CH₂). Beside it, a cobalt complex is illustrated, where three ethylenediamine ligands coordinate with a central cobalt ion, resulting in a coordination number of 6 with the notation '2+'.

In the bottom part, the structure of the EDTA ligand is shown with its extended molecular configuration, featuring multiple nitrogen and oxygen atoms set to form coordinate bonds. Adjacent to it, a cobalt complex illustrates how one EDTA ligand bonds with a central cobalt ion, also resulting in a coordination number of 6, indicated by the notation '2-'.

The image illustrates the difference in binding when using ethylenediamine versus EDTA, highlighting how many coordinate bonds each ligand can form with the cobalt ion.

[Generated by AI]

International Mindedness

EDTA and its ability to form six coordinate bonds to metal ions has extremely useful applications, particularly in medicine. EDTA is used around the world to treat heavy metal poisoning. Ingestion of food or water that has been contaminated with toxic heavy metal ions such as lead can be lethal. Physicians commonly treat patients who have high levels of heavy metal ions in their bloodstream with EDTA. The EDTA binds with heavy metal ions it encounters, preventing the ion from reacting with other substances in the body. After a few hours, the EDTA — metal complex is excreted in urine and is removed from the body. The process of using EDTA medicinally is often referred to as 'chelation therapy'. The word chelate is derived from the Greek word for claw and refers to any ligand capable of forming more than one coordinate bond with a metal ion, encasing the ion like a claw.

While the formation of complex ions is not exclusive to, it is very common in transition elements and they can form a wide variety of complexes. This is why the formation of complexes is considered an important characteristic of transition elements.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Being curious about the natural world
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Individual activity

Part 1: Transition metal crossword

Individually, complete the following transition metal crossword. While you should try to avoid it, you may need to research some specific information about a few of the metals.

**Across**

3 Disputes between scientists about whether this element should be a transition metal or not (8)

6 Common metal catalyst used in industry, used for the hydrogenation of oils to make spreadable fats such as margarine (6)

7 Primary metal in steel and ferromagnetic (4)

8 Many uses in the medical field for implants for joint

Check

Show solution

Retry

Interactive 1. Transition Metal Crossword.

 More information for interactive 1

The interactive contains a crossword puzzle with Across and Down clues related to transition metals. The crossword puzzle shuffles on refreshing, and the arrangement of the puzzle and clues varies but the clue and answer remain the same. The grid is arranged with 10 clues spilt into Across and Down. The number of Across and Down clues varies on shuffling. The clues given on the right side are:

Ferromagnetic and commonly forms orange-red complexes with 6 letters,

Many uses in the medical field for implants for joint and bone replacement with 8 letters,

Not technically a transition metal with 4 letters,

Has a melting point just slightly lower than vanadium with 8 letters,

Forms a blue complex in water with 6 letters,

Disputes between scientists about whether this element should be a transition metal or not with 8 letters,

First five successive ionization energies are relatively close together with 8 letters,

Five unpaired electrons with 9 letters,

Common metal catalyst used in industry, used for the hydrogenation of oils to make spreadable fats such as margarine with 6 letters,

Primary metal in steel and ferromagnetic with 4 letters.

The crossword grid has white squares for input and dark blue background squares separating each word, with numbered boxes indicating clue positions. The interface includes buttons labelled Check, Show solution, and Retry. The Check button displays the data as follows.

Read below for solution:

Ferromagnetic and commonly forms orange-red complexes with 6 letters: Cobalt

Many uses in the medical field for implants for joint and bone replacement with 8 letters: Titanium

Not technically a transition metal with 4 letters: Zinc

Has a melting point just slightly lower than vanadium with 8 letters: Chromium

Forms a blue complex in water with 6 letters: Copper

Disputes between scientists about whether this element should be a transition metal or not with 8 letters: Scandium

First five successive ionization energies are relatively close together with 8 letters. Vanadium

Five unpaired electrons with 9 letters: Manganese

Common metal catalyst used in industry, used for the hydrogenation of oils to make spreadable fats such as margarine with 6 letters: Nickel

Primary metal in steel and ferromagnetic with 4 letters: Iron

Part 2: Transition elements used in industry

Pick one transition metal from the crossword and research an industrial process that the transition element is used for. Research the following:

- The catalyst (pure metal or compound)
- The industrial process that uses the catalyst
- Why is the industrial process important for society?

Share your research with a peer.

Formation of variable oxidation states in transition elements [HL]

Higher level (HL)

Learning outcomes

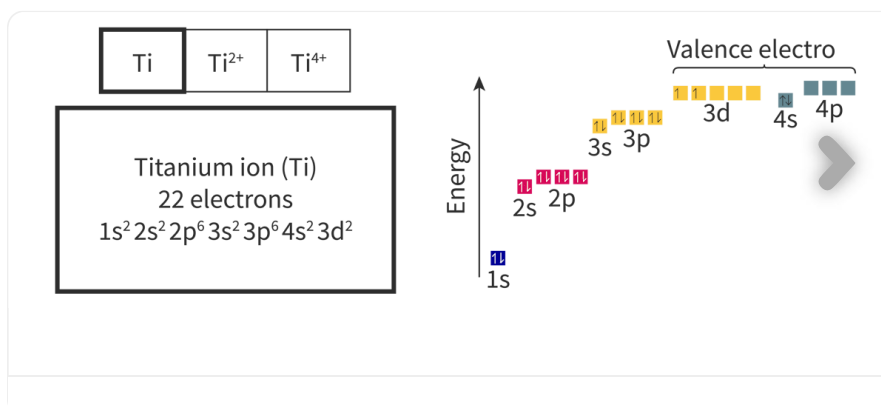
By the end of this section you should be able to:

- Explain why transition elements have variable oxidation states.
- Deduce the electron configurations of ions of the first-row transition elements.

Recall from section S2.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) that when we looked at the cations formed from transition elements, that a variety of different positive charges (oxidation states) were possible. This is why when we named ionic compounds involving transition elements in section S2.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) roman numerals were used to denote which oxidation state the metal was in (i.e. copper(II) sulfate to denote the +2 oxidation state of the copper ion). But why do transition elements form a variety of oxidation states? What is unique about the arrangement of electrons in transition elements that allow the loss of a variable number of electrons?

Electron configuration of ions of transition elements

Transition elements have valence electrons occupying both an s and d-sublevel. Recall from section S2.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) that when these elements lose electrons, they first lose valence electrons from the outer s-sublevel, before continuing to lose electrons from the outer d-sublevel. **Interactive 1** allows you to revisit the electron configurations and sublevels that electrons are lost from for Ti, Ti^{2+} and Ti^{4+} that you studied in section S2.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>).



Interactive 1. Electron Configurations and Orbital Diagrams for Ti, Ti^{2+} and Ti^{4+} .

 More information for interactive 1

An interactive slideshow with three slides displays the electronic configurations and orbital diagrams for Ti, Ti^{2+} , and Ti^{4+} .

Slide 1:

Titanium (Ti) has 22 electrons, with an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$. The energy levels increase in the following order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, and 4p. In the orbital diagram, the s orbital has a single box, the p orbital has three boxes, and the d orbital has five boxes. Electrons are represented by arrows inside these boxes. Each box corresponds to a specific orbital in a subshell. The valence orbitals include 3d, 4s, and 4p, and changes in electron occupancy illustrate how Ti loses electrons to form its common ions.

In Ti, 1s, 2s, 3s, and 4s orbitals have two electrons in opposite spins, 2p and 3p orbitals have six electrons, with each orbital containing a maximum of two electrons with opposite spins, and 3d orbital has two electrons in the first two boxes. 4p orbital is empty.

Slide 2:

Titanium (Ti^{2+}) has 20 electrons, with an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$. Two electrons are removed from the 4s orbital of the neutral titanium atom.

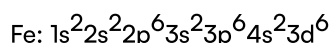
Slide 3:

Titanium (Ti^{4+}) has 18 electrons, with an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^6$. Two electrons are first removed from the 4s orbital, and the next two electrons are removed from the 3d orbital, leaving a stable noble gas configuration similar to argon (Ar). This slideshow helps users understand the electronic configurations of titanium ions, including the orbitals from which electrons are lost.

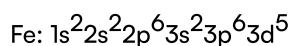
Worked example 1

Deduce the electron configuration for Fe^{3+} .

When writing the electron configuration for ions, it is helpful to start by writing the electron configuration for the neutral element first.



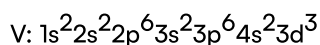
When iron loses 3 electrons it will first lose 2 electrons from the 4s-sublevel and then the third electron from the 3d-sublevel. This would make its electron configuration:



Worked example 2

Deduce the oxidation state of a vanadium ion with the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$.

A good first step would be to write the electron configuration of a neutral vanadium atom.



Comparing this to the electron configuration in the question we can see the ion has lost two 4s electrons and two 3d electrons for a total of four electrons lost.

Therefore, the oxidation state of the vanadium ion would be +4.

(Note that 4+ would be the ionic charge. Remember that oxidation states and ionic charges have a slightly different notation).

We can look at the successive ionisation energies of transition elements to help explain why they are able to form variable oxidation states. Recall from [section S1.3.7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/successive-ionisation-energies-hl-id-44045/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/successive-ionisation-energies-hl-id-44045/>) that successive ionisation refers to the process of removing successive electrons from an atom or positive ion. **Figure 1** shows the successive ionisation energies for removal of the first five electrons in calcium and vanadium. What do you notice about the energies required to remove successive electrons in vanadium in comparison to calcium?

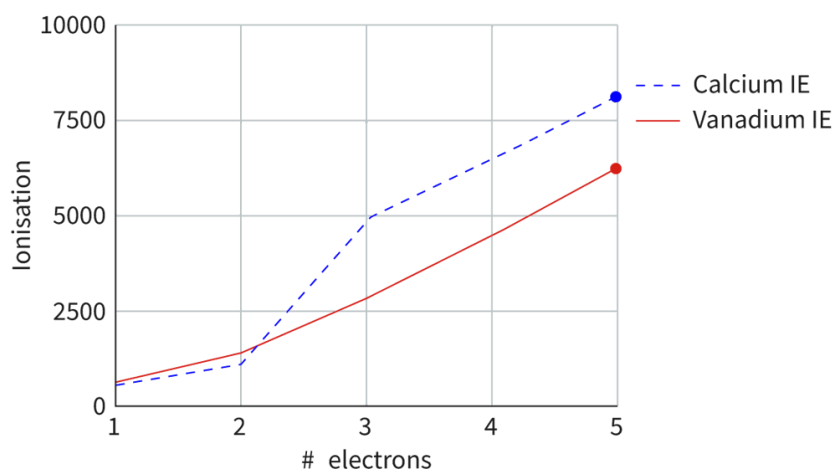


Figure 1. Successive ionisation energies for the first five electrons in calcium and vanadium.

 More information for figure 1

The graph displays the successive ionisation energies for the first five electrons of calcium and vanadium. The X-axis represents the number of electrons, ranging from 1 to 5. The Y-axis represents the ionisation energy in arbitrary units, ranging from 0 to 10000.

There are two lines on the graph: 1. A dashed blue line for calcium, showing an increase in ionisation energy, with a marked higher rise after the second electron. 2. A solid red line for vanadium, displaying a more gradual increase in ionisation energies throughout the removal of the five electrons.

Notably, vanadium's ionisation energy increases more gradually than calcium's, which shows a sharp increase after the second electron. This data visually supports the concept of variable oxidation states in vanadium.

[Generated by AI]

Notice that in vanadium, the ionisation energies required to remove the first five electrons increase at a lower rate when compared to calcium. The successive ionisation energies for calcium show a large increase after the removal of the outer valence shell electrons. Comparing the electron configurations of the two elements in **Figure 2**, why do you think this is? Technically, both elements have only two electrons in the $n = 4$ energy level, so why would vanadium be capable of easily losing more than two?

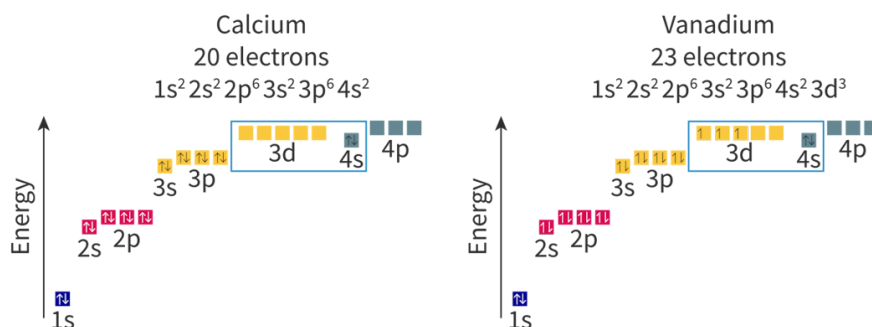


Figure 2. Electron configurations and orbital diagrams for calcium and vanadium.

 More information for figure 2

The image depicts two electron configuration diagrams, one for calcium and the other for vanadium, both showing energy levels and orbital occupancy.

On the left is calcium, with the configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. The diagram illustrates each orbital with arrows indicating the electrons' spin. The energy levels are arranged vertically, with arrows pointing upwards to show increasing energy. Orbitals are depicted as boxes where paired arrows indicate paired electrons. Notably, the 3d orbitals are empty in calcium.

On the right is vanadium, showing the configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$. Here, similar vertical structure is presented with visible electrons in the 3d orbitals depicted with arrows. This highlights the additional electrons in 3d compared to calcium, explaining vanadium's variable oxidation states and easier ionization from higher energy levels.

The overall layout emphasizes the energy differences and orbital filling in these two elements, relevant to their chemical properties.

[Generated by AI]

Since the 4s and 3d-sublevels are close in energy, electrons from both of these sublevels are able to act as valence electrons. Electrons are lost first from the 4s-sublevel; explaining why many transition metal ions form the +2 oxidation state. However, electrons can also easily be lost from the 3d-sublevel without applying much additional energy for removal. This is why transition elements can easily form

ions with variable oxidation states as there are a number of d electrons that can be lost in addition to s electrons. **Figure 3** shows the common oxidation states of the metals from groups 3–12 in period 4.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
								+1	
	+2	+2	+2	+2	+2	+2	+2	+2	+2
+3	+3	+3	+3	+3	+3	+3			
	+4	+4		+4					
		+5							
			+6	+6					
				+7					

Figure 3. Common oxidation states for period 4 metals in groups 3–12.

 More information for figure 3

The image is a table displaying common oxidation states for period 4 metals in groups 3 through 12. The horizontal axis lists the metals: Sc (Scandium), Ti (Titanium), V (Vanadium), Cr (Chromium), Mn (Manganese), Fe (Iron), Co (Cobalt), Ni (Nickel), Cu (Copper), and Zn (Zinc). The vertical axis represents the oxidation states: +1, +2, +3, +4, +5, +6, and +7.

- Sc shows the oxidation state of +3.
- Ti shows +2, +3, and +4.
- V shows +2, +3, +4, and +5.
- Cr shows +2, +3, +4, +5, and +6.
- Mn shows +2, +3, +4, +6, and +7.
- Fe shows +2 and +3.
- Co shows +2 and +3.
- Ni shows +2.
- Cu shows +1 and +2.
- Zn shows +2.

The table illustrates that, except for Zn, most metals can have multiple oxidation states, reflecting the variable oxidation capabilities of transition metals, primarily due to 3d sublevel electron interactions.

[Generated by AI]

Looking at **Figure 3** note that zinc only shows the +2 oxidation state. This is another reason to support the exclusion of zinc from transition elements. You may also notice that scandium is listed as having the +3 oxidation state. There are active disputes in the scientific community about whether scandium should be considered a transition element. Some argue it shouldn't be included because only the +3 oxidation state is common. Others argue it should be included because it is possible to form the +2 oxidation state, it is just uncommon to encounter.

Aspect: Science as a shared endeavour

Disputes in science are not uncommon. Can you think of ways that disputes between scientists help the production of knowledge in chemistry? Can you think of ways that disputes hinder the production of knowledge in chemistry? If an element does not fit neatly into a category that we, as humans, have created, should we exclude it or consider it an exception?

Figure 4 shows all the successive ionisation energies for vanadium, labelling the sublevels where electrons were lost from. Notice that the 4s and 3d electrons have successive ionisation energies close together, suggesting that they all behave as valence electrons. This also acts as further evidence that the 4s and 3d-sublevels are close in energy.

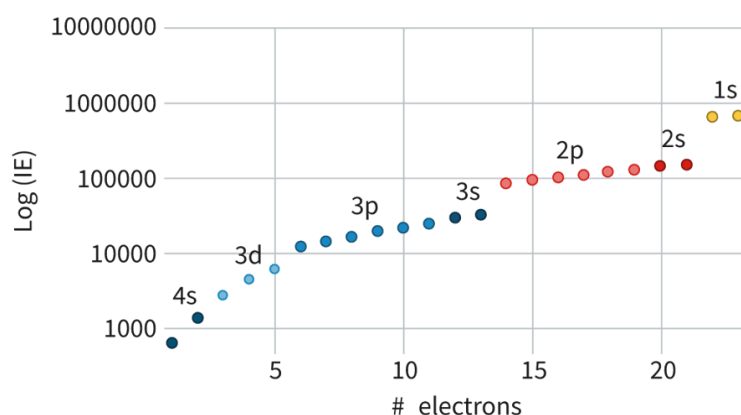


Figure 4. Successive ionisation energies for vanadium.

 More information for figure 4

The image is a graph illustrating the successive ionisation energies for vanadium. The X-axis represents the number of electrons, labeled from 0 to 25. The Y-axis is the logarithm of the ionisation energy (Log (IE)) and spans from 1,000 to 10,000,000 in logarithmic scale. Data points on the graph represent different sublevels from which electrons are removed.

The sublevels are labeled as 4s, 3d, 3p, 3s, 2p, 2s, and 1s, with each sublevel's electrons identified by colored dots (blue, red, yellow). The graph shows that the 4s and 3d sublevels are close in energy as their ionisation energies are grouped closely together, supporting the idea that they behave similarly as valence electrons. The ionisation energy increases steadily with electrons removed, and a distinct rise in ionisation energy can be noted as transitions to 3p and 3s occur, with further sharp increases at the 2p, 2s, and finally the 1s sublevels, indicating a higher energy required to remove core electrons.

[Generated by AI]

Knowledge of the oxidation number for transition elements is very important since they can lead to very different properties. Titanium(IV) oxide, for example, is utilised as an active ingredient in sunscreen because it reflects a high percentage of UV radiation, whereas titanium(II) and titanium(III) oxides do not. **Table 1** shows a comparison of these titanium oxides. Note that titanium does not have a +5 oxidation state as this would require the removal of an electron from the 3p sublevel.

Table 1. Titanium oxides with variable oxidation states and their properties.

Substance name	Chemical formula	Oxidation number of titanium	Properties
Titanium(II) oxide	TiO	+2	Brown solid. Melts at approximately 175
Titanium(III) oxide	Ti ₂ O ₃	+3	Black, semiconducting solid. Melts at approximately 2100 °C
Titanium(IV) oxide	TiO ₂	+4	White solid. Melts at approximately 180 Reflects UV radiation and is therefore used in sunscreen.

Activity

- **IB learner profile attribute:** Risk-taker
- **Approaches to learning:** Social skills — Appreciating the diverse talents and needs of others
- **Time required to complete activity:** 20 minutes
- **Activity type:** Group activity

Successive ionisation energy paper telephone

In groups of 5. Each person needs a stack with 5 paper squares. No talking with one another until all rounds have been completed. Each round will involve you writing/sketching on the top square and then passing the entire stack clockwise to the next person. When you receive a stack of squares, look only at the top square, move it to the bottom and complete the next round on the new square on top.

- Round 1 — select a transition element from period 4. Write it on the top sheet and pass it clockwise.
- Round 2 — Read the top square, move it to the bottom. Write the electron configuration of the neutral element on the top square.
- Round 3 — Read the top square, move it to the bottom. Sketch the plot of successive ionisation energies for this element based on its

electron configuration. Sketch on the top sheet and pass clockwise when finished.

- Round 4 — Look at the sketch on the top square, move to the bottom. Write down the electron configuration for the element having a +3 oxidation state. Pass clockwise when finished.
- Round 5 — Read the top square, move it to the bottom. Write the name of the element. Pass clockwise when finished.

You should now have your original stack. Look at the element written. Is it the same one you started with? One by one go through your entire stack with your group. If the elements at the beginning and end do not match as a group discuss where the misalignment occurred.

Coloured complexes of transition element (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Apply a colour wheel to deduce the wavelengths and frequencies of light absorbed and/or observed.
- Explain why transition metal complexes appear coloured.

Similar to coloured glass like you saw in [section S3.1.8](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>), many gemstones appear coloured due to the presence of

transition metal ions in the crystal structure. Transition metal ions interact with visible light in a unique manner causing the appearance of a specific colour for different gemstones. Gemstones that are more vivid in colour have higher value. This means the presence of transition metal impurities increase the value of the stone.

Interactive 1 shows some examples of gemstones, the chemical makeup of the gemstone and the transition metal ion responsible for its colour.

Interactive 1. Gemstones, Their Chemical Makeup and the Transition Metal Ions Responsible for Their Colour.

 More information for interactive 1

An interactive slideshow with seven slides presents photographs of gemstones, along with their chemical formulas and descriptions explaining the transition metal ions responsible for their colors.

Slide 1: Peridot

A cushion-shaped green gemstone.

Chemical formula: Mg_2SiO_4

Description: Color from Fe^{2+} ions replacing some magnesium ions in the crystal structure.

Slide 2: Garnet

A round-shaped red gemstone.

Chemical formula: $\text{Mg}_3\text{Al}_2(\text{SiO}_4)_3$

Description: Color from Fe^{2+} ions replacing some magnesium ions in the crystal structure.

Slide 3: Tourmaline

A rectangular pink gemstone.

Chemical formula: $\text{Na}_3\text{Li}_3\text{Al}_6(\text{BO}_3)_3(\text{SiO}_3)_6\text{F}_4$

Description: Color from manganese ions replacing some lithium and aluminum ions in the crystal structure.

Slide 4: Aquamarine

Trillion-shaped blue gemstones.

Chemical formula: $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$

Description: Color from Fe^{2+} and Fe^{3+} ions replacing some aluminum ions in the crystal.

Slide 5: Emerald

A deep, vivid green rectangular gemstone with step facets.

Chemical formula: $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$

Description: Color from manganese ions replacing some aluminum ions in the crystal structure.

Slide 6: Ruby

A pinkish-red, oval-shaped gemstone.

Chemical formula: Al_2O_3

Description: Color from chromium ions replacing some aluminum ions in the crystal structure.

Slide 7: Sapphire

A deep blue, round-shaped gemstone.

Chemical formula: Al_2O_3

Description: Color from titanium and iron ions replacing some aluminum ions in the crystal structure.

This interactivity helps users to understand the chemical formulas of common gemstones and explains how transition metal ions contribute to their characteristic color.

Similar to coloured gemstones, coloured solutions appear the colour they are because of the way they interact with visible light. We learned earlier in [section S3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>) that transition metal complexes appear coloured. But why is it that they are coloured but others are not? What is unique about the way that transition metal complexes interact with visible light? What ways can we quantitatively evaluate the colour of transition metal complexes?

Recall from [section S3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>) that complex ions are formed when ligands form coordinate bonds to the transition metal ion through the donation of a lone pair of electrons. As the ligand approaches the transition metal ion, the

electrons from the ligand will be closer to the electrons in some d-orbitals and further away from others. Let's look, in **Figure 1**, at the shapes of the d-orbitals and the shape of an octahedral complex to predict which d-orbitals the ligands would be closer to and which ones it would be further away from.

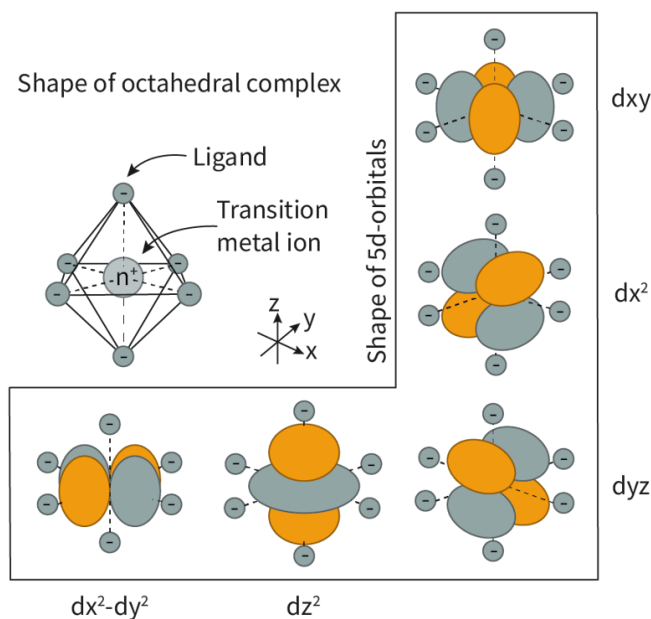


Figure 1. The shapes of an octahedral complex and the five d-orbitals.

 More information for figure 1

The image shows a diagram illustrating the shape of an octahedral complex and the five 3d orbitals: d_{xy} , d_{x^2} , d_{yz} , $d_{x^2-dy^2}$, and d_{z^2} . The octahedral complex is depicted in the top left, showing a central transition metal ion surrounded by six ligands, positioned along the x, y, and z axes.

Adjacent to this, the diagram portrays the shapes of the five 3d orbitals: - **D_{xy} Orbital:** Positioned towards the top right, with lobes oriented between the x and y axes. - **D_{x^2} Orbital:** Below the d_{xy} , with lobes lying directly along the x-axis. - **D_{yz} Orbital:** Further down, oriented between the y and z axes. - **$D_{x^2-Dy^2}$ Orbital:** Located at the bottom-left, with lobes stretching along the x and y axes. - **D_{z^2} Orbital:** To the right of the $d_{x^2-dy^2}$, with lobes elongated along the z-axis.

The d-orbitals are shown with grey and orange lobes indicating the regions of electron density. Some electrons are depicted as grey spheres positioned around the orbitals, emphasizing electron density regions. The orientation of the orbitals corresponds to how they're arranged within an octahedral complex, impacting electron repulsion in complex ion formation.

[Generated by AI]

You can see in **Figure 1** that two of the d-orbitals have electron density in the same region of space as the ligands need to occupy for the formation of a complex ion. This is the $d_{x^2-dy^2}$ and d_{z^2} -orbitals as the electron density within these orbitals aligns along the x, y and z axes as does the arrangement of ligands in an octahedral complex. As these ligands approach a transition metal ion, the electrons in the d-orbitals repel the lone pair electrons on the incoming ligand. This causes the d-

sublevel to split in energy where the orbitals that are closer in proximity to the ligands become higher in energy due to this repulsion and the remaining orbitals become slightly lower in energy. This is illustrated in **Figure 2**.

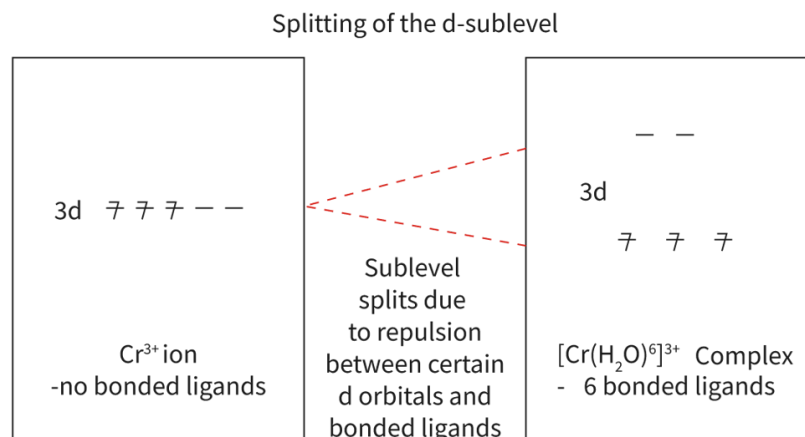


Figure 2. Splitting of the d-sublevel.

 More information for figure 2

The diagram illustrates the splitting of the d-sublevel energy levels in a chromium ion (Cr^{3+}) due to the presence of ligands. On the left, it shows an isolated chromium ion with 3d orbitals at the same energy level, labeled as " Cr^{3+} ion - no bonded ligands." The 3d orbitals are represented by horizontal lines, three of which are at the same energy level.

On the right, the diagram shows how these energy levels split when the ion is part of a complex, labeled as " $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ Complex - 6 bonded ligands." In this part, there are two levels of energy: one with two orbitals higher in energy and the other with three orbitals lower in energy. The splitting is indicated by red dashed lines pointing from the single energy level in the free ion to the two levels in the complex.

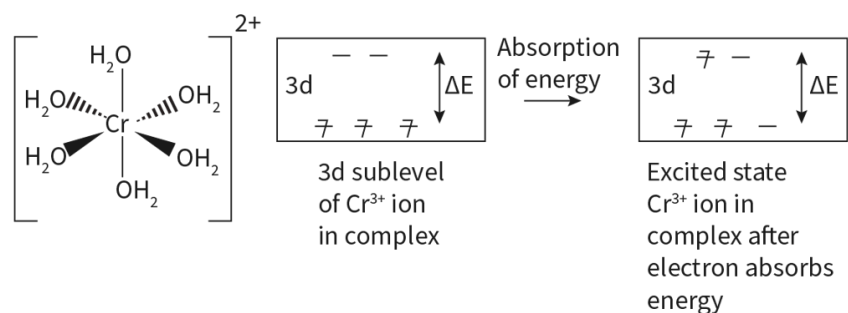
The text notes that the splitting occurs due to repulsion between certain d orbitals and bonded ligands.

[Generated by AI]

Study skills

You do need to know that the d-sublevel splits after the formation of a complex, but you are not required to know which d-orbitals are involved in splitting.

Since the d-sublevel splits into separate energies, this means electrons occupying d-orbitals in the lower level are able to absorb energy and transition to the higher level, forming an excited state. The formation of this excited state transition metal ion is shown in **Figure 3**.

Formation of an excited state Cr^{3+} ion in complex**Figure 3.** Formation of an excited state Cr^{3+} ion in complex.

More information for figure 3

The diagram illustrates the formation of an excited state Cr^{3+} ion in a complex through the absorption of energy. On the left, it shows Cr^{3+} ion surrounded by six water molecules, representing a coordination complex. The ion is under an oxidation state of 2+. Next to it, there are two energy boxes representing the 3d sublevel energy states. The first box shows the ground state, with three pairs of electron spins in lower energy, and an energy gap indicated by ΔE . An arrow labeled "Absorption of energy" points from this ground state box to the right box, which depicts the excited state. In this excited state, one of the pairs of electron spins has transitioned to a higher energy level, leaving two pairs in the lower level, indicating the movement of electrons after absorbing energy.

[Generated by AI]

The difference in energy of the splitting of the d-sublevel in transition metal ions correlates to the energy from photons of visible light. This is illustrated in **Figure 4**. The difference in energy between the split d-sublevels will differ depending on what transition metal ion makes up the complex, what oxidation state the transition element has, the ligands bonded to the ion, and the arrangement of the ligands around the ion.

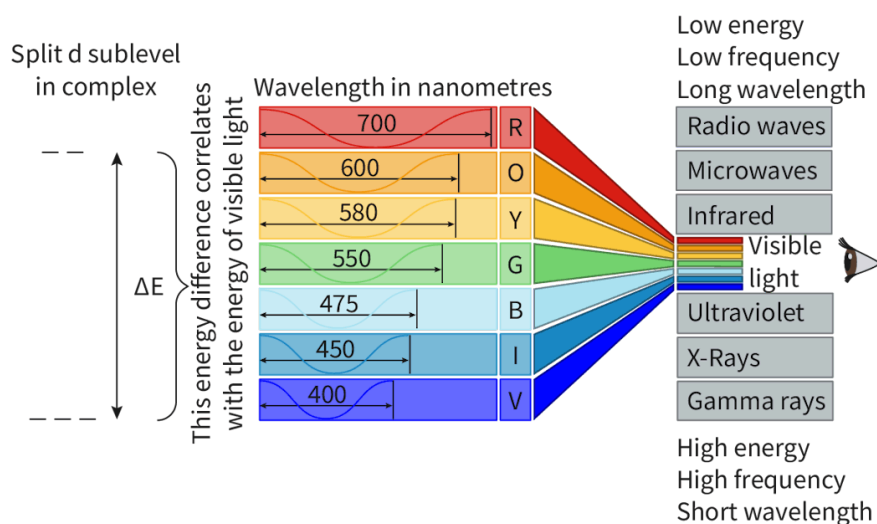


Figure 4. Energy difference between split d-sublevel correlates with visible light.

 More information for figure 4

The image is a diagram that illustrates the correlation between the energy difference in the split d-sublevel of transition metal complexes and the energy of visible light. On the left side, there's a vertical label "Split d sublevel in complex" with arrows indicating an increase in energy difference (ΔE). Beside this is a column labeled "Wavelength in nanometres" with color-coded sections indicating parts of the visible light spectrum: 700 nm (Red), 600 nm (Orange), 580 nm (Yellow), 550 nm (Green), 475 nm (Blue), 450 nm (Indigo), and 400 nm (Violet). Each color band shows double-headed horizontal arrows and sinusoidal wave patterns. On the right, there is a horizontal spectrum showing wavelengths ranging from low energy, low frequency, and long wavelength (Radio waves, Microwaves, Infrared) to high energy, high frequency, and short wavelength (Ultraviolet, X-Rays, Gamma rays). The visible light section is shown in the center of this spectrum.

[Generated by AI]

Since this d-sublevel splitting has the energy difference correlating with visible light, this means that when a solution containing a transition metal complex encounters visible light photons, electrons in the lower d-sublevel of the transition metal ion can absorb specific wavelengths of visible light to form excited state ions. Different complexes will have slightly different energies between the split sublevel resulting in visible light of a different wavelength absorbed for each complex. As specific wavelengths of visible light are absorbed, the remainder wavelengths of visible light are transmitted through the solution. This is illustrated in **Figure 5**.

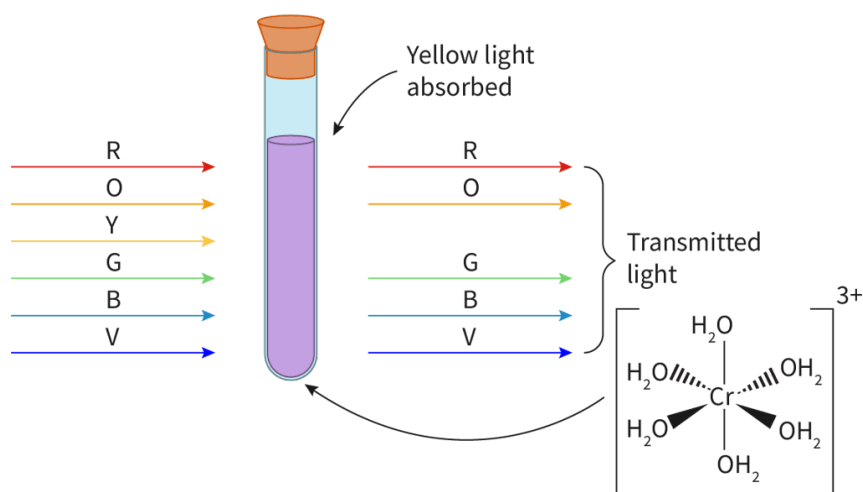


Figure 5. A solution of transition metal complexes absorbing specific wavelengths of visible light.

 More information for figure 5

The image is a diagram illustrating how a solution of transition metal complexes, specifically a chromium complex, absorbs specific wavelengths of visible light. The test tube contains a violet solution, indicating that yellow light (Y) has been absorbed. Arrows labeled as different colors of light, represented as R (Red), O (Orange), Y (Yellow), G (Green), B (Blue), and V (Violet), are shown entering the solution. The yellow light is absorbed, and the remaining

colors are transmitted as indicated by arrows exiting the test tube. The transmitted light refers to the visible light that the eye perceives, shown as an arc labeling the light as 'Transmitted light'. To the right, the chemical structure of the chromium complex $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is displayed. This diagram helps explain the concept of color absorption and transmission related to transition metal complexes.

[Generated by AI]

The transmitted light is what our eyes detect. If you notice in **Figure 5** yellow light was absorbed even though the solution colour appears violet. So how does the colour and wavelength of a transition metal complex relate to the colour and wavelength of light it absorbs? When we remove wavelengths of yellow light from the visible spectrum, our eye perceives the transmitted light as the complementary colour to yellow, which is violet. We can use the colour wheel to predict the wavelengths and colour a solution absorbs based on the colour a solution appears. A version of the colour wheel can be found in [section 15](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/colour-wheel-with-wavelengths-of-the-visible-id-45117/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/colour-wheel-with-wavelengths-of-the-visible-id-45117/>) of the DP chemistry data booklet and in **Figure 6**. The complementary colour and wavelengths can be found across the colour wheel.

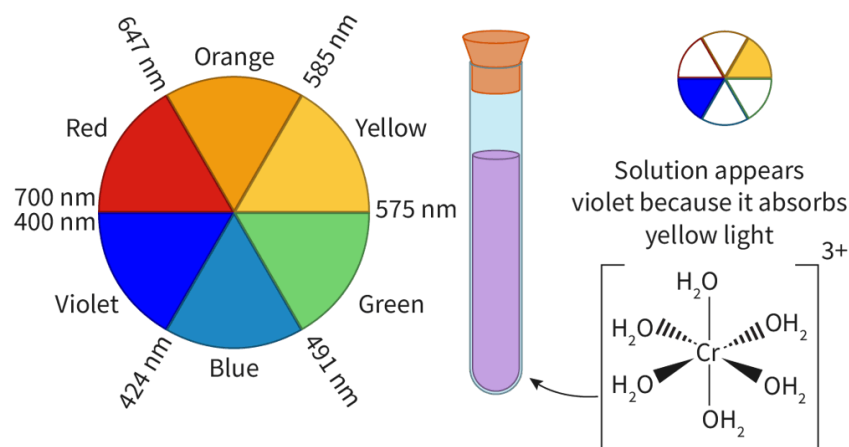


Figure 6. Using the colour wheel to correlate colours of visible light absorbed and perceived by the eye.

More information for figure 6

The image features a color wheel that is divided into six colored sections around a central point. The sections represent different colors of visible light: red, orange, yellow, green, blue, and violet. Each section is labeled with its corresponding wavelength in nanometers (nm). Specifically, red is 647 nm, orange is unlabeled, yellow is 585 nm, green is 575 nm, blue is 491 nm, and violet is 424 nm. The range from 700 nm to 400 nm is also noted.

Beside the color wheel, there is a test tube containing a violet-colored solution. The image explains visually why the solution appears violet: the solution absorbs yellow light, which is represented on the color wheel opposite violet. An additional small color wheel is shown to illustrate complementary colors.

There is also a molecular diagram representing a chromium complex with six water molecules $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ next to the test tube. The text below it states, "Solution appears violet because it absorbs yellow light."

[Generated by AI]

Study skills

It is important to note that transition metal complexes appear coloured because they absorb visible light. The colour they appear to the human eye is dependent on the wavelengths of light that are transmitted. A common misconception is that transition metal complexes emit light. Excited electrons in transition metal complexes tend to release energy in the form of heat when they return to the ground state rather than light. Ensure on assessments to discuss the colour of transition metal complexes in the context of light absorption and transmission, not emission.

We can relate the energy of splitting to the frequency and wavelength of the light absorbed using two simple equations, both found in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) of the DP chemistry data booklet. You learned how to use these equations in [section S1.3.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/>). A summary of these equations and the variables used within these equations is given in **Table 1**.

Table 1. Equations relating the difference in energy of orbitals to the frequency and wavelength of light absorbed.

Equation	Symbols	Units
$E = hf$	E = energy	joules (J)
	h = Planck's constant (6.63×10^{-34} J s)	joule seconds (J s)
	f = frequency	per second (s^{-1})
$c = f\lambda$	c = speed of light (3.00×10^8 m s $^{-1}$)	metres per second (m s $^{-1}$)
	λ = wavelength	metres (m)
	f = frequency	per second (s^{-1})

Looking at these equations you can see that the greater the energy difference in the splitting of the d-sublevel the higher frequency light that will be required to be absorbed since energy and frequency are directly proportional. In the second equation we can see that frequency and wavelength are inversely proportional, meaning that the higher frequency light absorbed will be from photos of light with smaller wavelengths. These equations allow us to calculate energy and frequency

from wavelength absorbed and vice versa. The worked example shows the calculation of frequency of light and energy difference of splitting from a solution absorbing yellow light with a wavelength of approximately $\lambda = 580 \text{ nm}$.

Worked example 1

A transition metal complex appears purple and absorbed yellow light with a wavelength of approximately $\lambda = 580 \text{ nm}$.

Calculate the following, expressing your answers to 3 significant figures:

1. The frequency of light absorbed
2. The energy of the d-sublevel splitting

1.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Change the units where necessary.

Step 3: Write out the equation.

Step 4: Rearrange the equation to make f the subject.

Step 5: Substitute the values given.

Step 6: State the answer with appropriate units and the number of significant figures used in rounding.

2.

Solution steps

Ca

Step 1: Write out the values given in the question.

$h =$

$f =$

Step 2: Write out the equation.

E

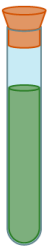
Step 3: Substitute the values given.

E

Solution steps	Ca
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	E

Try **Interactive 2** to match the d-sublevel splitting and the wavelength of absorption to the test tubes containing solutions of transition metal complexes.

d-sublevel splitting



Complex

Absorbs

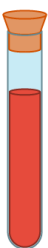
d-sublevel splitting



Complex

Absorbs

d-sublevel splitting



Complex

Absorbs

$\lambda = 450 \text{ nm}$

$\updownarrow \Delta E$

$\lambda = 520 \text{ nm}$

$\updownarrow \Delta E$

$\lambda = 650 \text{ nm}$

$\updownarrow \Delta E$

☒ Check

Interactive 2. Match the d-sublevel splitting and wavelength of absorption to the complex ions.

 More information for interactive 2

An interactive illustration features three test tubes containing coloured complex solutions: green, yellow, and red. Each test tube is closed with a rubber cork. Users need to drag and drop the correct wavelength of absorption and d-sublevel splitting into the correct test tube.

The drag-and-drop options are provided as follows:

Wavelength (λ):

$\lambda = 450 \text{ nm}$

$\lambda = 520 \text{ nm}$

$\lambda = 650 \text{ nm}$

d-orbital splitting diagrams:

Each diagram shows three lower energy orbitals at the bottom and two higher energy orbitals at the top. The vertical distance between them represents ΔE .

Diagram with maximum ΔE

Diagram with medium ΔE

Diagram with minimum ΔE

This interactivity helps users understand the colour of the transition metal complex with the wavelength of light it absorbs and the degree of d-orbital splitting.

Read below for the solution:

Green coloured solution:

Diagram with minimum ΔE

$\lambda = 650 \text{ nm}$

Yellow coloured solution:

Diagram with maximum ΔE

$\lambda = 450 \text{ nm}$

Red coloured solution:

Diagram with medium ΔE

$\lambda = 520 \text{ nm}$

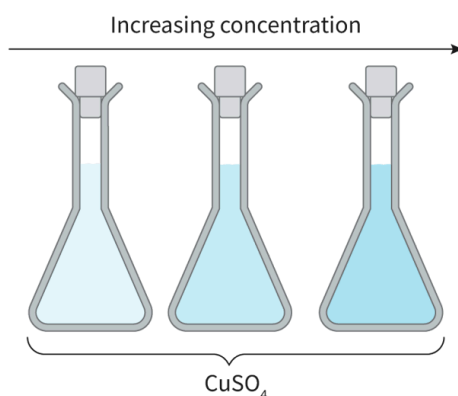
Theory of Knowledge

The colour wheel is an integral tool in the field of art and design to combine colours that are pleasing to the eye. There is evidence of artwork using paint made from ashes and plant materials going back 40 000 years. More recently, paints using lead, cadmium and chromium have been used to provide more vivid colours than natural pigments can provide. It is Sir Isaac Newton, a mathematician, who is credited with arranging colours into a wheel after his experiments with a prism in 1666. Does knowledge of experimental work with prisms validate concepts that artists instinctively understood thousands of years ago?

Quantifying transition metal ions in solution

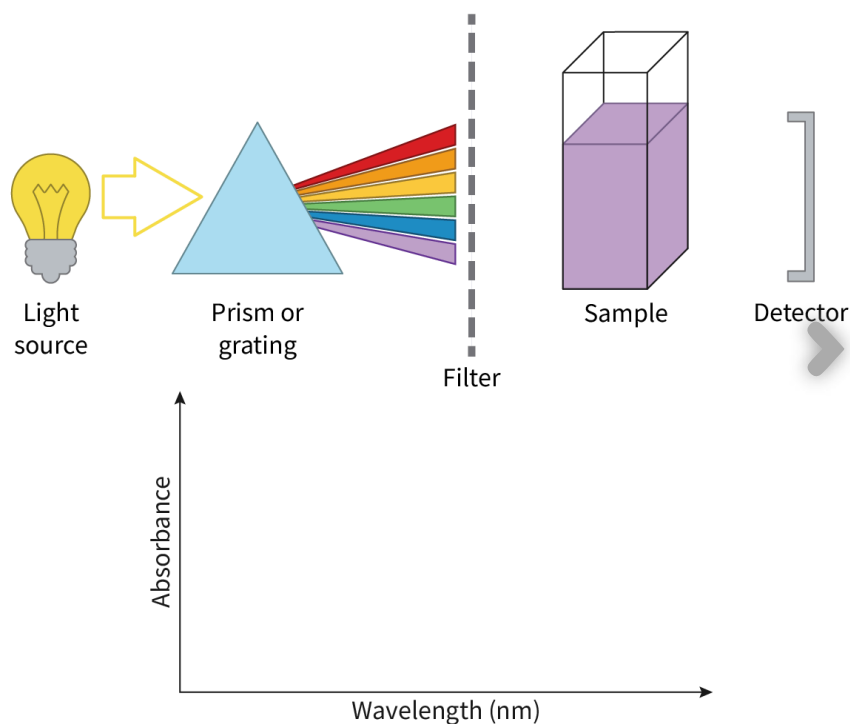
Since transition metal ions formed coloured complexes, we can utilise the colour of the solution to quantify the concentration of transition metal ions in solution. What do you think happens to the appearance of a complex when the concentration is high? What about when it is lower?

Two solutions of $\text{CuSO}_4(\text{aq})$ shown, note what happens to the appearance of the solution colour when the concentration of Cu^{2+} ions increases or decreases.



Two methods that can be used to quantify the concentration of transition metal ions in solution based on the colour they appear are spectrophotometry and colorimetry. **Interactive 3** shows generally how a spectrophotometer works to collect a spectrum of light absorbed.

General set up of spectrophotometer



Interactive 3. How a Spectrophotometer Works to Collect a Spectrum of a Coloured Solution.

 More information for interactive 3

An interactive slideshow presents the general setup of a spectrophotometer and its operation with different wavelengths of light.

Slide 1: Setup

Above: A schematic diagram of a spectrophotometer setup includes a light source, a prism or grating, a filter that allows a specific wavelength of light to pass, a cuvette containing the sample, and a detector that captures the transmitted light.

Below: A graph plots absorbance on the y-axis and wavelength in nm on the x-axis.

The following slides exhibit the graph changes for different colors.

Slide 2: Purple light passing through

Using the prism or grater and the filter, only purple light passes through the sample and reaches the detector.

The graph exhibits a small horizontal bar lying on the x-axis, indicating that no

violet wavelengths are absorbed.

Slide 3: Blue light passing through

Using the prism or grater and the filter, only blue light passes through the sample and reaches the detector.

The graph exhibits a medium-sized horizontal bar lying on the x-axis, indicating that no blue wavelengths are absorbed.

Slide 4: Green light passing through

Using the prism or grater and the filter, only green light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar on the x-axis, indicating that no green wavelengths are absorbed.

Slide 5: Yellow light passing through

Using the prism or grater and the filter, only yellow light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar on the x-axis, followed by a concave downward curve. The peak at $\lambda = 580 \text{ nm}$ on the curve indicates that yellow light is absorbed.

Slide 6: Orange light passing through

Using the prism or grater and the filter, only orange light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar lying on the x-axis, followed by a concave downward curve and a small horizontal bar. The small horizontal bar after the peak indicates that no orange light is absorbed.

Slide 7: Red light passing through

Using the prism or grater and the filter, only red light passes through the sample and reaches the detector.

The graph exhibits a long horizontal bar lying on the x-axis, followed by a concave downward curve and a small horizontal bar. The small horizontal bar after the peak indicates that no red light is absorbed.

This slideshow helps users understand the spectrometer setup and observe light absorption trends.

Both the spectrophotometer and colorimeter provide the absorbance of light at specific wavelengths. A spectrophotometer analyses all the wavelengths in the visible spectrum while a colorimeter only analyses specific wavelengths unique to the physical instrument. **Figure 7** shows how a spectrum produced by a spectrophotometer would change as the concentration of the solution changes.

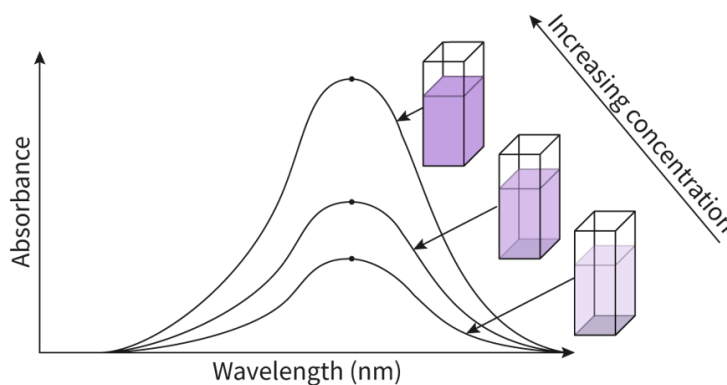


Figure 7. How a spectrophotometry spectrum changes with concentration.

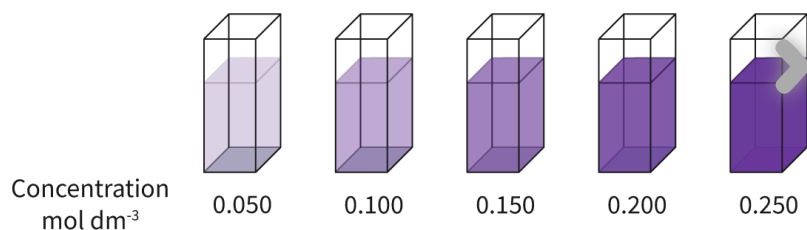
 More information for figure 7

The graph displays the relationship between absorbance and wavelength. The X-axis is labeled 'Wavelength (nm)' and the Y-axis is labeled 'Absorbance.' There are multiple curves representing different concentrations. Each curve shows that absorbance increases with wavelength up to a peak, and then decreases. The peaks of these curves are progressively higher with increasing concentration, which is visually represented by three purple boxes positioned above the curves. An arrow labeled 'Increasing concentration' points diagonally across the graph, illustrating that absorbance increases with concentration.

[Generated by AI]

But how exactly does knowing the absorbance of light help us quantify the concentration of transition metals in solution? This is where we need to utilise some known information for comparison. If we know the identity of the sample, we can make up solutions of known concentrations. We can analyse the absorbance of light for each of the concentrations and compare the absorbance of the unknown concentration to the known concentrations. This enables us to deduce the concentration. **Interactive 4** shows the process for how to do this.

Use spectrophotometer or colorimeter to deduce the absorbance at specified wavelength for each sample



Interactive 4. Deducing the Unknown Concentration of a Sample Using Spectrophotometry.

 More information for interactive 4

An interactive slideshow with three slides demonstrates the process of deducing the unknown concentration of a sample using spectrophotometry. Each slide contains explanatory text along with relevant visuals to illustrate key concepts.

Slide 1:

The text reads as follows: Use a spectrophotometer or colorimeter to deduce the absorbance at a specific wavelength for each sample.

Below are illustrations of five cuvettes containing a purple solution at different

concentrations.

The concentrations of each cuvette, in mol dm^{-3} , are as follows:

Cuvette 1- 0.050

Cuvette 2- 0.100

Cuvette 3- 0.150

Cuvette 4- 0.200

Cuvette 5- 0.250

Slide 2:

The text reads as follows: Plot the absorbance versus concentration for each sample and add the line of best fit.

A scatterplot displays absorbance at 580 nm on the y-axis and concentration on the x-axis. Five data points are marked in an increasing trend, with a best-fit line passing through them. This plot is referred to as the calibration curve.

Slide 3:

The text reads as follows: Use a spectrophotometer to analyze absorbance at the same wavelength of the sample with unknown concentration. Use the line of best fit to determine the concentration for the absorbance value.

An illustration displays a cuvette containing a solution with an unknown concentration on the left. A scatterplot on the right displays absorbance at 580 nm on the y-axis and concentration on the x-axis. Five data points are marked in an increasing trend, with a best-fit line passing through them. A red dashed line extends from the absorbance value on the y-axis to the best-fit line, then continues downward to mark the corresponding unknown concentration on the x-axis.

This slideshow helps users understand how spectrophotometry is used to determine the concentration of an unknown solution.

Practical skills

Tool 1: Experimental techniques — Applying techniques

Analysing complexes using spectrophotometry or colorimetry is an excellent way to study colourful solutions in a data rich manner. We can use these techniques to determine the concentration of solutions containing transition metal ions, to study the wavelength of maximum absorbance as binding ligands are exchanged, or even to test methods to remove metal ions from water sources. Read through the abstract of the following [journal article](https://pubs.acs.org/doi/10.1021/acssuschemeng.5b00207)  (<https://pubs.acs.org/doi/10.1021/acssuschemeng.5b00207>). Can you think of how you might use spectrophotometry in this investigation?

The following activity has some questions for you to consider relating to transition metal complexes.

Activity

- **IB learner profile attribute:** Knowledgeable

- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Individual activity

The following activity has some questions for you to consider. You will need to write out your answers to the questions asked along with accessing a simulation and graphing data.

Part 1: Coloured solutions

Figure 8 shows two beakers. One containing ZnSO_4 the other containing CuSO_4 . Explain, using electron configurations and d-sublevel splitting, why solutions containing the Cu^{2+} ion is coloured while the one containing Zn^{2+} ions is not coloured.

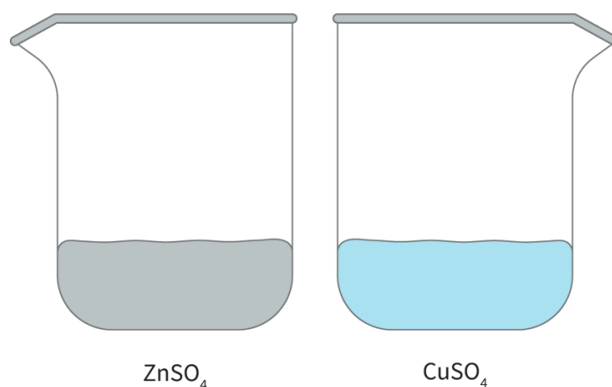


Figure 8. Beakers containing ZnSO_4 and CuSO_4 .

Part 2: Spectrophotometry to deduce unknown concentration

A sample of water, believed to be contaminated with cobalt(II) chloride was found. You can find the concentration of cobalt(II) chloride in your unknown sample by first analysing solutions of five different concentrations of cobalt(II) chloride. Using spectrophotometry, you collect the data shown in **Table 2**.

Table 2. Data collected using spectrophotometry

Concentration (mol dm^{-3})	Absorbance
0.0550	
0.100	
0.135	
0.170	
0.185	

Using this data complete the following:

- Construct a plot with values for concentration on the x-axis and values for the absorbance at $\lambda = 549$ nm on the y-axis.
- Add a trendline and an equation of the line.
- Solve for the concentration of a solution of CoCl_2 with an absorbance of 1.19 at $\lambda = 549$ nm.
- Compare your calculated concentration with a peer.

Part 2: Ligand exchange

By adding different substances, it is common for the ligands bonded to transition metal ions to exchange for different ligands. The ability for ligands to exchange depends on the strength of the interaction between the lone pair of electrons on the ligand and the transition metal ion and the quantity of ligand molecules surrounding transition metal ions (**Table 3**).

Table 3. Ligand exchange data.

Substance	Substance added	Ligand(s)	Wavelength
$\text{CuCl}_2(\text{aq})$	none	Cl^-	
	H_2O	H_2O	
	KSCN	SCN^-	
	NH_3	NH_3	

- Rank the ligands in order of increasing energy between the split d-sublevels of the Cu^{2+} ion. Explain how you used the data in **Table 3** to do so.
- Calculate the frequency (in s^{-1}) of light absorbed for each Cu^{2+} complex.
- Calculate the energy (in J) of the splitting between the d-sublevels for each Cu^{2+} complex.